76.1 1. Conventions and weak estimates at every actual scale

Fix a compact matrix gauge group, its local complexification, a physical four-torus of side T, a fixed odd blocking factor L, and

$$8 < p < \infty, \qquad 0 < s < \tfrac{1}{2}. \tag{WB1}$$

Write $h_j = L^j a$, $h_k = r \le 1$, and $\ell_j = h_{j+1}$. All pairings and norms use physical volume. Transposes in holomorphic identities are invariant bilinear transposes; estimates use absolute Hermitian norms.

The earlier actual raw-coarsening result WP gives the following implication without a curvature-supremum assumption:

$$\|u\|_{W_a^{1,p}} \le M \quad \Longrightarrow \quad U_j = \mathcal{M}^{(j)}(e^{au}) = e^{h_j u_j}, \qquad \sup_{j \le k} \|u_j\|_{W_{h_j}^{1,p}} \le CM. \tag{WB2}$$

The radius is fixed sufficiently small below all the WP and CK radii. Set $m = M + M^2$; constants can absorb higher powers of M because $M \le 1$.

The plaquette BCH expansion gives

$$\|F_{h_j}(u_j)\|_{L^p} \le Cm.$$

For a site or plaquette field the elementary physical-volume inverse inequality is $\|f\|_\infty \le h_j^{-4/p} \|f\|_{L^p}$. Hence

$$\alpha_j := \|F_{h_j}(u_j)\|_\infty \le C h_j^{-4/p} m, \qquad \ell_j^2 \alpha_j \le C_L \ell_j^{2-4/p} m. \tag{WB3}$$

In particular the dimensionless one-step loops remain in a common small logarithmic branch. Since $2 - 4/p > 0$, this branch condition is independent of the number of steps. Equation (WB3) is an inverse lattice estimate with its scale retained; it is not a uniform physical supremum estimate.

Morrey's inequality applied to the multilinear interpolants of the component fields gives

$$\sup_j \|u_j\|_\infty \le CM. \tag{WB4}$$

Every one-step transport therefore has norm and inverse norm at most $\exp(C_L \ell_j M)$. No change of the global frame is made between levels.

76.3 3. Exact Wilson errors without a bounded curvature premise

The actual logarithmic plaquette curvature has the expansion

$$F_a(u) = d_a u + \mathcal{B}(u,u) + a\,\mathcal{R}_a(u, D_a u), \tag{WB11}$$

with the definitions and exact shifts of NK9. The analytic remainder starts with $uD_a u$ and u^3. The weak product estimate gives

$$\|\mathcal{R}_a(u, D_a u)\|_{L^p} \le C(M^2 + M^3).$$

One fine inverse difference on the term already multiplied by a, together with the $W^{1,p}$ algebra estimate on $\mathcal{B}$, proves

$$\|G_a(u)\|_{W_a^{1,p}} \le CM^2, \qquad G_a(u) = F_a(u) - d_a u. \tag{WB12}$$

No second derivative of u is required.

There is also a direct verification of (WB12) that does not separate the constant bracket. Subtract the linear term in the four-link BCH series before dividing by a^2. Every remaining word of degree $n \ge 2$ in shifted values of u has coefficient a^{n-2}. The $W^{1,p}$ algebra estimate bounds its norm by $C_n a^{n-2} M^n$. The analytic word series converges on the common $a\|u\|_\infty$ ball, giving CM^2. This confirms that no hidden curvature supremum or second potential derivative enters (WB12).

For the fractional estimates put $K_u = \|u\|_{H_a^{1+s}}$. WP5–7 give

$$\|G_a(u)\|_{H_a^{1+s}} \le CmK_u, \qquad \|F_a(u)\|_{H_a^s} \le C(1+m)K_u. \tag{WB13}$$

These are the first two conclusions of the earlier curvature-remainder argument. Its separate supremum conclusion is not used.

Write the exact Wilson incidence as

$$\mathcal{C}_U = \delta_a + \mathcal{T}_U,$$

where the zero-order shifted coefficients of $\mathcal{T}_U$ have $W^{1,p}$ norm $O(m)$. The Wilson flux satisfies

$$\Phi_a(F) = F + a^4 \mathcal{P}_a(F).$$

On the small dimensionless plaquette branch,

$$\|\mathcal{P}_a(F)\|_{H_a^s} \le C\|F\|_\infty^2 \|F\|_{H_a^s}.$$

Using (WB3) at the fine mesh, the one inverse difference in $\mathcal{C}_U$ costs a^{-1}, and consequently

$$\begin{aligned} a^4\|\mathcal{C}_U\mathcal{P}_a(F_a(u))\|_{H^s_a} &\le Ca^{\,3-8/p}m^2(1+m)K_u \\ &\le CM^2K_u. \end{aligned} \tag{WB14}$$

The exponent $3-8/p$ is positive under (WB1). Higher terms in the analytic flux series have additional powers of the small dimensionless quantity a^2F, and obey the same bound after summation.

The identical calculation in L^p, with $\|F\|_{L^p} \le Cm$, gives a bound CM^3 for the flux correction. Since $\mathcal{T}_U F$ has L^p norm CM^2 and (WB12) controls $\delta_a G_a(u)$, the exact identity

$$\delta_a d_a u = J_a(u) + f_a(u)$$

has the two estimates

$$\boxed{\|f_a(u)\|_{L^p} \le CM^2, \qquad \|f_a(u)\|_{H^s_a} \le CmK_u.} \tag{WB15}$$

All terms are those of the actual Wilson force, including its flux correction. No continuum Euler equation has been substituted.

76.4 4. A microscopic-cutoff-independent curvature bound for CK

Theorem WB-A. On the weak analytic branch of CK, with $b = \|B\|_{W^{1,p}_r}$ below one fixed radius,

$$\boxed{\begin{aligned} \|F_a(w(B))\|_{L^p} &\le Cb, \\ \|F_a(w(B))\|_{W^{1,p}_a} &\le Cr^{-1}b, \\ \|F_a(w(B))\|_{\infty} &\le Cr^{-4/p}b. \end{aligned}} \tag{WB16}$$

The estimates hold on the same common complex weak ball. Their constants are independent of a, r, k, with the powers of r displayed.

Proof. CK gives $\|w\|_{W^{1,p}} + \|\lambda_B\|_{W^{-1,p}_r} \le Cb$. The coarse inverse inequality, proved by duality from $\|\varphi\|_{W^{1,p'}_r} \le Cr^{-1}\|\varphi\|_{L^{p'}}$, is

$$\|\lambda_B\|_{L^p_r} \le Cr^{-1}\|\lambda_B\|_{W^{-1,p}_r}.$$

Apply (WB8), at exponent p, to the genuine coarse Ward multiplier. This gives

$$\|J_a(w)\|_{L^p} \le Cr^{-1}b. \tag{WB17}$$

The flat curl-energy equation in (WB15), on nonzero modes, implies

$$\|d_a w\|_{W^{1,p}_a} \le C_{p,T}(\|J_a(w)\|_{L^p} + \|f_a(w)\|_{L^p}).$$

This is the usual periodic flat elliptic multiplier applied to the transverse part; it does not require a gauge condition on w. A constant potential has zero flat curl. Equations (WB12), (WB15), and (WB17) prove the second line of (WB16). The first follows directly from the weak plaquette expansion.

For completeness, the uniform discrete Gagliardo–Nirenberg estimate on the fixed torus is

$$\|f\|_\infty \le C_{p,T}\left(\|f\|_{L^p} + \|f\|_{L^p}^{1-4/p}\|f\|_{W^{1,p}}^{4/p}\right).$$

It follows by periodic multilinear interpolation and the continuum inequality, with nodal values unchanged. Apply it to each plaquette component and use $r \le 1$. This proves the last line of (WB16).

At fixed physical retained spacing $r > 0$, this is a uniform matrix-curvature bound as $a \downarrow 0$. It is stronger than the direct fine inverse bound $a^{-4/p}b$. It is not a bound uniform as $r \downarrow 0$. Choosing a larger fixed finite p makes the displayed power smaller only when the corresponding input $W^{1,p}$ bound is actually available.

Uniform parameter Cauchy estimates follow from (WB16) on the common weak polydisc, with the same retained-scale factor. No complex gauge-anchor estimate is invoked.

76.5 5. The fractional periodic theorem with no curvature-supremum premise

Theorem WB-B. Fix the parameters of (WB1). Suppose the actual periodic fields satisfy

$$\begin{aligned} &\|A\|_{W^{1,p}_a} + \|u\|_{W^{1,p}_a} \le \varepsilon_*, \\ &\mathcal{M}_r(e^{au}) = \mathcal{M}_r(e^{aA}) = V, \\ &J_a(u) = D\mathcal{M}_r(e^{au})^{\mathsf{t}}\lambda, \qquad d_V^{\mathsf{t}}\lambda = 0. \end{aligned} \tag{WB18}$$

There is no separate bound on $\|F_a(A)\|_\infty$ or $\|F_a(u)\|_\infty$. Then

$$\boxed{\|\lambda\|_{H^s_r} + \|J_a(u)\|_{H^s_a} + \|F_a(u)\|_{H^{1+s}_a} \le C\left(\|A\|_{H^{1+s}_a} + \|F_a(A)\|_{H^{1+s}_a}\right).} \tag{WB19}$$

There is a holomorphically chosen gauge equal to the identity at every retained node for which the transformed potential $\widetilde{u}$ also satisfies

$$\|\widetilde{u}\|_{H^{1+s}_a} \le C\left(\|A\|_{H^{1+s}_a} + \|F_a(A)\|_{H^{1+s}_a}\right). \tag{WB20}$$

The initial small radius and all constants are independent of a, r, k.

Proof. First work in the exact flat smoothed Landau slice. The actual transverse constraint comparison WP17 and the full raw remainder WP18 depend only on the weak potential chart and on the displayed high norms. Their bounds remain

$$\|e_c\|_{H^s_r} \le Cr^2 m(K_A + K_u), \qquad \|N_{\mathrm{raw}}(v)\|_{H^{1+s}_r} \le Crm\|v\|_{H^{1+s}_a}. \tag{WB21}$$

Replace the former actual-current estimate by (WB10) and the former Wilson error estimate by (WB15). The coarse Ward correction is a zero-order multiplier with $W^{1,p}$ coefficients of size $O(m)$, so its H^s estimate uses WP7 and no curvature bound.

All remaining KKT steps are flat Fourier identities and bounds: the transverse Schur estimates NK19–22, the longitudinal Ward estimate NK25–26, and the mean Euler equation determining the harmonic multiplier. Repeating those identities now gives

$$\|\lambda\|_{H^s_r} + \|J_a(u)\|_{H^s_a} + \|F_a(u)\|_{H^{1+s}_a} \le C\left[\|F_a(A)\|_{H^{1+s}_a} + \|A\|_2 + m(K_A + K_u)\right]. \tag{WB22}$$

The r^{-2} in the transverse Schur inverse is canceled by the r^2 in (WB21). The current error contributes only $Cm\|\lambda\|_{H^s_r}$, by the last inequality in (WB10).

The slice closure PL25–33 uses the mixed biharmonic lift, the full raw remainder in (WB21), and the harmonic mean constraint. It therefore gives

$$K_u \le C\left(\|A\|_{H^{1+s}_a} + \|F_a(A)\|_{H^{1+s}_a}\right) + CmK_u. \tag{WB23}$$

At each finite cutoff K_u is finite. Choose the fixed weak radius so that the final coefficient is at most one half, and absorb it. This proves (WB19)–(WB20) in the slice.

For a field initially outside that slice, GS constructs the retained-node-identity gauge using only its weak $W^{1,p}$ norm. The gauge parameter is bounded in $W^{2,p}$, and the transformed weak potential is still small after decreasing the initial radius once. The raw target and physical equations are preserved exactly. Apply the preceding slice argument with the original input A unchanged.

Finally choose $s < \alpha < 1 - 4/p$. The gauge and its inverse have bounded $C^{1,\alpha}$ norms, so conjugation transfers curvature H^{1+s} and current H^s back to the original representative. This multiplication estimate needs no curvature supremum. The coarse multiplier is unchanged because the gauge fixes retained nodes. This proves the theorem.

The theorem controls the high potential norm in the constructed slice. It does not silently transfer that potential norm through a gauge for which only $W^{2,p}$, rather than H^{2+s}, has been proved.

76.6 6. A common complex fractional hull for the actual critical map

Define the linear space

$$\|A\|_{\mathcal{Z}_s} = \|A\|_{W_a^{1,p}} + \|A\|_{H_a^{1+s}} + \|d_a A\|_{H_a^{1+s}}. \tag{WB24}$$

On a small weak ball, (WB13) makes its strong part equivalent, up to fixed constants, to $\|A\|_{H^{1+s}} + \|F_a(A)\|_{H^{1+s}}$.

Compose the actual raw map, CK, and the retained-node gauge section:

$$A \longmapsto B = q_{r,\mathrm{raw}}(A) \longmapsto w(B) \longmapsto \widetilde{w}(A). \tag{WB25}$$

All three maps are holomorphic on one common weak neighborhood. They preserve the actual target, and CK supplies the physical KKT and Ward equations. Theorem WB-B therefore gives

$$\|\widetilde{w}(A)\|_{\mathcal{Z}_s} \le C\|A\|_{\mathcal{Z}_s}. \tag{WB26}$$

Here the $d\widetilde{w}$ term follows by subtracting $G_a(\widetilde{w})$ from the controlled curvature and using (WB13).

If $A(z)$ is holomorphic on a fixed polydisc, remains in that weak neighborhood, and has $\sup_z \|A(z)\|_{\mathcal{Z}_s} \le K$, then

$$\|\partial_z^\gamma \widetilde{w}(0)\|_{\mathcal{Z}_s} + \|\partial_z^\gamma F_a(\widetilde{w})(0)\|_{H_a^{1+s}} \le CK\,\gamma! \prod_i \rho_i^{-\gamma_i}. \tag{WB27}$$

Use smaller open radii and take their limit when needed.

The linear hull is stable under bounded scalar interpolation and smooth spatial cutoffs with the explicitly required derivative norms. The proof uses the exact difference-product rule for $d_a(\chi A)$, the H^{1+s} multiplication estimate, and (WB13). No claim that the interpolated field remains stationary is made.

This closes the simultaneous weak-potential/fractional-curvature chart for the actual periodic critical map on this input hull. It does not establish the hull premise for an arbitrary original paired-core input: the original $C^{1,\beta}$ potential allowance alone need not bound dA in H^{1+s}.

76.7 7. Compatible Euler and constraint residuals in the weak chart

The same removal of the curvature-supremum premise applies to CR. Suppose

$$J_a(u) = D\mathcal{M}_r(e^{au})^{\mathsf{t}}\lambda + E, \qquad V = \mathcal{M}_r(e^{au}), \qquad q = d_V^{\mathsf{t}}\lambda. \tag{WB28}$$

No separate Ward condition is imposed. Exact gauge invariance gives

$$d_U^{\mathsf{t}} E = -N^{\mathsf{t}} q.$$

The atomic embedding estimate CR18–21 and the weak $W^{1,p}$ multiplier bound use no curvature supremum. Therefore

$$r^2 a^{-1-s}\|q\|_2 + \|q\|_{H_r^{s-1}} \le C\|E\|_{H_a^s}. \tag{WB29}$$

Propagate complete actual currents even when their divergences do not vanish. The exact defect at spacing h_j is $q_j = N_{j,r}^{\mathsf{t}} q$, of norm $(r/h_j)^2\|q\|_2$. Before imposing any Ward restriction the one-step identity is

$$DM_j^{\mathsf{t}} = T_j^0 + B_j - \rho_j^{\mathsf{t}} d_{U_{j+1}}^{\mathsf{t}},$$

where

$$\|B_j\|_{2\to 2} \le C(M\ell_j + m\ell_j^{2-4/p}), \qquad \|\rho_j^{\mathsf{t}}\|_{2\to 2} \le C\ell_j. \tag{WB30}$$

The bounded coefficient product from (WB7), followed by the endpoint geometric sum, gives

$$\|\lambda_j\|_2 \le C\|\lambda\|_2 + Cr^2 h_j^{-1}\|q\|_2. \tag{WB31}$$

Unroll with flat tails, as in CR27. The endpoint contribution in H^s is bounded by $Cr^2\ell_j^{-1-s}\|q\|_2$. The additional defect part from $B_j\lambda_{j+1}$ is bounded by

$$Cr^2\big(M\ell_j^{-s} + m\ell_j^{1-s-4/p}\big)\|q\|_2.$$

The first sum is dominated by the smallest scale. The second exponent is strictly positive because $p > 8$ and $s < 1/2$, so that sum is bounded by its largest-scale term. Both are bounded by a fixed multiple of $r^2 a^{-1-s}\|q\|_2$. Hence (WB29) proves

$$\boxed{\|D\mathcal{M}_r^{\mathsf{t}}\lambda - Q_{\mathrm{bl}}^{\mathsf{t}}\lambda\|_{H_a^s} \le C(Mr^{1-s} + mr^{2-s-4/p})\|\lambda\|_2 + C\|E\|_{H_a^s}.} \tag{WB32}$$

This is an unrestricted-current estimate derived before taking the Ward defect to zero.

Let the actual logarithmic constraint residual be

$$q_{r,\mathrm{raw}}(u) - q_{r,\mathrm{raw}}(A) = b$$

and define

$$\mathfrak{B}_{s,r}(b) = \|\Pi_r^0 b\|_{H_r^{2+s}} + \|P_{L,r} b\|_{H_r^{1+s}} + \|P_{H,r} b\|_2. \tag{WB33}$$

The transverse, longitudinal and harmonic projections are the flat coarse Hodge projections, with the same conventions as CR36.

Theorem WB-C. Under the small weak potential hypothesis in (WB18), the actual residual equation (WB28), and the displayed constraint residual, there is the retained-node-identity gauge of GS for which

$$\boxed{\begin{aligned}&\|\lambda\|_{H_r^s} + \|J_a(u)\|_{H_a^s} + \|F_a(u)\|_{H_a^{1+s}} + \|\widetilde{u}\|_{H_a^{1+s}} \\ &\quad\le C\left[\|A\|_{H_a^{1+s}} + \|F_a(A)\|_{H_a^{1+s}} + \|E\|_{H_a^s} + \mathfrak{B}_{s,r}(b)\right].\end{aligned}} \tag{WB34}$$

No curvature-supremum hypothesis and no independent coarse Ward hypothesis are used.

To prove the statement, insert (WB32) and (WB15) into the same coarse Hodge and transverse Schur equations as in CR31–37. The longitudinal multiplier is controlled by the second term in (WB29), the harmonic multiplier by the mean Euler equation, and the transverse multiplier by the Schur

inverse. The constraint error has exactly the three norms in (WB33). The slice closure absorbs the same $CmK_{\widetilde{u}}$ term. The GS gauge transports E in H^s, while fixing both raw targets and the coarse multiplier. Every remaining step is therefore the one already displayed in the proof of WB-B or CR37.

The theorem controls a residual once it has been assembled in the actual physical equation. It does not establish that an auxiliary formation residual belongs to H^s, is small there, or is supported outside a protected region. Those statements require the actual formation formula.

76.8 8. Hypotheses still outside these results

The microscopic curvature bound (WB16) retains $r^{-4/p}$. It cannot be reported as a two-scale-uniform matrix bound. The fractional theorem (WB19), by contrast, is uniform in both scales and does not require that matrix bound.

For localized formation, the field must satisfy the actual periodic equations or an appropriate proved local/residual version. An original independent pair (U, J) is not necessarily physical; a response map centered at that pair cannot simply be replaced by CK.

For a mixed-boundary or wall-dependent field, one must still prove the common weak chart and the exact protected equation or compatible residual. For an original low-derivative input, one must still obtain the required strong input norm by an actual equation or another smoothing estimate.

None of these estimates controls interactions near an arbitrary renormalized operator insertion or constructs the common continuum field hierarchy. Those remain separate quantum construction tasks.

77.1 1. Statement

Let a divide r with n=r/a odd, and let Q=Q_bl,a,r be the actual direct centered box-and-line average. Let H_bl be the unique flat curl-energy reconstruction in the exact slice of FW, with prescribed coarse datum B. For every 1<=p<=infinity,

$$\boxed{\|d_a H_{\mathrm{bl}} B\|_{L^p_a} \le C \|d_r B\|_{L^p_r}.} \tag{FE1}$$

The constant is independent of a,r,n and the compatible periodic volume. In particular,

$$\boxed{\|d_a H_{\mathrm{bl}} Q_{\mathrm{bl}} A\|_{L^\infty_a} \le C \|d_a A\|_{L^\infty_a}.} \tag{FE2}$$

No bound for a potential in L infinity is required on the right. The same curvature bounds hold for H_raw, because FW25 changes H_bl by a fine gradient. Moreover d_rQ_raw=d_rQ_bl, so the same conclusion holds for the actual raw constrained curvature projection.

Here d denotes the exact forward lattice exterior derivative, with the corresponding adjoint delta. A componentwise form norm is understood; finite-dimensional equivalent norms only change C.

77.2 2. The affine first jet of the full-gradient reconstruction

Rescale r=1 and a=1/n. Write

$$H_0 = G_a Q^*(QG_aQ^*)^{-1}, \qquad H_{\mathrm{bl}} = H_0 P_C + \text{a fine gradient.} \tag{FE3}$$

The second equality is FW21. Index fine sites by a coarse site y and an offset xi in n^{-1}Z^4 intersected with a fixed half-open unit cube. Let h_mu,xi(z) be the offset multiplier of H_0 in spatial component mu. FW8 gives

$$h_{\mu,\xi}(z) = \frac{e^{iz\cdot\xi}\overline{q}_{\mu,0}(z) + \lambda_0(z)\sum_{\nu\neq 0} e^{i(z+2\pi\nu)\cdot\xi}\overline{q}_{\mu,\nu}(z)\lambda_\nu(z)^{-1}}{|q_{\mu,0}(z)|^2 + \lambda_0(z)\sum_{\nu\neq 0}|q_{\mu,\nu}(z)|^2\lambda_\nu(z)^{-1}}. \tag{FE4}$$

Bars and squared absolute values in this formula have their analytic conjugate extensions when z is complex. The alias estimates FW9–13 imply uniform analyticity and bounded fixed-order derivatives both for h and for its physical fine first differences, on a fixed complex strip. These are the exact uniform estimates established in FW14; no new derivative of a rough coefficient is assumed here.

For the actual centered filters, in the bond-midpoint Fourier convention, the principal alias is explicitly

$$q_{\mu,0}(z) = \left\{\prod_{j=1}^{4}\frac{\sin(z_j/2)}{n\sin(z_j/(2n))}\right\}\frac{\sin(z_\mu/2)}{n\sin(z_\mu/(2n))} = 1 + O(|z|^2), \tag{FE4a}$$

with all fixed-order bounds uniform in n. Each nonprincipal alias has at least one zero box numerator at z=0, so q_mu,nu(0)=0 for nu different from zero. The derivative estimates FW9 and the summable first-difference majorant FW11 therefore show that the nonprincipal numerator in FE4, before its lambda_0 factor, is O(|z|), uniformly in xi,n, and remains O(|z|) after a physical fine first difference. Since lambda_0=O(|z|^2), the entire nonprincipal contribution to h and to its first fine difference is O(|z|^3). This verifies directly that no alias contributes to the affine jet. Changing from midpoint phases to site phases adds only a fixed component-dependent first moment to q_mu,0 and to the output offset; the cancellation below treats that convention as well.

At z=0, lambda_0 and its first derivatives vanish, while q_mu,0(0)=1. The numerator and denominator in FE4 therefore give the exact first jet

$$h_{\mu,\xi}(0) = 1, \qquad \partial_{z_j} h_{\mu,\xi}(0) = i\xi_j - \partial_{z_j} q_{\mu,0}(0). \tag{FE5}$$

The possible component-dependent first moment of the averaging convention is independent of xi. It cancels when a fine difference is taken. If a neighboring offset wraps across a coarse cell, its value multiplier has the extra coarse phase e^{iz_j}; the same calculation still gives

$$D_{a,j}h_{\mu,\xi}(0) = 0, \qquad \partial_{z_k}(D_{a,j}h_{\mu,\xi})(0) = i\delta_{jk}. \tag{FE6}$$

Indeed the offset changes by 1/n and the physical difference multiplies by n. A coarse-cell wrap adds precisely the missing integer to the offset first moment. This proves affine reproduction at the level of the first difference even in conventions where the value has an offset.

Let L_xi(z) denote the multiplier from a coarse one-form B to the fine two-form d_aH_0B at a fine plaquette with offset xi. Let d_1(z) be the coarse exterior derivative from one-forms to two-forms. Since its own first jet is i z wedge, FE6 proves

$$L_\xi(z) = iz \wedge + O(|z|^2), \qquad D_\xi(z) := L_\xi(z) - d_1(z) = O(|z|^2). \tag{FE7}$$

More precisely, for each fixed multi-index alpha,

$$|\partial_z^\alpha D_\xi(z)| \le C_\alpha |z|^{2-|\alpha|}, \qquad 0 < |z| \le 1, \tag{FE8}$$

uniformly in xi and n. For derivative orders at least two this follows from uniform analyticity and |z|<=1; the two lower orders use the exact vanishing first jet. Away from zero all those derivatives are uniformly bounded.

77.3 3. Remove the angular singularity before estimating the kernel

Let d_0(z) be the coarse gradient and let P_C be the exact weighted Hodge projection FW16. Its uniform multiplier bounds and algebra are

$$|\partial^\alpha P_C(z)| \le C_\alpha |z|^{-|\alpha|}, \qquad P_C d_0 = 0, \qquad d_1 P_C = d_1. \tag{FE9}$$

The last equality follows because I-P_C has gradient range. At the harmonic mode P_C acts as the identity. Define

$$E_\xi(z) = L_\xi(z) P_C(z) - d_1(z) = D_\xi(z) P_C(z). \tag{FE10}$$

This is the crucial subtraction. It yields

$$|\partial^\alpha E_\xi(z)| \le C_\alpha |z|^{2-|\alpha|}, \qquad E_\xi d_0 = 0, \tag{FE11}$$

and E_xi has value zero on constants. Thus, by the exact coarse Hodge decomposition of a one-form,

$$E_\xi B = E_\xi \delta_2 G_r (d_r B), \tag{FE12}$$

where delta_2 is the adjoint from two-forms to one-forms and G_r is the mean-zero scalar lattice Green operator acting componentwise. The gradient and harmonic pieces of B vanish in this identity. No projection of an arbitrary two-form in L infinity is being bounded.

In retained units the multiplier of delta_2G_r has order minus one. Consequently

$$T_\xi(z) := E_\xi(z) \delta_2(z) \lambda_r(z)^{-1}, \qquad T_\xi(0) = 0, \qquad |\partial^\alpha T_\xi(z)| \le C_\alpha |z|^{1-|\alpha|}. \tag{FE13}$$

All derivatives are uniformly bounded away from zero. Restoring r does not change T: E acquires a factor r^{-1}, while delta_2G_r acquires r.

77.5 5. The box average commutes with curl through a positive filter

Let R be the centered site box filter before decimation, and let L_mu be the normalized line filter of length n in direction mu. In physical connection units the mu component of Q_bl is R L_mu. The exact telescoping identity for differences is

$$D_{r,\mu} = L_\mu D_{a,\mu} \tag{FE17}$$

when evaluated on retained nodes. All the flat filters commute before decimation. Hence

$$(d_r Q_{\mathrm{bl}} A)_{\mu\nu} = R L_\mu L_\nu (d_a A)_{\mu\nu} =: (Q_{\mathrm{bl}}^{(2)} d_a A)_{\mu\nu}. \tag{FE18}$$

The two-form filter has nonnegative coefficients summing to one. It is therefore an L infinity contraction, and the same physical averaging argument gives its Lp contraction for 1<=p<=infinity. Applying FE1 to B=Q_bl A proves FE2, and in fact its Lp version.

Finally, Q_raw=Q_bl-d_rK, so d_rQ_raw=d_rQ_bl. FW25 says that H_raw-H_bl has fine gradient range. Since d_aH_bl kills coarse gradients by FE1,

$$d_a H_{\mathrm{raw}} Q_{\mathrm{raw}} A = d_a H_{\mathrm{bl}} Q_{\mathrm{bl}} A. \tag{FE19}$$

This confirms that the endpoint estimate concerns the actual constrained flat curvature, not an unrelated full-gradient surrogate. It supplies the missing flat lemma in the proposed nonlinear comparison; that comparison still has to verify its own nonlinear forcing, flux, and constraint estimates.

78.1 1. Statement and the precise domain

Fix a physical four-torus of side T, a fixed odd blocking factor L, 8<p<infinity, and compatible spacings a and r=Lˆk a with r<=1. All norms use physical volume. Constants below can depend on p,T,L and the compact matrix group, but not on a,r or k. Transposes in the field equations are the invariant complex-bilinear transposes. Norm inequalities use absolute Hermitian norms.

Let q_raw be the actual k-step raw logarithmic mean, let

$$B = q_{r,\mathrm{raw}}(A), \qquad u = w(B), \qquad V = e^{rB}, \tag{CC1}$$

where w is the CK branch. Thus the actual group constraint and physical Euler equation hold:

$$\mathcal{M}_r(e^{au}) = \mathcal{M}_r(e^{aA}) = V, \qquad J_a(u) = D\mathcal{M}_r(e^{au})^{\mathsf{t}}\lambda, \qquad d_V^{\mathsf{t}}\lambda = 0. \tag{CC2}$$

The weak CK and WP bounds give, after increasing one fixed constant,

$$\|A\|_{W_a^{1,p}} + \|u\|_{W_a^{1,p}} + \|B\|_{W_r^{1,p}} + \|\lambda\|_{W_r^{-1,p}} \le M, \qquad M \le \varepsilon_*. \tag{CC3}$$

Here M can be chosen to be C times the input Wˆ{1,p} norm. Negative Sobolev norms are dual to Wˆ{1,p'} with the physical-volume pairing; they are inhomogeneous norms on the fixed torus, including constants.

Theorem CC. There is a fixed positive epsilon_* for which

$$\boxed{\|F_a(u)\|_\infty \le C(\|F_a(A)\|_\infty + M^2).} \tag{CC4}$$

The estimate holds for complex A on the common weak branch, with the right side finite at each cutoff. In particular, define the linear input norm

$$\|A\|_{\mathcal{Y}_a} = \|A\|_{W^{1,p}_a} + \|d_a A\|_\infty. \tag{CC5}$$

On one common small complex Y_a ball, the same-target branch has uniformly small potential W^{1,p} norm and uniformly small matrix curvature supremum. This statement includes flat and reducible input configurations. It does not claim that a W^{1,p} potential alone has bounded curvature uniformly in a.

78.2 2. The two multiplier errors are genuinely of quadratic size

Write Q=Q_bl,a,r. Propagate the complete Ward-compatible current through the endpoint-removed chain as in WB5–8. Unroll its difference from the flat chain at exponent p, using only flat tails on each error. WB6 and the geometric sums give

$$\begin{aligned} e_A &= D\mathcal{M}_r(e^{au})^{\mathsf{t}}\lambda - Q^{\mathsf{t}}\lambda, \\ \|e_A\|_{L^p_a} &\le C(Mr + (M+M^2)r^{2-4/p})\|\lambda\|_{L^p_r}. \end{aligned} \tag{CC6}$$

The first identity uses the exact Ward equation to identify the actual and endpoint-removed currents; it does not assign a Ward equation to an isolated error. The elementary coarse inverse estimate, by duality, is

$$\|\lambda\|_{L^p_r} \le Cr^{-1}\|\lambda\|_{W^{-1,p}_r}. \tag{CC7}$$

Since 1-4/p>0, equations CC3, CC6 and CC7 imply

$$\|e_A\|_{L^p_a} \le CM^2. \tag{CC8}$$

The longitudinal part of lambda needs a different estimate; applying CC7 too early would lose a power of r. Let delta_r be the flat divergence and write its Hodge decomposition as

$$\lambda = \lambda_T + \lambda_L + \lambda_H, \qquad \lambda_L = d_r G_r \delta_r \lambda, \tag{CC9}$$

where lambda_T is mean-zero and flat co-closed, lambda_H is harmonic, and G_r is the mean-zero scalar Green operator.

The exact coarse Ward equation gives

$$\delta_r \lambda = R_B \lambda, \qquad R_B = \delta_r - d_V^{\mathsf{t}}. \tag{CC10}$$

R_B is a finite sum of zero-order shifted multipliers with W_r^{1,p} coefficient norm at most CM. For example its link coefficient is a shifted version of r^{-1}(Ad(exp(rB))-I), an analytic function equal to ad(B)+O(rB^2). Multiplication by a W^{1,p} coefficient acts on W^{1,p'} with its W^{1,p} norm for p>4, uniformly in the lattice spacing. Indeed the differentiated coefficient is in L^p and its partner is in L^{p/(p-2)}, which follows from the W^{1,p'} Sobolev embedding. The other product term uses the coefficient supremum. Duality therefore proves

$$\|R_B\lambda\|_{W^{-1,p}_r} \le CM\|\lambda\|_{W^{-1,p}_r} \le CM^2. \tag{CC11}$$

On the fixed torus d_rG_r maps mean-zero W_r^{-1,p} to L_r^p uniformly for 1<p<infinity. This is the ordinary discrete elliptic multiplier of order minus one. The right side of CC10 has mean zero because it equals a flat divergence. Hence

$$\|\lambda_L\|_{L^p_r} \le CM^2, \qquad \|Q^{\mathsf{t}}\lambda_L\|_{L^p_a} \le CM^2. \tag{CC12}$$

The last inequality uses the physical Lp contraction of the normalized box-and-line adjoint. Both CC8 and CC12 are thus uniform as r tends to zero.

78.3 3. Separate a divergence of a Holder-small tensor

Let F=F_a(u) be the logarithmic plaquette curvature and define

$$G_u = F - d_a u, \qquad \Phi_a(F) = F + a^4\mathcal{P}_a(F). \tag{CC13}$$

Here Phi is the exact Wilson flux, including the Lie-algebra projection where necessary. The analytic function P starts at field degree three. These definitions retain the exact shifts and matrix orders of the Wilson force. On the common small dimensionless plaquette branch, WB12 and the elementary inverse estimate give

$$\|G_u\|_{W^{1,p}_a} \le CM^2, \qquad \|F\|_{L^p_a} \le CM, \qquad \|F\|_\infty \le Ca^{-4/p}M. \tag{CC14}$$

The last bound is used only with explicit positive powers of a; it is not the conclusion being proved. Choose once and for all

$$0 < \alpha < 1 - 4/p. \tag{CC15}$$

Uniform discrete Morrey gives $\|G_u\|_{C^\alpha} \le CM^2$. For the flux correction the elementary discrete Holder inverse inequality and CC14 give

$$a^4\|\mathcal{P}_a(F)\|_{C^\alpha} \le Ca^{4-\alpha}\|F\|^3_\infty \le Ca^{4-\alpha-12/p}M^3 \le CM^3. \tag{CC16}$$

The exponent is positive: CC15 makes it greater than 3-8/p. Higher terms in P contain further powers of the uniformly small quantity $a^2\|F\|_\infty$. Thus they obey the same estimate after summation.

Write the exact incidence operator as C_U=delta_a+T_U. The shifted zero-order coefficients of T_U have supremum at most CM. Using the physical Euler equation CC2 and CC8, define

$$K = G_u + a^4\mathcal{P}_a(F), \qquad h = -\mathcal{T}_U\Phi_a(F) + e_A + Q^{\mathsf{t}}\lambda_L. \tag{CC17}$$

All signs are determined by J=(delta_a+T_U)Phi. We have the exact identity

$$\delta_a d_a u = Q^{\mathsf{t}}\lambda_T + Q^{\mathsf{t}}\lambda_H - \delta_a K + h, \qquad \|K\|_{C^\alpha} + \|h\|_{L^p_a} \le CM^2. \tag{CC18}$$

For the h estimate, $\|\Phi_a(F)\|_p \le C\|F\|_p$ follows from $a^2\|F\|_\infty$ small, and the other two summands use CC8 and CC12.

Let P_a^0 be the fine flat mean-zero co-closed projection and put

$$v = G_a P_a^0(-\delta_a K + h). \tag{CC19}$$

The curl removes the gradient and constant parts of this projection, so

$$d_a v = -d_a G_a \delta_a K + d_a G_a h. \tag{CC20}$$

The order-zero periodic operator d_aG_a delta_a is bounded on Cˆalpha uniformly in a, and the order-minus-one operator d_aG_a maps Lˆp to Lˆinfinity for p>4, again uniformly in a. To see the constants directly, decompose the discrete Green multipliers into physical dyadic annuli. The order-zero kernels have cancellation and uniformly bounded rescaled derivatives, giving the usual Cˆalpha geometric sum; the order-minus-one kernel has the four-dimensional bound C(|x|+a)ˆ{-3} plus a bounded fixed-volume term. Its Lˆ{p'} norm is uniformly finite precisely for p>4. The diagonal lattice cell has the same bound after including its physical volume. The zero mode is removed by definition of G. These arguments apply componentwise to forms and to complex coefficients. Consequently

$$\|d_a v\|_\infty \le C M^2. \tag{CC21}$$

This is why it is useful to retain delta K as a divergence rather than estimating every nonlinear term as an arbitrary rough force.

78.4 4. The nonlinear raw constraint has the required extra r

Let Q_raw denote the actual linearization of the k-step raw mean and write

$$N_{a,r}(z) = q_{r,\mathrm{raw}}(z) - Q_{\mathrm{raw}} z. \tag{CC22}$$

For z in the common weak chart,

$$\|N_{a,r}(z)\|_{W_r^{1,p}} \le C r \|z\|_{W_a^{1,p}}^2. \tag{CC23}$$

Here is a direct verification, including the raw endpoints. At one fixed-L step with output spacing ell, the actual finite analytic series CM6 is

$$q_{\ell,\mathrm{raw}}(z) = Q_{\ell,\mathrm{raw}} z + \sum_{n\ge 2} \ell^{n-1} B_n(z,\dots,z). \tag{CC24}$$

Each B_n is a fixed-stencil multilinear map, with its analytic coefficient sum bounded geometrically in n. The lattice Wˆ{1,p} algebra estimate and a coarse difference telescoped into L fine differences give

$$\|q_{\ell,\mathrm{raw}}(z) - Q_{\ell,\mathrm{raw}} z\|_{W_\ell^{1,p}} \le C_L \ell \|z\|_{W_h^{1,p}}^2. \tag{CC25}$$

This weaker first-derivative bound does not need the extra quadratic constant-field cancellation. Fixed-L sampling changes only C_L. The actual intermediate z_j obey WB2. Unroll the composition exactly using linear raw tails on each nonlinear error. Every such full raw tail is uniformly bounded on Wˆ{1,p} by FW23–24 and Q_raw=Q_bl-dK. Thus its total norm is bounded by C sum_j ell_j

M^2<=CrM2, proving CC23. This is the actual fixed-L composition; no associativity of the nonlinear group mean is assumed.

Let Pi_r^0 project to mean-zero co-closed coarse one-forms and set

$$e_c = \Pi_r^0 Q(u - A) = \Pi_r^0(N_{a,r}(A) - N_{a,r}(u)). \tag{CC26}$$

The equality follows from the exact nonlinear constraint and Q_raw=Q-d_rK: the extra linear connector is a coarse gradient. Since Pi_r^0 is uniformly bounded on W_r^{1,p}, CC23 proves

$$\|e_c\|_{W_r^{1,p}} \le CrM^2, \qquad \|d_r e_c\|_\infty \le Cr^{-4/p}\|d_r e_c\|_{L_r^p} \le Cr^{1-4/p}M^2 \le CM^2. \tag{CC27}$$

One could also omit the projection in the curl calculation because d_r Pi_r^0=d_r. The displayed inverse power has been paid for by the actual r in the nonlinear constraint defect.

78.5 5. An exact transverse identity completes the estimate

On coarse mean-zero co-closed fields define

$$C_\Pi = \Pi_r^0 Q G_a Q^{\mathsf{t}} \Pi_r^0, \qquad H_T = G_a Q^{\mathsf{t}} \Pi_r^0 C_\Pi^{-1}. \tag{CC28}$$

These are the actual flat transverse Schur covariance and its reconstruction. Their curvature is the one of H_bl in FW and FE. Indeed they solve the same constrained curl-energy minimization; any change in the fine gauge slice changes only a fine gradient.

Q maps gradients to gradients, and both Q and Q^t map harmonic constant one-forms to harmonic constant one-forms. For Q^t this also follows from its normalized partition-of-unity weights, or from the vanishing nonprincipal aliases at zero in FE4a. Also delta_a Q^t lambda_T=0 by the exact flat scalar averaging chain. Project CC18 to fine mean-zero co-closed fields and apply G_a. It follows exactly that

$$P_a^0 u = G_a Q^{\mathsf{t}} \lambda_T + v. \tag{CC29}$$

There is no surviving harmonic forcing in this equality. Applying Pi_r^0 Q and using CC26 now gives

$$\lambda_T = C_\Pi^{-1}\{\Pi_r^0 Q(A - v) + e_c\}. \tag{CC30}$$

Taking the fine curl therefore yields the exact comparison

$$\boxed{d_a u = d_a H_T \Pi_r^0 Q(A - v) + d_a H_T e_c + d_a v.} \tag{CC31}$$

The flat endpoint theorem FE1–2 applies to both reconstruction terms:

$$\|d_a H_T \Pi_r^0 Q z\|_\infty \le C\|d_a z\|_\infty, \qquad \|d_a H_T e\|_\infty \le C\|d_r e\|_\infty. \tag{CC32}$$

The coarse gradient and harmonic parts removed in the first expression have zero reconstructed curl. Equations CC21, CC27, CC31 and CC32 give

$$\|d_a u\|_\infty \le C\|d_a A\|_\infty + CM^2. \tag{CC33}$$

Finally $\|F_a(z) - d_a z\|_\infty \le C\|z\|^2_{W^{1,p}}$ for both z=A and z=u, by WB12 and Morrey. This proves CC4.

78.6 6. Analytic consequences, and what is still outside the theorem

CK already constructs the holomorphic map on a common weak Wˆ{1,p} ball. Restrict it to a common Y_a ball as in CC5. Every expression used above is the actual finite-cutoff holomorphic expression, and the absolute bounds are uniform there. Parameter families staying a fixed distance inside this ball therefore have uniform matrix-curvature Cauchy estimates, including all fixed orders of parameter derivatives. No conclusion is inferred from mere real-axis boundedness.

On real WC inputs the CK identification theorem already identifies this critical branch with the original regular orbit. CC supplies its missing complex curvature bound on the actual same-target branch. Together with the retained-node gauge section GS and WB-B it also gives the strong fractional estimates whenever the input strong norms are supplied:

$$\|\widetilde{u}\|_{H_a^{1+s}} + \|F_a(u)\|_{H_a^{1+s}} + \|J_a(u)\|_{H_a^s} \le C(\|A\|_{H_a^{1+s}} + \|F_a(A)\|_{H_a^{1+s}}), \quad 0 < s < \tfrac{1}{2}. \tag{CC34}$$

The small complex GS gauge has uniformly bounded adjoint action and preserves the matrix curvature bound. This does not identify its individual potential values with an independently prescribed source gauge; that separate bridge statement is unnecessary for CC4.

The scope of the closure is exact:

- The periodic same-target common complex weak-plus-curvature domain is established for the CK branch, without an irreducible holonomy gap.
- Weak inputs alone also have the microscopic-cutoff bound with the explicit retained-scale loss in WB-A. CC4 improves it when the input curvature is uniformly bounded; the two statements have different hypotheses.
- Original paired domains that assume only first-derivative Holder control do not automatically supply the stronger input norms in CC34. That matching obligation remains.
- This proof does not identify off-shell current coordinates, actual wall-modified maps on arbitrary determining regions, sourceful large-field normalizations, or the quantum hierarchy of arbitrary local insertions. Those obligations cannot be replaced by the periodic curvature theorem.

The earlier CX/CK statement that simultaneous complex control was still missing is superseded for this precise branch and input class by CC4. No interacting four-dimensional Yang–Mills existence theorem follows without the remaining measure and observable constructions.

79.1 1. Statement

Use the periodic conventions of CC, with a fixed physical torus, p>8, fixed odd L, and r=Lˆk a<=1. Suppose

$$\|A\|_{W_a^{1,p}} + \|u\|_{W_a^{1,p}} + \|B\|_{W_r^{1,p}} + \|\lambda\|_{W_r^{-1,p}} \le M \le \varepsilon_*, \qquad B = q_{r,\mathrm{raw}}(u), \quad V = e^{rB}. \tag{RC1}$$

Define the actual residuals, rather than prescribing unrelated errors:

$$E = J_a(u) - D\mathcal{M}_r(e^{au})^{\mathsf{t}}\lambda, \qquad b = q_{r,\mathrm{raw}}(u) - q_{r,\mathrm{raw}}(A), \qquad q = d_V^{\mathsf{t}}\lambda. \tag{RC2}$$

Then

$$\boxed{\|F_a(u)\|_\infty \le C(\|F_a(A)\|_\infty + M^2 + \|E\|_{L_a^p} + \|d_r b\|_\infty).} \tag{RC3}$$

The constant is independent of a,r,k. This theorem requires the displayed weak bound on lambda; it does not construct lambda for an arbitrary approximate field. At k=0 the conclusion follows directly from b=u-A and the BCH remainder, so the proof below may take k>=1.

79.2 2. Noether gives both the atomic and the coarse negative norm

Gauge invariance of the Wilson action and the exact raw gauge chain give

$$d_U^{\mathsf{t}} E = -N^{\mathsf{t}} q, \qquad U = e^{au}. \tag{RC4}$$

Here N is literal restriction of a fine scalar gauge parameter to the retained sites. Its transpose uses physical volume. Put

$$\gamma = 3 - 4/p > 0, \qquad A_{a,r,p} = r^{4-4/p} a^{-3+4/p}. \tag{RC5}$$

The sparse-site test in RC4 proves

$$A_{a,r,p}\|q\|_{L_r^p} \le C\|E\|_{L_a^p}. \tag{RC6}$$

For completeness, choose arbitrary coarse scalar values z and define a fine scalar test phi to equal z at retained sites and zero at all other sites. Since r>=La and L>=3, the stars of retained sites have uniformly bounded overlap. The exact physical-volume norm and the one-step difference estimate give

$$\|\phi\|_{L_a^{p'}} = (a/r)^{4/p'}\|z\|_{L_r^{p'}}, \qquad \|d_U\phi\|_{L_a^{p'}} \le Ca^{-1}(a/r)^{4/p'}\|z\|_{L_r^{p'}}. \tag{RC7}$$

The covariant-minus-flat term is bounded by CM times the undifferentiated test and is absorbed because a<=1. Pair RC4 with phi, use N phi=z, and take the coarse Lp dual supremum. The reciprocal of the coefficient in RC7 is exactly A_{a,r,p}. This proof works for complex fields using dual absolute norms; it does not use positivity of a complex bilinear square.

There is also the separate bound

$$\|q\|_{W_r^{-1,p}} \le C\|E\|_{L_a^p}. \tag{RC8}$$

To prove it, replace the sparse test by the multilinear nodal interpolation I_r z, sampled on the fine mesh. It has N I_r z=z and

$$\|I_r z\|_{W_a^{1,p'}} \le C\|z\|_{W_r^{1,p'}} \tag{RC9}$$

for every p'>1, uniformly in the mesh ratio. Indeed every fine difference is a convex combination of coarse slopes, and the physical sum on each coarse cube is controlled by its integral with a fixed finite overlap. The zeroth-order covariant term has coefficient O(M). Pairing RC4 with I_rz and taking the W^{1,p'} dual supremum proves RC8. No point trace of an arbitrary limiting Sobolev function is asserted by this explicit interpolant construction.

Both bounds are needed. RC8 controls a low-frequency longitudinal correction; RC6 pays the potentially large raw endpoint terms.

79.3 3. Unrestricted raw current propagation, with the endpoint retained

Write h_j=L^j a and ell_j=h_{j+1}. Propagate

$$\lambda_k = \lambda, \qquad \lambda_j = DM_j(U_j)^{\mathsf{t}}\lambda_{j+1}, \qquad q_j = d^{\mathsf{t}}_{U_j}\lambda_j. \tag{RC10}$$

The raw gauge identity gives exactly

$$q_j = N_j^{\mathsf{t}} q_{j+1}, \qquad \|q_j\|_{L^p_{h_j}} = (r/h_j)^{4-4/p}\|q\|_{L^p_r}. \tag{RC11}$$

For each fixed-L step the full-space endpoint decomposition is

$$\lambda_j = T_j^0\lambda_{j+1} + \mathcal{B}_j\lambda_{j+1} - \rho_j^{\mathsf{t}} q_{j+1}, \tag{RC12}$$

where the earlier actual-mean estimates give

$$\|\mathcal{B}_j\|_{p\to p} \le C(M\ell_j + (M+M^2)\ell_j^{2-4/p}), \qquad \|\rho_j^{\mathsf{t}}\|_{p\to p} \le C_L\ell_j. \tag{RC13}$$

The second assertion is only for a fixed-L step, where row and column counts are fixed. It is not an all-depth norm claim for a point trace. The first is WB6 with its weak curvature inverse bound.

The product of 1 plus the first error in RC13 is bounded uniformly, because its positive powers sum geometrically. The inhomogeneous recursion therefore gives

$$\|\lambda_j\|_{L^p_{h_j}} \le C\|\lambda\|_{L^p_r} + Cr^{4-4/p}h_j^{-3+4/p}\|q\|_{L^p_r}. \tag{RC14}$$

Unroll RC12 against the whole flat tails, which are Lp contractions. The terminal-multiplier part is the same as CC6. The q-dependent part is at most a constant times

$$r^{4-4/p}\|q\|_{L^p_r}\sum_j \left\{M\ell_j^{-2+4/p} + (M+M^2)\ell_j^{-1} + \ell_j^{-3+4/p}\right\}. \tag{RC15}$$

All exponents are negative for p>8. The last sum is at most C aˆ{-3+4/p}; the first and second are no larger after multiplication by M<=1 and using a<=1. Thus RC6 and the coarse inverse bound $\|\lambda\|_p \le Cr^{-1}\|\lambda\|_{W^{-1,p}}$ prove

$$\|D\mathcal{M}_r(e^{au})^{\mathsf{t}}\lambda - Q_{\mathrm{bl}}^{\mathsf{t}}\lambda\|_{L_a^p} \le CM^2 + C\|E\|_{L_a^p}. \tag{RC16}$$

This is the required comparison without falsely assigning a Ward identity to any intermediate error. The fine endpoint power is paid by the actual Wilson Noether identity.

79.4 4. The longitudinal term and the exact curvature comparison

Coarse Hodge decomposition and RC2 give

$$\lambda_L = d_r G_r \delta_r \lambda = d_r G_r (q + R_B \lambda), \qquad \|\lambda_L\|_{L_r^p} \le C\big(M^2 + \|E\|_{L_a^p}\big). \tag{RC17}$$

This is CC10–12 with RC8 added. The sum q+R_B lambda is mean zero because it is a flat divergence; G_r removes any individual constant parts without needing them to vanish separately.

Use the same tensor K=F_a(u)-d_au+aˆ4 P_a(F_a(u)) as in CC17. Its Cˆalpha norm is at most CMˆ2. The exact equation now has

$$\delta_a d_a u = Q^*\lambda_T + Q^*\lambda_H - \delta_a K + h, \qquad h = E + (DM^t\lambda - Q^*\lambda) + Q^*\lambda_L - \mathcal{T}_U \Phi_a(F_a(u)), \tag{RC18}$$

and $\|h\|_p \le C(M^2 + \|E\|_p)$. Consequently the same fine transverse particular solution v=G_aP_aˆ0(-delta K+h) obeys

$$\|d_a v\|_\infty \le C\big(M^2 + \|E\|_{L_a^p}\big). \tag{RC19}$$

The nonlinear constraint defect becomes

$$e_c = \Pi_r^0 Q(u - A) = \Pi_r^0\{b + N_{a,r}(A) - N_{a,r}(u)\}, \qquad \|d_r e_c\|_\infty \le \|d_r b\|_\infty + CM^2, \tag{RC20}$$

by CC23–27. The exact transverse Schur identity CC31 is unchanged. Apply the two endpoint estimates FE1–2, then add back the bounded BCH remainder F-du. This proves RC3.

80.2 2. The divergence decomposition retains the required regularity

The exact logarithmic plaquette word gives G_u=F_a(u)-d_au with Wˆ{1,p} norm CM squared. Each word after the linear one has degree at least two in shifted potential values and coefficient aˆ{n-2}; one physical difference differentiates a potential factor only once. No second potential derivative is hidden here.

With alpha between zero and 1-4/p, this gives a Cˆalpha-small tensor. The flux error is controlled separately by

$$a^4\|\mathcal{P}_a(F)\|_{C^\alpha} \le Ca^{4-\alpha-12/p}M^3, \qquad 4-\alpha-12/p > 3-8/p > 0. \tag{CA3}$$

Consequently the K in CC17 is Cˆalpha-small. The incidence correction has coefficients of supremum CM, so its action on the flux is Lp-small; the two multiplier errors already have that same bound. This validates the exact separation -delta K+h rather than treating delta K as a generic rough force.

The lattice Hölder singular-integral bound and the integrable order-minus-one Green kernel then give CC21 uniformly in a. The use of a cutoff-dependent inverse bound for F occurs only with the displayed positive powers of a.

80.3 3. The actual raw constraint retains one full factor of r

At a fixed-L step, the nonlinear raw logarithmic mean has a convergent finite-stencil expansion starting at ell times a quadratic potential word. A coarse first difference telescopes into exactly L fine differences. Hence its W^{1,p} defect is bounded by C_L ell times the square of the input W^{1,p} norm.

The composition is unrolled with **linear raw tails**, not with box tails substituted for them. FW23–24 and Q_raw=Q_bl-dK bound each such whole tail on W^{1,p}, uniformly in the ratio of its two spacings. The actual intermediate WP/WB weak chart is used on each nonlinear defect. Therefore

$$\|N_{a,r}(z)\|_{W_r^{1,p}} \le C \sum_j \ell_j \, \|z\|_{W_a^{1,p}}^2 \le Cr \|z\|_{W_a^{1,p}}^2. \tag{CA4}$$

This proves the precise factor required in CC27. Its single coarse inverse costs r^{-4/p}, and 1-4/p is positive. The raw connector disappears only after applying coarse curl or the flat co-closed projection, where it is exactly a gradient.

80.4 4. The transverse identity is exact

The normalized site-box adjoint sends a coarse constant to the identical fine constant: the centered boxes partition the fine lattice in physical volume. The normalized line adjoint also preserves constants. Thus Q_bl^* sends harmonic constants to harmonic constants. They are removed by the **mean-zero** transverse projection, or by G itself.

For co-closed mean-zero lambda_T, Q_bl^* lambda_T is also co-closed and mean-zero, by the exact scalar averaging chain. Therefore the projected Euler equation gives CC29 directly. Moreover Pi_r^0 Q P_a^0 u=Pi_r^0 Q u, because Q sends fine gradients to coarse gradients and fine constants to coarse constants. Applying the actual constraint gives precisely

$$C_\Pi \lambda_T = \Pi_r^0 Q(A - v) + e_c. \tag{CA5}$$

Its sign agrees with the definition e_c=Pi_r^0Q(u-A). This proves CC31 with no missing gauge or harmonic term.

The curvature of H_T Pi_r^0 agrees with that of H_bl. Equivalently, the FW weighted projection solves C^{-1}P_C B=lambda_T with Pi C lambda_T=Pi B and delta lambda_T=0; this is the same transverse Schur system as CC28. Thus FE applies to both terms of CC31. A change of the representative's flat gauge cannot change this curvature.

81 A uniform Banach inverse for the boxed augmented system

This note proves the flat negative-Sobolev inverse needed for the local augmented construction, then checks a specific covariant perturbation row by row. The retained spacing is one. The constraint is the actual flat box-and-line Q, with Qd=d_cS and S the site box average. It is not the raw point-restriction constraint responsible for FI.

The result is periodic. It supplies a uniform inverse on a periodic localization tile in one small reference chart. A covariant Dirichlet or clamped version, or an equivariant construction of periodic reference extensions on overlapping tiles, is a separate gluing issue; neither is asserted at the end of this proof.

81.1 1. The full flat theorem and its spaces

Let a=1/n, with n odd, and let the periodic torus have an integer number of retained cells in every direction. Fine norms use physical volume. Let a_c be a coarse scalar weight with fixed positive upper and lower bounds. Write the five flat equations in second-order form:

$$\begin{aligned}(\Delta_1 + Q^* a_c Q)A - d\psi + Q^*\mu &= f,\\ \Delta\psi + S^*\nu &= g,\\ -\delta A + \psi + \Delta\phi &= h,\\ S\phi &= k,\\ QA &= B.\end{aligned} \tag{BI1}$$

This is exactly the AL five-type matrix. Define, for 2<=p<infinity,

$$\begin{aligned}X_p &= W_a^{1,p}(A) \oplus L_a^p(\psi) \oplus W_a^{2,p}(\phi) \oplus \ell_c^p(\nu) \oplus \ell_c^p(\mu),\\ Y_p &= W_a^{-1,p}(f) \oplus W_a^{-2,p}(g) \oplus L_a^p(h) \oplus \ell_c^p(k) \oplus \ell_c^p(B).\end{aligned} \tag{BI2}$$

The negative norms are dual to the indicated inhomogeneous positive Sobolev norms with exponent p'. Constants and harmonic modes are included. At retained spacing one all fixed finite coarse derivative norms are bounded by the coarse Lp norm, so no inverse retained-scale factor has been hidden in the coarse types.

Theorem BI. The full inverse obeys

$$\|\mathbb{K}_0^{-1} F\|_{X_p} \le C_p \|F\|_{Y_p}. \tag{BI3}$$

The constant is independent of the fine cutoff and of the number of retained cells. The inverse has an exponentially summable operator majorant between retained unit neighborhoods in these types. In particular its physical one-form block maps arbitrary W^{-1,p} forcing to W^{1,p} with this uniform bound.

There is also the stronger scalar variant

$$\begin{aligned}X_p^+ &= W_a^{1,p}(A) \oplus W_a^{1,p}(\psi) \oplus W_a^{2,p}(\phi) \oplus \ell_c^p(\nu,\mu),\\ Y_p^+ &= W_a^{-1,p}(f) \oplus W_a^{-1,p}(g) \oplus L_a^p(h) \oplus \ell_c^p(k,B), \qquad \|\mathbb{K}_0^{-1} F\|_{X_p^+} \le C_p \|F\|_{Y_p^+}.\end{aligned} \tag{BI4}$$

The proof uses the established full five-type weighted energy inverse AL14–19. It does not attempt to infer a Banach inverse merely from an unweighted spectral gap.

81.2 2. Localizing negative norms correctly

For m=1 or 2, Hahn–Banach applied to the derivative graph v -> (D^alpha v)_{|alpha|<=m} gives the exact quotient representation

$$f = \sum_{|\alpha|\le m} (D^*)^\alpha f_\alpha, \qquad \left(\sum_{|\alpha|\le m} \|f_\alpha\|_p^p\right)^{1/p} \le C_m \|f\|_{W_a^{-m,p}}. \tag{BI5}$$

One may make the constant one for the corresponding derivative-graph norm. The fixed finite number of derivatives only changes C_m for equivalent norms. This argument is valid on each finite lattice and has no dimension-of-the-lattice factor. The transpose differences and the physical pairing fix the normalization.

Choose a bounded-overlap smooth partition chi_z of unity on retained unit neighborhoods, and sample it on each fine component lattice. Partition the coefficient fields in BI5 **before** differentiating:

$$f_z = \sum_{|\alpha|\le m} (D^*)^\alpha(\chi_z f_\alpha), \qquad f = \sum_z f_z. \tag{BI6}$$

Each f_z is supported in a fixed enlargement of the cell at z. It has H^{-m} norm bounded by the local L2 norms of its coefficient fields. For p>=2 these are at most a fixed constant times their local Lp norms, because that enlargement has fixed physical volume. No derivative of a cutoff is estimated by a fine inverse power; the derivatives in BI6 are kept as negative-Sobolev derivatives.

This localization is different from falsely presuming that an arbitrary global Hahn–Banach representation already has the support of f. If a localized distribution is needed separately, multiply its representation by a smooth cutoff equal to one on its support and use the exact finite Leibniz identity. Only fixed unit-scale derivatives of that cutoff occur, and the coefficients then lie in a fixed enlargement with the same norm bound. Equivalently, for a distribution supported there, $\|f\|_{H^{-m}} \le C\|f\|_{W^{-m,p}}$ follows by testing against chi v and using the fixed-volume inclusion H^m -> W^{m,p'} on that support.

Apply BI5–6 to the f and g rows of BI1. Partition the h and coarse rows in the ordinary way. There are nonnegative cell numbers c_z with

$$\|F_z\|_{Y_E} \le Cc_z, \qquad \|(c_z)\|_{\ell^p} \le C\|F\|_{Y_p}. \tag{BI7}$$

The energy space Y_E is exactly the five-type extension of AL14.

81.3 3. Weighted energy followed by literal Poisson equations

Let Gamma_0=K_0^{-1}. AL19 applied to BI7 gives, on a fixed enlargement D_x of the unit cell at x,

$$\|\Gamma_0 F\|_{X_E(D_x)} \le Cb_x, \qquad b_x = \sum_z e^{-c\,\mathrm{dist}(x,z)} c_z, \qquad \|(b_x)\|_{\ell^p} \le C\|F\|_{Y_p}. \tag{BI8}$$

This controls local A in H1, psi in L2, phi in H2, and each nearby coarse value of nu and mu. The finite number of coarse sites in D_x turns their local ell2 norm into a pointwise bound without a fine-mesh factor.

The averaging terms have a particularly simple local bound. At retained spacing one, each Q average has a fine physical L2 kernel of bounded norm, on a fixed determining carrier. Therefore

$$|QA(y)| \le C\|A\|_{L^2(D_y^+)}, \qquad \|Q^*(a_c QA + \mu)\|_{L^\infty(D_x)} + \|S^*\nu\|_{L^\infty(D_x)} \le C\sum_{y\sim x} b_y. \tag{BI9}$$

The adjoint estimates use the literal normalized block hold and line average; no singular projection acts here. Finite-range support fixes the neighbor relation y similar to x.

Use the following cutoff-uniform interior flat Poisson bounds on fixed physical collars D compactly contained in D+:

$$\begin{aligned} \|v\|_{L^p(D)} &\le C\{\|\Delta v\|_{W^{-2,p}(D^+)} + \|v\|_{L^2(D^+)}\}, \\ \|v\|_{W^{1,p}(D)} &\le C\{\|\Delta v\|_{W^{-1,p}(D^+)} + \|v\|_{L^2(D^+)}\}, \\ \|v\|_{W^{2,p}(D)} &\le C\{\|\Delta v\|_{L^p(D^+)} + \|v\|_{L^2(D^+)}\}. \end{aligned} \tag{BI10}$$

Here a local negative norm can be defined through coefficients as in BI5 in the larger collar. One direct proof solves the cutoff source by the constant-coefficient lattice Green operator on a larger fixed periodic box, treating its scalar mean by a smooth local quadratic particular solution. The usual discrete multipliers of orders minus two, minus one, and zero give the displayed source norms. The difference is discrete harmonic on D+ and has all interior Lp bounds controlled by its local L2 norm. The same argument, or the lattice harmonic mean value estimates, gives constants uniform as a decreases. In particular the first line does not demand an unproved a priori Lp norm of v on the outer collar.

Now apply the first line to

$$\Delta\psi = g - S^*\nu. \tag{BI11}$$

It gives the local Lp bound for psi from the local g-coefficients, BI8, and BI9. The first row of BI1 is then

$$\Delta_1 A = f + d\psi - Q^*(a_c Q A + \mu). \tag{BI12}$$

Its right side has the required local W^{-1,p} bound, so the second line of BI10 gives A in W^{1,p}. Finally

$$\Delta\phi = h + \delta A - \psi \tag{BI13}$$

and the last line of BI10 give phi in W^{2,p}. Use finitely nested unit collars, whose enlargement only changes the fixed convolution constant in BI8. Summing the local estimates to the pth power proves BI3. If g is in W^{-1,p}, the middle line of BI10 applied to BI11 also gives psi in W^{1,p}; this proves BI4.

For a source localized in one fixed unit neighborhood, the same proof retains the exponential from BI8 outside finitely many neighboring cells. The local source terms vanish there. Hence the asserted exponentially summable Banach-space majorant follows as well. This is the version needed for later local parametrix estimates.

81.4 4. The equivalent first row used in the covariant system

The covariant construction uses the symmetric first-row form

$$(\delta d + Q^* a_c Q)A + d\Delta\phi + Q^*\mu = f_{\mathrm{sym}}, \tag{BI14}$$

with the other four rows of BI1 unchanged. Since Delta phi=h+delta A-psi, it is equivalent to BI1 with f=f_sym-dh. This is a bounded invertible change of input types in both BI2 and BI4. All conclusions

therefore hold for the precise flat first row BI14 as well. For arbitrary forcing its gauge auxiliary need not vanish; its response is part of the augmented inverse.

This distinction is material when comparing with FI: no raw node constraint has been introduced, and an arbitrary force has not been declared to preserve a strict gauge slice after suppressing its scalar response.

81.5 5. Reference coefficient assumptions and multiplier facts

For the perturbation let p>4 and U=eˆ{a u} lie in one small physical potential chart. A convenient volume-uniform hypothesis is

$$\sup_z \|u\|_{W_a^{1,p}(D_z^+)} \le M, \qquad M \le M_*. \tag{BI15}$$

The collars have a fixed radius large enough to contain the retained- scale determining sets. A small global Wˆ{1,p} norm on a fixed torus implies this condition. A small uniform Cˆ{1,beta} reference chart also implies it, for every fixed finite p. Constants may depend on these fixed collar sizes, but not on the fine cutoff or total volume.

The uniform unit-cell Sobolev inequalities imply $\|u\|_\infty \le CM$. They also prove, by the exact product difference and summation over cells, that a coefficient c with uniformly local Wˆ{1,p} norm CM is a multiplier on Wˆ{1,p'} and therefore on Wˆ{-1,p}, with operator norm CM. The differentiated product uses Wˆ{1,p'} into Lˆ{p/(p-2)}, valid in four dimensions for p>4.

In physical tangent coordinates write

$$d_U = d + C_U, \qquad \delta_U = \delta + \widetilde{C}_U, \qquad \Delta_U = \delta_U d_U. \tag{BI16}$$

C_U and its transpose are finite shifted zero-order coefficients of size CM in the uniformly local Wˆ{1,p} multiplier norm. The coefficient is, up to its endpoint convention, (Ad exp(a u)-I)/a. Thus

$$\begin{aligned} \|\Delta_U - \Delta\|_{W^{2,p}\to L^p} &\le CM, \\ \|\Delta_U - \Delta\|_{W^{1,p}\to W^{-1,p}} &\le CM, \\ \|\Delta_U - \Delta\|_{L^p\to W^{-2,p}} &\le CM, \\ \|d_U\Delta_U - d\Delta\|_{W^{2,p}\to W^{-1,p}} &\le CM. \end{aligned} \tag{BI17}$$

For the first line expand delta C_U phi. The derivative of the coefficient is in local Lp, while phi is locally bounded by its W2p norm; the other terms use the coefficient supremum. The second line keeps the outer divergence as a negative derivative. The third follows by the same product rules in duality against Wˆ{2,p'}, or by the Wˆ{-1,p} multiplier bound on C_U d phi. The last line is d(Delta_U-Delta)+C_U Delta_U. No second derivative of u is used.

81.6 6. The actual endpoint-removed averaging coefficients are small

Let U_j be the actual raw hierarchy of U, at spacings h_j=Lˆj a, with ell_j=h_{j+1} and h_k=1. Define the actual operators CK4–6,

$$\mathcal{A}_j = DM_j(U_j) + d_{U_{j+1}}\rho_j, \quad S_j = N_j + \rho_j d_{U_j}, \qquad \mathcal{A}_U = \mathcal{A}_{k-1}\cdots\mathcal{A}_0, \quad S_U = S_{k-1}\cdots S_0. \quad \text{(BI18)}$$

Their exact gauge chain is A_U d_U=d_V S_U. It is not needed as an additional coercivity assumption in the following full-space bound.

The small intermediate weak chart can be verified for p>4 directly. At a fixed-L step the actual analytic reference-log map has

$$\|q_{\ell,\mathrm{raw}}(v) - Q_{\ell,\mathrm{raw}}v\|_{W^{1,p}_{\ell}} \le C_L\ell\|v\|^2_{W^{1,p}_{h}}. \quad \text{(BI19)}$$

This is CC24–25: the finite-stencil expansion starts at ell times a quadratic potential word, the W1p algebra holds for p>4, and a coarse difference telescopes into L fine differences. The whole linear raw tails are uniformly W1p bounded by FW23–24 for p>2. Unrolling the nonlinear composition and applying a first-exit argument gives

$$\sup_{j\le k}\|u_j\|_{W^{1,p}_{h_j}} \le CM, \qquad \max_{i\le j} M_i \le C_0M + C_1\sum_{i<j}\ell_i M_i^2. \quad \text{(BI20)}$$

Indeed sum ell_i is bounded at retained spacing one, and the right side stays strictly below 2C_0 M if all earlier M_i do and M is small enough. This proves the bootstrap without iterating an independent sampling constant. The same argument holds in uniformly local norms: every whole tail has determining diameter O(sum ell_i)=O(1), so a single fixed enlargement of the cell is used before its uniform norm bound. No fractional p>8 premise is needed for this weak step.

The curvature estimate at each level is $\|F_{h_j}(u_j)\|_{p,\mathrm{loc}} \le CM$ and hence $\|F_{h_j}(u_j)\|_\infty \le Ch_j^{-4/p}M$. The literal full-space row/column estimates of PG/WB therefore give, for every finite q and also at q=1,infinity,

$$\|\mathcal{A}_j - Q_{\mathrm{bl},j}\|_{q\to q} + \|\mathcal{A}_j^{\mathsf{t}} - Q^*_{\mathrm{bl},j}\|_{q\to q} \le C_L(\ell_j M + \ell_j^{2-4/p}M), \quad \text{(BI21)}$$

after harmless higher powers of M are absorbed. The scalar S_j is a normalized transported site average; its corresponding difference from the flat site average is O_L(ell_j M). The flat products are contractions and these positive powers sum geometrically. Thus

$$\|\mathcal{A}_U - Q\|_{q\to q} + \|\mathcal{A}_U^{\mathsf{t}} - Q^*\|_{q\to q} + \|S_U - S\|_{q\to q} + \|S_U^{\mathsf{t}} - S^*\|_{q\to q} \le C_qM. \quad \text{(BI22)}$$

Every operator still has range at most a fixed retained-scale distance. These are full-space estimates. No Ward residual is required to vanish. They concern the actual endpoint-removed A_U, not the unrestricted raw differential with its thin endpoint term left in place.

81.7 7. The actual Wilson Hessian has the required smallness

Normalize the Wilson action by its fixed representation factor so its flat physical Hessian is delta d, as in AL. Let H_log,W(U) be its actual Hessian in the relative chart a_var -> exp(a a_var)U. Then

$$\|H_{\log,W}(U) - \delta d\|_{W^{1,p}_a \to W^{-1,p}_a} \le CM. \quad \text{(BI23)}$$

Here the current-dependent coordinate term is included. A derivation is as follows. In ordinary logarithmic potential coordinates,

$$F(u) = du + G(u), \qquad \Phi_a(F) = F + a^4 P_a(F). \tag{BI24}$$

G is the convergent sum of shifted potential words of degree at least two with coefficient a^{n-2}. Consequently DG(u) is a shifted zero-order operator with coefficient supremum CM, and D2G(u)[v,w] is a sum of bounded products of shifted v and w. At every unit cell, $\|F\|_p \le CM$ and $\|F\|_\infty \le Ca^{-4/p}M$. The exact flux Hessian therefore obeys

$$\|D\Phi_a(F) - I\|_\infty \le Ca^{4-8/p}M^2 \le CM^2. \tag{BI25}$$

The action's second variation is the sum of the exact pairings <D Phi(F) DF[v],DF[w]> and <Phi(F),D2F[v,w]>, with the fixed form normalization. Compare the first pairing with <dv,dw>. Its difference is bounded by $CM\|v\|_{W^{1,p}}\|w\|_{W^{1,p'}}$. For the second, use F in local Lp, v in local Linfinity by W1p, and w in local Lp', then sum cells with finite overlap. This gives the same bound.

Finally pass to the relative chart Xi_u(v)=a^{-1}log(exp(a v)exp(a u)). Its first derivative is I plus a W1p-small multiplier, bounded on both W1p and W1p'. Its second derivative maps W1p times W1p' into W1p' with a uniform O(a) bound. The additional Hessian term is the actual first action derivative paired with D2Xi_u. Its divergence-plus-zero-order representation has coefficient Lp norm at most CM on every fixed unit collar: the flux is locally Lp-small and DG is locally bounded. Under the uniformly local hypothesis BI15 this is a uniformly local W^{-1,p} statement, not a claim that the global current norm is volume independent. Pair this representation locally with D2Xi_u[v,w]. Its local W1p' norm is at most Ca times the local W1p norm of v times the local W1p' norm of w. Summing the products over unit cells by ellp–ellp' duality gives the global bilinear bound $CM\|v\|_{W^{1,p}}\|w\|_{W^{1,p'}}$. Thus the coordinate term is O(M) in the required operator norm. This proves BI23 without assuming that the Hessian annihilates gauge directions away from a stationary point.

81.8 8. The covariant five-type inverse

Consider exactly the following reference matrix, with no suppression of the scalar or coarse types:

$$\begin{pmatrix} [H_{\log,W}(U) + \mathcal{A}_U^{\mathsf{t}} a_c \mathcal{A}_U]A + d_U \Delta_U \phi + \mathcal{A}_U^{\mathsf{t}}\mu \\ \Delta_U \psi + S_U^{\mathsf{t}}\nu \\ -\delta_U A + \psi + \Delta_U \phi \\ S_U \phi \\ \mathcal{A}_U A \end{pmatrix} = \mathbb{K}_U(A,\psi,\phi,\nu,\mu). \tag{BI26}$$

At U=I it is the flat system with first row BI14. Equations BI17, BI22, and BI23 prove, on both choices of types,

$$\|\mathbb{K}_U - \mathbb{K}_0\|_{X_p \to Y_p} + \|\mathbb{K}_U - \mathbb{K}_0\|_{X_p^+ \to Y_p^+} \le C_p M. \tag{BI27}$$

For example, the new mass term is bounded by its two differences (A_U^{t-Q})*a_c A_U and Q*a_c(A_U-Q), using the Lp row/column bounds. The scalar rows use BI17 and the Lp bound for S_U-S. The constraint row uses the Lp bound for A_U-Q. These estimates act on the complete unrestricted spaces, not only on a gauge-compatible force subspace.

Choose M once so that C_p M times the flat inverse bound is less than one half. The Neumann series proves

$$\|\mathbb{K}_U^{-1}\|_{Y_p \to X_p} + \|\mathbb{K}_U^{-1}\|_{Y_p^+ \to X_p^+} \le C_p. \tag{BI28}$$

The inverse is holomorphic in an admitted complex reference family. The same estimates apply after conjugation by a smooth physical exponential weight of sufficiently small fixed slope. Differential commutators have only fixed lower orders, and every averaging entry has bounded retained range; its conjugation costs a fixed factor. The perturbation remains O(M) in the weighted types. Thus BI28 has the same exponentially summable locality majorant, with a possibly smaller fixed exponent.

82.1 1. Local hypotheses and statement

Let X be a rectangular fine-lattice box of fixed physical side comparable to R>=1, and let X_0 lie a fixed positive physical distance inside X. Choose X large enough to contain the complete retained-scale stencils of every row to be used on X_0. The fine spacing is a<=1. We work with a compact matrix gauge group and its local complexification.

Assume the real/complex field U on X satisfies

$$U_b = e^{a z_b} \bar{U}_b, \qquad \bar{U}_b \in G, \qquad \|z\|_\infty \le \alpha, \qquad \sup_{p \subset X} \|F_a(U)_p\| \le \epsilon, \qquad R\alpha \le c_0. \tag{CE1}$$

The representation in CE1 is used to bound transports. It need not be chosen as part of the extension, and no uniqueness of this representation is asserted. On the real orbit alpha=0. Fix the group logarithm branch near the identity; the smallness ceilings below guarantee its validity on all plaquettes and canonical forward links.

There is an extension U_X on a compatible periodic box of side O(R), constructed only from U restricted to X, with the following properties.

1. U_X equals U on X_0, including the complete stencils there.
2. It is holomorphic in the admitted complex field variables.
3. If U is changed by an arbitrary real gauge h, there is a periodic gauge h_X equal to h on X_0 for which Ext_X($U^{h)=Ext_X(U)}${h_X} exactly.
4. It has a gauge representative w_X with

$$\|w_X\|_{W_a^{1,p}} + \|F_a(w_X)\|_\infty \le C_{R,p}(\epsilon + \epsilon^2), \qquad p > 4, \tag{CE2}$$

and all gauges used between the extension and this representative have bounded adjoint and inverse adjoint norms, uniformly in a. The ceiling epsilon_*(R,p) is a fixed positive number after R is chosen. The theorem does not claim a bound independent of R before that ceiling is chosen.

82.2 2. A canonical tree trivialization

Fix a root x_* and the usual coordinate-ordered tree of X. Its root paths have physical length O(R). Define g_U(x_*)=1 and recursively along each oriented tree edge b=(x,y)

$$g_U(y) = g_U(x) U_b. \tag{CE3}$$

In the convention U_bˆg=g(x)U_b g(y)ˆ{-1}, every tree edge of the canonical field hat U=Uˆ{g_U} is the identity. CE1 gives

$$\sup_{x \in X} \left(\|g_U(x)\| + \|g_U(x)^{-1}\| \right) \le C e^{CR\alpha}. \tag{CE4}$$

The original real tree transports have norm one. The complex perturbation is paid by its physical path length, not by its number of links.

The loop made by two adjacent coordinate-ordered root paths and their connecting edge has a ribbon filling with at most C R/a plaquettes. Peel it across those plaquettes. Each plaquette is transported along a path of length O(R), so CE1 and CE4 imply

$$\|\widehat{U}_b - I\| \le CaR\epsilon e^{CR\alpha} e^{CaR\epsilon e^{CR\alpha}}. \tag{CE5}$$

The ribbon estimate is specific to adjacent coordinate-ordered paths: their filling has physical area O(aR), not O(Rˆ2). It follows by moving the connecting edge successively through the later coordinate legs. Every move contributes one plaquette and there are at most C R/a moves. No commutativity of

the plaquettes is used; the exact ordered product is bounded by the product of 1 plus their absolute deviations.

For a fixed small epsilon_*(R), every forward link therefore has a canonical logarithmic potential

$$\widehat{U}_b = e^{aA_b}, \qquad \|A\|_\infty \le C_R\epsilon. \tag{CE6}$$

Curvature transforms by exact conjugation at the basepoint. The local plaquette BCH series, after subtracting its linear curl, contains only words of field degree at least two with coefficient a^{n-2}. Thus

$$\|F_a(A)\|_\infty \le C_R\epsilon, \qquad \|F_a(A) - d_aA\|_\infty \le C\|A\|_\infty^2, \qquad \|d_aA\|_\infty \le C_R(\epsilon + \epsilon^2). \tag{CE7}$$

This calculation does not claim a bound for all first derivatives of A. An axial potential may have uncontrolled transverse derivatives even when its curvature is bounded. Only its curl is used here.

82.3 3. Cut off the canonical potential, rather than the gauge

Choose a real scalar cutoff chi on X, equal to one on a neighborhood of all required X_0 stencils and zero on a neighborhood of the boundary. Its physical first differences are bounded by a fixed constant depending on the chosen collar. On each positively oriented link use the cutoff at its basepoint, and define

$$A^\chi_\mu(x) = \chi(x)A_\mu(x). \tag{CE8}$$

Set it to zero outside X and periodize on a larger compatible box. The boundary collar makes this extension literal at every fine stencil. The exact discrete product rule gives

$$\begin{aligned}(d_aA^\chi)_{\mu\nu}(x) = \chi(x)(d_aA)_{\mu\nu}(x) \\ + (D_\mu\chi)(x)A_\nu(x + a\hat{\mu}) - (D_\nu\chi)(x)A_\mu(x + a\hat{\nu}).\end{aligned} \tag{CE9}$$

Consequently

$$\|A^\chi\|_{L^p_a} \le C_{R,p}\epsilon, \qquad \|d_aA^\chi\|_\infty + \|F_a(A^\chi)\|_\infty \le C_R(\epsilon + \epsilon^2). \tag{CE10}$$

No uncontrolled derivative of the axial potential appears in CE9. Let hat U_X=exp(a A^chi). Extend g_U to all periodic sites by choosing the identity at sites outside X; denote this arbitrary sitewise extension by g_{U,X}. It need not have small derivatives. Define

$$U_X = \widehat{U}_X^{,g_{U,X}^{-1}}. \tag{CE11}$$

On the complete inner carrier this is exactly U. The discontinuity of the extending gauge at the outer boundary is a pure gauge effect and does not spoil curvature bounds. Its adjoint norm is bounded by CE4.

82.4 4. Covariance is exact, including the extension boundary

For a gauge h on X, the canonical tree recursion gives

$$g_{U^h}(x) = h(x_*)g_U(x)h(x)^{-1}, \qquad \widehat{U}(U^h) = \widehat{U}(U)^{,h(x_*)}. \tag{CE12}$$

A constant conjugation commutes with the canonical logarithm and the scalar cutoff. Define on the periodic extension

$$h_X(x) = g_{U^h,X}(x)^{-1}\, h(x_*)\, g_{U,X}(x). \tag{CE13}$$

Then h_X=h inside X, and h_X=h(x_*) at exterior sites where both extended tree gauges were set to one. Equations CE11–13 give

$$U_X(U^h) = U_X(U)^{h_X}. \tag{CE14}$$

This is an exact identity, even on links crossing the artificial extension boundary. For real U,h every gauge is unitary. The algebraic identity also holds for admitted complex gauges whenever both sides stay in the specified logarithm branches.

All choices of tree, root, cutoff and periodic box are fixed geometric choices, independent of the field. The extension is a holomorphic function of the original link matrices on its admitted complex domain: it uses products, inverses, the uniform local logarithm, and exponentials.

82.5 5. A weak analysis gauge exists on the extension

Apply the low-Lp Coulomb construction WC17–24 to the canonical periodic potential A^chi. Its small-potential Lp hypothesis follows from CE10 after epsilon_*(R,p) is chosen. The construction and its estimates use absolute norms and holomorphic BCH expressions, so they apply on the same complex domain. It produces a gauge exp(phi) with

$$\|\phi\|_{W_a^{1,p}} + \|w_X\|_{L_a^p} \le C\|A^\chi\|_{L_a^p}, \qquad \delta_a w_X = 0. \tag{CE15}$$

Morrey gives a bounded adjoint action of this gauge. Curvature covariance and CE10 bound F_a(w_X) in supremum and hence in Lp. The subsequent div-curl bootstrap in WC proves

$$\|w_X\|_{W_a^{1,p}} \le C(\|F_a(w_X)\|_{L_a^p} + \|w_X\|_{L_a^p}), \tag{CE16}$$

after the uniformly small lower-order term is absorbed. That proof does not assume a W^{1,p} bound on the initial axial potential. Equations CE10 and CE15–16 prove CE2.

There is no coarse-target normalization problem here: this gauge is used to estimate a covariant local operator. All of its field types and coarse targets transform together. It is not being used to claim that two unrelated fixed-target gauge representatives have a bounded bridge.

82.6 6. Use in a common carrier expansion

Let K_U be an intrinsically covariant augmented matrix whose entries depend on complete retained-scale stencils, and suppose its periodic inverse has the proved small-Wˆ{1,p}-reference bound in the appropriate covariant Banach norms. Define the local inverse on X by

$$\Gamma_X(U) = K^{-1}_{,U_X(U)}. \tag{CE17}$$

Estimate it in the weak analysis gauge of Section 5. Gauge covariance transports the result back with bounded adjoint factors **in covariant Sobolev norms**. For clarity, the one-form type uses one covariant difference, the scalar phi type uses two, and the negative types are their duals. These differences intertwine exactly under a change of frame. Bounded adjoints by themselves would not transport ordinary first or second differences through the rough tree gauge. In the weak analysis frame w_X the covariant norms are uniformly equivalent to the ordinary BI norms: the first-order difference from D has coefficient O(w_X), while the second-order difference has coefficients of the forms w_X D, D w_X and w_X squared. The product bounds for p>4 and CE2 give the equivalence, including its dual negative norms, without estimating derivatives of the axial-to-Coulomb gauge. On the inner carrier K_{U_X}=K_U exactly. CE14 and uniqueness of the inverse imply exact gauge covariance of the restricted kernels of Gamma_X. The extension, all of its tree paths, and the periodic inverse are filed on the **whole box X**. No part of a tree is silently placed outside the declared carrier.

For the usual cutoff parametrix, this gives the exact identity

$$K_U(\chi_X \Gamma_X \chi_X) = \chi_X^2 + [K_U, \chi_X]\Gamma_X \chi_X, \tag{CE18}$$

provided the support of chi_X and its entry stencils lies in X_0. Thus the same complete-carrier weighting used in AL is compatible with gauge covariance. This last section is conditional only on the stated inverse theorem for the specified covariant matrix; it does not prove that theorem by covariance alone.

One chooses the localization geometry first and then the field ceiling. When the small-reference inverse is an O(M) perturbation of the volume-uniform boxed inverse, its bound is uniform after CE2 is made smaller than its fixed perturbation radius. Growth of C_{R,p} therefore changes the admissible field ceiling rather than the microscopic cutoff or the length of the carrier expansion.

This construction addresses local gauges and covariant periodization. It does not identify arbitrary mixed determining-set constraints with the full periodic boxed constraints, or by itself prove a sourced renormalization estimate.

83.2 2. The earliest literal endpoint mismatch

Bałaban II, CMP 116 (1988), (1.2)–(1.5), defines the extended response by fixed-point equations of the form

$$\mathcal{H}(U,J;B) = H_0 B + A_0 - HD(H_0 B + A_0), \qquad D(0)=0, \quad A_0(U,J;0)=0. \tag{NL1}$$

Consequently

$$\mathcal{H}(U,J;0) = 0 \tag{NL2}$$

on the entire admitted independent-pair domain. The field reconstructed at zero fluctuation is therefore the given U. This remains true after merely replacing the propagators in these equations. I (3.1) explicitly constructs the extension by replacing the physical current in its operators by the independent variable J. It does not assert that this extended response is a physical Wilson minimizer at every independent pair.

Here is a finite-cutoff obstruction to identifying (NL1) with a physical KKT solution for every pair. Choose a nonzero Cartan generator T and a Cartan-valued one-form $v = v_{\mathrm{real}} T$ with

$$Q_{\mathrm{raw}} v = 0, \qquad d_a v \neq 0. \tag{NL3}$$

Such hidden nongauge directions exist whenever a blocking step has genuine fluctuation degrees of freedom. At any fixed cutoff the raw mean is a submersion at the identity. Choose a right inverse of its finite-dimensional derivative and apply the ordinary implicit-function theorem to obtain an exact fiber path

$$a(t) = tv + O(t^2), \qquad U_t = \exp(a\, a(t)), \qquad \mathcal{M}(U_t) = 1. \tag{NL3a}$$

This construction avoids any extra assumption that a chosen nonlinear group mean is exactly its flat derivative on all commuting input tuples. Since the Wilson action has zero first jet at the identity and a positive quadratic curl form on v,

$$\frac{d}{dt} S_{\mathrm{W}}(U_t) = c_T t \|d_a v\|_2^2 + O(t^2), \qquad c_T > 0. \tag{NL4}$$

For sufficiently small nonzero t, U_t is not stationary on that very constraint fiber. Nevertheless (NL2) leaves it unchanged at $B = 0$. At any fixed cutoff, t can be chosen so small that the regularity and reconstructed-background conditions of the original paired core hold.

This is not a counterexample to the published formation theorem or to CK. It proves a narrower negative statement:

> A replacement which is a physical Wilson critical configuration for every independent-pair input cannot have the original response as its endpoint on the whole independent-pair domain.

One may construct a different analytic extension which agrees on the physical critical graph, or retain the actual independent-current/background defect in an augmented equation. Either choice requires an argument; equality on the physical graph alone is not equality of the original extended maps.

83.2.1 2.1 The stronger-input regularity issue is also real

I (3.31)–(3.32) assumes a first-derivative Hölder norm for the outer field. The discussion immediately before I (4.18) explicitly distinguishes this from a second-derivative bound. II (1.21) preserves these admissible norms under the wall parameters; it does not state a curvature $W^{1+s,p}$ bound.

For example, on a fixed torus take a resolved Cartan-valued transverse mode

$$A_{N,2}(x) = \epsilon\, \lambda_a(N)^{-1} \sin(Nx_1)\, T, \qquad \lambda_a(N) = 4a^{-2} \sin^2(aN/2), \tag{NL5}$$

with $aN \to 0$. The other components vanish. Then, for any fixed $\beta < 1$,

$$\|A_N\|_{C^{1,\beta}} + \|\Delta_a A_N\|_\infty \le C_\beta \epsilon, \qquad \|F_a(A_N)\|_\infty \le C\epsilon/N.$$

The Wilson current has leading term $\epsilon \sin(Nx_1)T$, with a smaller analytic lattice correction. Hence

$$\|J_{\mathrm{W}}(A_N)\|_{W^{s,p}} \ge c\epsilon N^s, \qquad \|F_a(A_N)\|_{W^{1+s,p}} \ge c\epsilon N^s \tag{NL6}$$

along a sufficiently fine sequence. This example also obeys the displayed low current supremum bound. If desired choose N to be a retained reciprocal frequency so that its coarse averages vanish. Thus the stronger fractional norm cannot be inferred from the original low core alone.

83.4 4. A nonlinear local augmented system

For this section and the wall construction rescale the retained spacing to one, and put $a = L^{-k}$. Constants may depend on L, β, the fixed wall geometry, and later interior collars, but not on a, k, or the periodic volume.

Write $q^{\#} = q^{\#}_{k-1} \circ \cdots \circ q^{\#}_0$, and $Q = Dq^{\#}(0) = Q_{\mathrm{bl}}$. Let S be site averaging, and use exactly the flat gauge auxiliaries of AL. The unknowns are

$$\mathbf{u} = (u, \psi, \phi, \nu, \mu).$$

Here ν is a coarse scalar and μ a coarse one-form. Let $a_c > 0$ be the fixed bounded coarse mass weight of AL. Consider the equations

$$\begin{aligned} \mathcal{E}_a(u)^{\mathsf{t}} J_{\mathrm{W}}(u) + d\Delta\phi + Dq^{\#}(u)^{\mathsf{t}}\{a_c q^{\#}(u) + \mu\} &= 0, \\ \Delta\psi + S^*\nu &= 0, \\ -\delta u + \psi + \Delta\phi &= 0, \\ S\phi &= 0, \\ q^{\#}(u) &= b. \end{aligned} \tag{NL16}$$

The first term is the exact logarithmic gradient of the Wilson action. There is no independent current substituted for it. The $Dq^{\#\mathsf{t}} a_c q^{\#}$ term is a coarse constraint penalty; on the constraint it is absorbed into the coarse multiplier. Using a fixed $Q^* a_c Q u$ in the full nonlinear equation instead would not have this property.

At zero, (NL16) is precisely AL's five-type boxed saddle matrix:

$$\begin{gathered} (\Delta_1 + Q^* a_c Q)u - d\psi + Q^*\mu = 0, \\ \Delta\psi + S^*\nu = 0, \quad -\delta u + \psi + \Delta\phi = 0, \quad S\phi = 0, \quad Qu = b. \end{gathered} \tag{NL17}$$

Denote this flat matrix by $\mathbb{K}$.

The apparently all-depth nonlinear constraint is local at retained scale. Each one-step determining patch has diameter $C_L\ell_j$, and $\sum \ell_j \le C_L$. Therefore every complete determining patch of $q^{\#}$, its first derivative, and its transpose has diameter at most a fixed C_L. One may introduce each intermediate field as an auxiliary variable to see the layer-by-layer locality, but no infinite-range operator is hidden in (NL16).

83.4.1 4.1 The nonlinear remainder has the old weak norm

Use the AL norm

$$\|\mathbf{u}\|_{\mathcal{X}_\beta} = \|u\|_{C^{1,\beta}} + \|\Delta_1 u\|_\infty + \|\psi\|_{C^{1,\beta}} + \|\Delta\psi\|_\infty + \|\phi\|_{C^{2,\beta}} + \|\nu\|_\infty + \|\mu\|_\infty. \tag{NL18}$$

The corresponding $\mathcal{Y}_\beta$ is AL17 with the additional coarse constraint component. All nonlinear components except the first and last are zero.

Let

$$N_{\mathrm{W}}(u) = \mathcal{E}_a(u)^{\mathsf{t}} J_{\mathrm{W}}(u) - \delta du.$$

Then, uniformly for $\|u\|_{C^{1,\beta}} \le M \ll 1$,

$$\|N_{\mathrm{W}}(u)\|_\infty \le CM^2. \tag{NL19}$$

A direct calculation establishes the fine-cutoff dependence. The plaquette logarithm has the finite local convergent expansion

$$F_a(u) = d_a u + G_a(u), \qquad G_a(u) = \sum_{n\ge 2} a^{n-2} B_{n,p}(u,\dots,u), \qquad \|G_a(u)\|_{W^{1,\infty}} \le CM^2. \tag{NL20}$$

The linear plaquette word is exactly $d_a u$. Differentiating a nonlinear word uses an exact finite product difference, so its first physical difference is bounded by $CM\|\nabla u\|_\infty$, with no second derivative of u.

The Wilson flux is $F + a^4 R(F)$, where $R(F) = O(F^3)$ on the small plaquette branch. The covariant incidence differs from the flat incidence by bounded shifted coefficients of size CM. One inverse fine difference therefore bounds the differentiated flux correction by Ca^3M^3. Finally $\mathcal{E}_a(u) - I = O(aM)$, while $\|\delta du\|_\infty \le Ca^{-1}\|\nabla u\|_\infty$. Their product is $O(M^2)$. These are all terms in (NL19). The estimates apply to the holomorphic complexification of the Wilson action, not to a nonholomorphic matrix absolute square.

Equations (NL9)–(NL11), and the endpoint exponential differentials, give

$$\|q^{\#}(u) - Qu\|_\infty \le CM^2, \qquad \|(Dq^{\#}(u) - Q)^{\mathsf{t}}\|_{\infty\to\infty} \le CM. \tag{NL21}$$

Thus, writing (NL16) as

$$\mathbb{K}\mathbf{u} + \mathbf{N}(\mathbf{u}) = \mathbf{b}, \qquad \mathbf{b} = (0,0,0,0,b), \tag{NL22}$$

one obtains

$$\|\mathbf{N}(\mathbf{u})\|_{\mathcal{Y}_\beta} \le CM^2, \qquad \|D\mathbf{N}(\mathbf{u})\|_{\mathcal{X}_\beta\to\mathcal{Y}_\beta} \le CM. \tag{NL23}$$

The second bound follows either by differentiating these finite local expressions or by Banach-valued Cauchy on a fixed larger ball. This Cauchy argument is dimension independent: it bounds the nonlinear map in the displayed Banach norms and does not replace a coefficient norm by a dimension-dependent coordinate supremum.

83.5 5. The common nonlinear wall expansion

Use AL20–AL25 to construct one common family $\Gamma_{\mathbf{s}}$ for every block of $\mathbb{K}^{-1}$. Fix a wall polydisc $|s_\Delta| < R_w$, $R_w > 1$. First choose the localization boxes large enough for the linear walk series on this disc; only afterward choose the weak field ceiling.

File each nonlinear output on a retained unit cell together with its entire determining neighborhood of radius C_L. This gives bounded-overlap local pieces $\mathbf{N}_z$. Multiply a piece by the product of the wall parameters over its complete carrier. Call the resulting map $\mathbf{N}_{\mathbf{s}}$. At $\mathbf{s} = 1$ it is the original $\mathbf{N}$. There are only a fixed number of wall cells per such vertex, hence

$$\|\mathbf{N}_{\mathbf{s}}(\mathbf{u})\|_{\mathcal{Y}_\beta} \le C_{R_w} M^2, \qquad \|D\mathbf{N}_{\mathbf{s}}(\mathbf{u})\| \le C_{R_w} M. \tag{NL24}$$

The first nonlinear output norm is a supremum norm; no derivative of the piecewise wall coefficient is taken there. The last output is at retained spacing one. Thus this filing does not introduce a fine-interface derivative loss.

Solve

$$\mathbf{u}_{\mathbf{s}} = \Gamma_{\mathbf{s}}[\,\mathbf{b} - \mathbf{N}_{\mathbf{s}}(\mathbf{u}_{\mathbf{s}})\,]. \tag{NL25}$$

If $K_{R_w} = \sup \|\Gamma_{\mathbf{s}}\|$, choose $2K_{R_w} C_{R_w} M < 1/2$, and then $\|b\|_\infty < M/(2K_{R_w})$. The closed M-ball is invariant and the map is a contraction. Its fixed point is holomorphic in b and all wall parameters, with a common domain and uniform bound.

For a protected set B_+, declare unweighted all linear seed carriers which have a row there, and all nonlinear vertex carriers which have an output there. AL's same-complete-carrier identity and (NL25) then give the **exact** equations (NL16) on B_+. This includes all scalar and coarse rows.

At zero walls the linear walks and nonlinear vertices which meet the removed cells vanish. Convergent fixed-point iteration therefore gives exact component dependence. The associated expansions retain the actual complete carriers of every vertex and walk. Standard Cauchy/FTC localization now applies on this family; it does not assume that independently weakened nonlocal projections continue to satisfy a PDE.

The nonlinear tree series may contain the same wall parameter in several vertices. This is harmless: its holomorphic bound is paid in (NL24) and by the small contraction factor. The carrier-once convention needed for the **linear protected-row telescope** is still the one in AL23–AL25.

83.6 6. Protected fractional regularity follows from these exact equations

Fix finite $p > 4$ and

$$0 < s < \min(\beta, 1/p). \tag{NL26}$$

Let $B \Subset B_1 \Subset B_+$ have fixed positive retained-scale collars. Enlarge B_+ to contain every complete determining patch and retained cell needed by a row in B_1. Then the nonlinear family in (NL25) satisfies

$$\boxed{\|u_{\mathbf{s}}\|_{W_a^{2+s,p}(B)} + \|F_a(u_{\mathbf{s}})\|_{W_a^{1+s,p}(B)} + \|J_{\mathrm{W}}(u_{\mathbf{s}})\|_{W_a^{s,p}(B)} \le C_{B,B_+,s,p} M.} \tag{NL27}$$

All constants are independent of the microscopic cutoff and the depth. No curvature supremum is an extra hypothesis: the common weak $C^{1,\beta}$ bound already controls it by (NL20).

Here are the details of the nonlinear force estimate needed in the proof. The first two terms in

$$J_{\mathrm{W}} - \delta du = \delta G_a(u) + (\delta_U - \delta) F_a(u) + \delta_U [a^4 R(F_a(u))] \tag{NL28}$$

have uniformly bounded C^β norm CM^2. This follows by applying the exact product-difference identity to (NL20), using $u \in C^{1,\beta}$, and using that $(\operatorname{Ad} e^{au} - I)/a$ is a uniformly bounded analytic $C^{1,\beta}$ coefficient. A C^β function has a local $W^{s,p}$ bound for $s < \beta$.

For the flux term, $F \in C^\beta$ gives a supremum bound $Ca^{3+\beta}M^3$ after its one difference. The fine fractional inverse cost is at most Ca^{-s}, still leaving a nonnegative power of a. The extra logarithmic tangent factor in N_{W} is treated similarly:

$$\|\delta du\|_\infty \le Ca^{\beta-1}M, \qquad \|(\mathcal{E}_a(u)^{\mathsf{t}} - I)\delta du\|_{W^{s,p}(B_1)} \le Ca^{\beta-s}M^2. \tag{NL29}$$

The remaining products have better powers. Smooth physical cutoffs contribute only bounded lower-order terms. Consequently

$$\|N_{\mathrm{W}}(u)\|_{W^{s,p}(B_1)} \le CM^2. \tag{NL30}$$

This estimate does not infer a fractional norm for the full linear δdu from its supremum. Its potentially rough second differences occur only in the displayed lattice-suppressed nonlinear term.

The protected scalar equation gives $\Delta\psi = -S^*\nu$. Block hold maps retained L^p to fine $W^{s,p}$ for $s < 1/p$, as proved in (NL13). Interior lattice elliptic regularity therefore bounds ψ in $W^{2+s,p}$, with the weak norm controlling its local constant.

Put $\lambda' = a_c q^{\#}(u) + \mu$. The protected first equation, using the third, is exactly

$$\Delta_1 u = d\psi - Dq^{\#}(u)^{\mathsf{t}}\lambda' - N_{\mathrm{W}}(u). \tag{NL31}$$

The coarse field λ' is bounded. Only a fixed number of retained cells can influence B_1; cut it off on those cells before applying (NL15). At retained mesh one its $W^{s,p}$ norm is bounded by its local L^p norm. Equations (NL15), (NL30), and the bound on ψ give a uniform $W^{s,p}$ right side in (NL31). Interior elliptic regularity gives $u \in W^{2+s,p}$. Formula (NL20) and the uniform fractional product bound give the curvature estimate. Formula (NL28) gives the current estimate. This proves (NL27).

No exact raw Ward equation is claimed for a wall-deformed field. In fact, restoring a raw multiplier from a non-Ward straightened multiplier can reintroduce thin endpoint terms. The proof of (NL27) does not do that: it uses the full controlled adjoint of the genuine straightened constraint and the coupled scalar equations. The physical current in (NL27) is nevertheless the actual Wilson current of the constructed field.

83.7 7. The unweakened endpoint is physically critical

This step is global and periodic; it is not applied to a partial protected set with walls.

The straightened average has an exact gauge cocycle. Infinitesimally, with

$$\mathfrak{d}_u\zeta = \mathcal{E}_a(u)^{-1}d_U\zeta,$$

one has

$$Dq^{\#}(u)\mathfrak{d}_u = \mathcal{E}_1(q^{\#}(u))^{-1}d_{V^{\#}}\,S^{\#}(u), \qquad V^{\#} = e^{q^{\#}(u)}. \tag{NL32}$$

At one step, $S_j^{\#}(u)$ is the derivative of the actual coarse gauge cocycle

$$g_j(u^{e^{t\zeta}})\,(N_j e^{t\zeta})\,g_j(u)^{-1}.$$

It is a finite local analytic matrix, with $S_j^{\#}(0) = S_j$, the site average. Its physical row and column remainder is $O_L(\ell_j M)$, by the same finite-step argument as (NL11). Thus

$$\|S^{\#}(u) - S\|_{2\to 2} \le CM. \tag{NL33}$$

Let R_0 be the orthogonal projector onto $\Delta \ker S$. The last scalar rows in (NL16) imply

$$\Delta\phi = R_0\delta u. \tag{NL34}$$

The map

$$\zeta \longmapsto (R_0\delta\mathfrak{d}_u\zeta,\ S^{\#}(u)\zeta) : H^2_a \longrightarrow \operatorname{ran}(R_0)\oplus \ell^2_c \tag{NL35}$$

is onto for sufficiently small M, with a cutoff-uniform inverse. At zero this is the scalar AL problem: given (h,k), use $\zeta = \Delta^{-1}h + E_{\mathrm{box}}k$, treating constants separately. The first term belongs to $\ker S$; the second is the box biharmonic lift. AL13 gives the uniform H^2 estimate. For the perturbation, $\mathfrak{d}_u - d$ is a finite shifted zero-order coefficient of size CM, with first differences CM. It follows that $\|\delta(\mathfrak{d}_u - d)\zeta\|_2 \le CM\|\zeta\|_{H^2}$. Combine this with (NL33) and use a Neumann argument.

Pair the first equation of (NL16) at $\mathbf{s}=1$ with $\mathfrak{d}_u\zeta$, choosing $S^{\#}(u)\zeta = 0$. The Wilson term vanishes by exact gauge invariance and the constraint term vanishes by (NL32). The remaining pairing is

$$\langle R_0\delta\mathfrak{d}_u\zeta,\ R_0\delta u\rangle = 0.$$

The first component in (NL35) can be arbitrary while the second is zero. Nondegeneracy of the complex bilinear pairing therefore implies

$$R_0\delta u = 0, \qquad \phi = 0. \tag{NL36}$$

For complex fields one uses arbitrary test h, not positivity of a complex bilinear square.

The first equation now is the genuine stationarity equation for the straightened constraint. Pairing all gauge variations and using the surjectivity of $S^{\#}$ yields its coarse Ward equation.

The iterated straightened output is exactly

$$\mathcal{M}_{a,1}(e^{au})^{\,G(u)} = e^b. \tag{NL37}$$

The accumulated coarse gauge satisfies $\|G(u) - I\|_\infty \le CM$: its successive factors have size $C\ell_j M$, and retained-node restriction does not increase the supremum. Ward cancellation in the derivative of (NL37) converts stationarity to the original raw KKT equation at raw target $\mathcal{M}(e^{au})$.

To obtain the exact displayed raw target e^b, choose a fine gauge with retained values $G(u)$. One quantitative choice is

$$h = \exp[E_{\mathrm{mix}} \log G(u)]. \tag{NL38}$$

The LF mixed lift satisfies $NE_{\mathrm{mix}} = I$. Thus $\mathcal{M}((e^{au})^h) = e^b$ exactly. In retained units a coarse supremum controls its fixed finite differences. The LF offset-kernel proof also gives a fine $C^{3,\gamma}$ bound for the lift, for every $\gamma < 1$: a fine derivative of order $t < 4$ has dyadic alias majorant

$$C\,2^{(t-4)m}(m+1)^3, \tag{NL39}$$

after the same principal-pole removal. This is summable. Hence (NL38) preserves a slightly smaller weak $C^{1,\beta}$, Laplacian-supremum chart and the strong local norms above. The corrected endpoint is real for real data and holomorphic for complex data.

For real small periodic data it lies in the regular orbit of the published variational theorem. Its uniqueness there identifies the **gauge-invariant readout** with the original physical minimal background. Equality extends to the common complex data neighborhood by the identity theorem. This last identification imports the published regular-orbit uniqueness theorem; it does not identify the two chosen gauge representatives as matrices.

83.8 8. The exact gauge-averaged coarea identity

There is a useful integration identity which is stronger than equality of critical values, but weaker than equality of the old canonical integrands. Let U range over a real finite lattice, let $F(U)$ be any integrable carrying function, and let $G(U)$ be any measurable coarse gauge for which $\mathcal{M}^{\#}(U) = \mathcal{M}(U)^{G(U)}$. With normalized coarse Haar measure,

$$\begin{aligned} &\int dU\, dh\, F(U)\, \delta(\mathcal{M}^{\#}(U)^h V^{-1}) \\ &\qquad = \int dU\, dh\, F(U)\, \delta(\mathcal{M}(U)^h V^{-1}). \end{aligned} \tag{NL40}$$

For each fixed U, change only the independent variable h by multiplication with $G(U)$, in the order fixed by the convention for gauge composition. Haar invariance gives unit Jacobian. Fubini justifies the field-dependent change because U itself is not changed. The identity extends to the usual coarea distributional notation by testing against a continuous target function.

If F and its retained tests are gauge invariant and the fine gauge extension of an arbitrary coarse gauge is admitted, the raw pushforward density is coarse-gauge invariant. Its gauge average on the right then equals that raw density. Without those qualifications, only (NL40), or equality against coarse gauge-invariant tests, is asserted.

This is also the correct setting for the endpoint-removed differential frame. For a joint raw constraint with an independent coarse gauge variable, its tangent has the form

$$(v, \sigma) \longmapsto Q_U v + d_V \sigma.$$

The pointwise frame substitution

$$\sigma = \tau + \rho_U v$$

has block determinant one and changes it to

$$\mathcal{A}_U v + d_V \tau. \tag{NL41}$$

It supplies the compatible horizontal and vertical stationarity equations. It does not assert that ρ_U is an integrable gauge coordinate. Nor does it say that the bare coarea Gram determinant is unchanged under a nonorthogonal frame: the fiber volume form and the determinant must be transformed together.

83.9 9. Why the full original formation is still not identified

Suppose $\Phi_o(b, \xi_o)$ and $\Phi_n(b, \xi_n)$ are two actual gauge-fixed coordinate charts for the same physical fiber, with all required gauge compensations retained. If $\Phi_n(b, \xi) = \Phi_o(b, \Psi_b(\xi))$ modulo that admitted gauge, their full coarea densities obey

$$j_n(b, \xi) = j_o(b, \Psi_b(\xi))\, \det D_\xi \Psi_b(\xi), \tag{NL42}$$

with the orientation or real absolute-value convention appropriate to the original integral. Every physical cutoff must also be pulled back: $\chi_n(b, \xi) = \chi_o(b, \Psi_b(\xi))$. At a common saddle, the Hessians transform by congruence; the Jacobian and Gaussian determinant cancel in the usual exact finite-dimensional way. That algebra alone gives no uniform analytic bound on the full transition.

In the present problem, (NL40) provides an exact gauge-averaged starting integral, and (NL16) supplies an actual local nonlinear saddle for it. What has **not** been proved is a common-domain, localized version of (NL42) between this chart and every original fluctuation formation, including:

- the fixed original Gaussian coordinates and characteristic functions;
- the field-dependent coarea/FP factors and their localized logarithms;
- the full independent-(U, J) analytic extension;
- mixed determining sets and the stored live-island boundary maps.

There is also a chart-covariance issue. The flat inverse in (NL25) is fixed in one identity potential chart. Although a trace readout is gauge invariant as a function of its output field and its physical endpoint agrees with the original readout, the individual wall values have not been proved covariant under an arbitrary coarse gauge change of the input. Such a change can leave the chosen potential chart. Gauge invariance of the sum at $\mathbf{s} = 1$ does not make every FTC activity gauge invariant. A covariant chart/gluing construction, or a legitimate gauge average on a common orbit domain, must be included before using these terms as the original gauge-invariant local activities.

This is a precise missing comparison, not a demand for equality of the old and new wall values. If (NL42), its common domain, and the required local majorants are proved, AL37 permits the new interpolation while keeping the physical endpoint integral unchanged. There is no need for a new physical change of variables at every complex wall parameter.

One must not infer that comparison from the $W^{1,p}$, $p > 4$, gauge bounds. Raw endpoint connectors involve retained-node restrictions. On ambient H^2 scalar gauge parameters in four dimensions, point restriction is not uniformly bounded as the cutoff is removed; the borderline biharmonic Green example gives a growing point evaluation. Thus estimates adequate for smooth classical backgrounds do not automatically control the transition on arbitrary Gaussian fluctuation coordinates. Gauge quotient cancellations may remove this issue in a complete coarea calculation, but they have to be exhibited.

The new nonlinear wall theorem therefore closes a limited and useful subproblem: **a protected nonlinear Wilson field/current hull with exact raw physical endpoint readouts in the periodic weak chart**. It does not close the full nonlinear formation or the physical large-field source transport theorem.

83.11 11. An exact source-centering chart on one bounded reference component

There is a useful positive statement even before that last old/new-fiber identification. On the bounded current-scale component used in R-II (1.2)–(1.11), source-dependent coordinates can be introduced without moving or differentiating its original reference cutoff. This is a finite-dimensional statement. It is not a new identification of all fine fluctuation coordinates or of a mixed-boundary nonlinear wall map.

After eliminating the relative tree coordinates, write the actual reference integral as

$$I(z) = \int_D j(B) \exp H_z(B)\, dB, \qquad D = \prod_{\ell=1}^{m} \{B_\ell \in \mathbb{R}^{d_\ell} : |B_\ell| < R_\ell\}. \tag{NL44}$$

The Euclidean product-ball formulation is the statement below; it applies when this is the norm convention of the original reference cutoff. If another norm or an additional characteristic function is present, its exact domain must be used instead. No replacement of that domain by an inscribed ball is implicit. The total number of retained real coordinates $n = \sum d_\ell$ is bounded by $C(100M)^4$; it is independent of the fine spacing. Haar-coordinate and gauge factors are part of j.

Assume that j is holomorphic and nonzero on a fixed complex tube around $\overline{D}$, and that for real $B \in D$ and small complex v

$$\big|\log j(B+v) - \log j(B)\big| \le C_M\big[(1+|B|)|v| + |v|^2\big]. \tag{NL45}$$

Assume also the one-joint-integral majorant

$$\int_D |j(B)| e^{t_M(1+|B|^2)}\, dB \le C_M, \qquad |I(0)| \ge c_M > 0. \tag{NL46}$$

These are exactly bounded-component assumptions, rather than independent estimates multiplied over fine coordinates. For the original canonical reference form, (NL45) follows from a bounded Gaussian precision and linear term, the analytic Haar density, and a Wilson remainder $g^{-2}V(gB)$ with V of order at least three: its gradient is $O(g|B|^2)$, hence $O(\epsilon|B|)$ on $g|B| \le \epsilon$. The requisite uniform analytic remainder bounds are those of the original bounded-component expansion, not the new periodic theorem. The positive comparison and analytic lower bound used in (NL46) are the bounded-component inputs isolated in S30–S35.

Let a holomorphic phase P_0 have a real critical point b_0, with

$$|b_{0,\ell}| \le R_\ell/4, \qquad \sigma_{\min}(D^2P_0(b_0)) \ge c_M, \tag{NL47}$$

and a uniform local third-derivative bound. A source H_z on an ordinary complex source polydisc is assumed small in the corresponding local analytic norm and to obey

$$|H_z(B)| \le h_M(1+|B|^2) \tag{NL48}$$

on the same tube. The finite-dimensional analytic implicit-function theorem gives a holomorphic critical point b_z of $P_0 - H_z$, with $b_z - b_0 = O_M(h_M)$. This does not require the source-free critical point to be exactly the origin of the original Gaussian coordinates; the small boundary linear term may be retained. Assume explicitly that $H_0 = 0$, that P_0, H_z are real for real arguments and real source parameters, and that $\min_\ell R_\ell \ge R_* > 0$ uniformly in the cutoff. The local derivative smallness of H_z near b_0 is a separate hypothesis from its global quadratic-growth bound; the latter alone would not control derivatives at a center allowed to grow with the cutoff. This is the ordinary quadratic-growth source class of S30–S35, not an ordinary exponential source theorem for an unbounded cubic composite.

Use the complex-bilinear polynomial

$$\vartheta_\ell(B_\ell) = 1 - \frac{B_\ell \cdot B_\ell}{R_\ell^2}, \qquad \Psi_{z,\ell}(B_\ell) = B_\ell + \frac{\vartheta_\ell(B_\ell)}{\vartheta_\ell(b_{0,\ell})}(b_{z,\ell} - b_{0,\ell}). \tag{NL49}$$

Thus $\Psi_z(b_0) = b_z$. The map is holomorphic in B, z, and fixes each corresponding ball boundary. If

$$\sup_{z,\ell} |b_{z,\ell} - b_{0,\ell}|/R_\ell < c \tag{NL50}$$

with a sufficiently small fixed c, then for real sources it is a diffeomorphism of D onto D. Indeed its derivative differs from the identity by less than $1/2$, giving injectivity. For one factor, the bound

$$t + \kappa(1-t^2) \le 1 \quad (0 \le t \le 1,\ \kappa \le 1/2)$$

shows that the ball maps into itself. Boundary identity, local invertibility, and degree one give surjectivity; the product follows. Its exact determinant is

$$\det D\Psi_z(B) = \prod_{\ell=1}^{m} \left(1 - \frac{2(b_{z,\ell} - b_{0,\ell}) \cdot B_\ell}{R_\ell^2 \vartheta_\ell(b_{0,\ell})}\right). \tag{NL51}$$

It has a holomorphic logarithm on the common polydisc, bounded by $C_M \sup_z |b_z - b_0|/\min R_\ell$. This includes the determinant; there is no omitted coordinate Jacobian.

For real z, ordinary change of variables gives the exact identity

$$I(z) = \int_D j(\Psi_z(B))\, e^{H_z(\Psi_z(B))} \det D\Psi_z(B)\, dB. \tag{NL52}$$

Both sides are holomorphic on a common smaller complex source polydisc, so the identity continues there. To see uniform domination, put $\Delta_z(B) = \Psi_z(B) - B$; it is uniformly $O_M(h_M)$ on D. Equations (NL45), (NL48), and (NL51), followed by $c|B| \le \theta|B|^2 + c^2/(4\theta)$, dominate the integrand by the single positive majorant in (NL46), after taking h_M, θ small enough. This proof is uniform in the fine cutoff. Every additional holomorphic carrying observable is composed with the same Ψ_z. At b_0, the source phase has zero first derivative in the new coordinates and its Hessian transforms by the exact congruence $D\Psi_z(b_0)^t D^2(P_0 - H_z)(b_z) D\Psi_z(b_0)$.

The importance of (NL52) is its support statement: the **actual reference product cutoff is preserved**, so no complex differentiation of a characteristic function occurs. This does not license replacing an arbitrary numerator history test χ_{old} by the same function of B. Such a test becomes $\chi_{\mathrm{old}}(\Psi_z(B))$, and that expression is not generally holomorphic. The bounded reference component and the entire old fluctuation/history integral therefore remain distinct cases.

This construction resolves an elementary source-centering/coarea issue on the fixed bounded reference integral. It uses the original reconstruction inside j. It neither asserts equality of that reconstruction with the periodic NL family at every point of D, nor supplies the still-missing local chart transition for all fine Gaussian variables.

83.12 12. Alternative source extensions on the actual critical graph

A source-only replacement has a less demanding endpoint condition than replacing the whole old fluctuation integrand. The following elementary nesting statement specifies precisely when it is available.

Let the actual iterated averages satisfy $\mathcal{M}^k = \mathcal{M}^{k-j} \circ \mathcal{M}^j$. Suppose that U_k is a critical point of the physical Wilson action on the k-constraint fiber, and that the j-reconstruction has a unique critical orbit in the admitted small chart with the same boundary data. Then

$$U_j(\mathcal{M}^j U_k) \simeq U_k. \tag{NL53}$$

Here $\simeq$ denotes exactly the gauge equivalence allowed by those boundary data. The proof is fiber inclusion: every variation with fixed $\mathcal{M}^j U$ also has fixed $\mathcal{M}^k U$. Thus U_k is critical on the smaller fiber, and uniqueness identifies its orbit. For minimizing reconstructions the same argument is simply restriction of the minimizing class. This uses actual iteration, not an associativity or direct-block-collapse identity for a nonlinear group mean.

Consequently a gauge-invariant readout may be extended in a weak chart by

$$F_{\mathrm{new}}(U, J) = \mathcal{O}\Big(U^{\mathrm{new}}_{\mathrm{crit}}(\mathcal{M}^j U)\Big), \tag{NL54}$$

independently of J, provided that the new critical branch has that same fiber and boundary condition. It agrees with the physical readout on the graph in (NL53). There is no need to change Gaussian coordinates merely to insert this identical function on that graph.

The qualifications are substantive. A raw fluctuation field $\exp(gB)U_k$ is not in general critical. The inner reconstruction must be completed before using (NL53). A mixed determining set or local box with different boundary data does not have the needed fiber inclusion merely because its scale label is j. Nor does (NL54) agree with the original source function for all independent paired-core arguments. Those arguments were never required by this limited equality.

There is a precise local gauge-gluing target. At a fixed reference connection U, CK gives

$$A_U = Q_U + d_V \rho_U, \qquad S_U = N + \rho_U d_U, \qquad A_U d_U = d_V S_U. \tag{NL55}$$

A genuine local nonlinear relative constraint with first derivative A_U is obtained by following the actual raw average of $\exp(a)U$ by the coarse gauge $\exp(\rho_U a)$, with signs and mesh factors in the physical tangent convention. Only its first derivative is being prescribed. This construction does not integrate the generally nonintegrable one-form ρ along arbitrary paths.

On overlaps, local rooted gauges have an exact transition cocycle. If both representatives have small physical potentials, their transition has a small first difference after one constant frame is

removed. Covariant derivatives, the actual Wilson Hessian, and the operators in (NL55) then have exact conjugacy. This suggests a common covariant augmented matrix with the same five types as AL, using $d_U, \delta_U, \Delta_U, A_U, S_U$. However, the needed analytic estimate is a genuine new one:

$$\|\Gamma_{U,X}\|_{Y^{-1,p}\longrightarrow X^{1,p}} + \text{uniform common-carrier collar bounds} \leq C, \tag{NL56}$$

on the original regular orbit domain and all required local boxes. The already proved fixed identity-chart $C^{1,\beta}$ inverse does not imply (NL56) for an arbitrary regular reference. In particular, small real curvature does not give a small physical current supremum; the current is naturally small only in a negative Sobolev norm. After (NL56) and a covariant nonlinear fixed-point construction, the same-gauge overlap identities could glue the source extension.

Independently gauge-fixing each already localized NL term does not provide that construction: the wall values need not be invariant under the resulting nonconstant transition. Endpoint invariance alone does not preserve the FTC sum when a different gauge is substituted in each term. Thus (NL53)–(NL54) removes the need for a full fluctuation-coordinate replacement in this source-only route, but (NL56) and compatible local gauge gluing remain explicit obligations.

84.1 1. A genuine relative gauge at a single actual blocking step

Let U, V be two connections on a lattice of spacing a_j, with $a_{j+1} = \ell_j = La_j$. Write $P_U(y,x)$ for the actual endpoint transport from the retained center y to the fine site x in its block. This includes the inner averaged transport used by the original mean, if present; it is not replaced by an unrelated direct path or a collapsed all-depth mean. Its covariance is

$$P_{U^g}(y,x) = g(y)P_U(y,x)g(x)^{-1}. \tag{CL1}$$

Use the analytic symmetric bi-equivariant group mean $\mathfrak{m}$ near the identity, normalized so that

$$\mathfrak{m}(1,\dots,1) = 1, \qquad D\mathfrak{m}_{(1,\dots,1)}(X_x) = \frac{1}{L^4}\sum_x X_x. \tag{CL2}$$

The mean used for the inner endpoint construction has these properties. No associativity of this mean is needed.

Define

$$R_x(U,V) = P_V(y,x)P_U(y,x)^{-1}, \qquad H_y(U,V) = \mathfrak{m}_x R_x(U,V), \qquad h_y(U,V) = H_y(U,V)^{-1}, \tag{CL3}$$

and the new one-step connection

$$\mathcal{B}_U(V) = \mathcal{M}(V)^{\,h(U,V)}. \tag{CL4}$$

Here the gauge convention is $W_b^g = g(x)W_b g(x')^{-1}$. The inverse in (CL3) fixes the sign of the endpoint correction in that convention. Reversing the sign convention for the gauge generator reverses the sign of ρ below as well, leaving $d\rho$ unchanged.

For $V_b = e^{a_j b_b} U_b$, the arguments R_x are near the identity on a fixed small physical supremum ball for b. Each path has physical length $O_L(\ell_j)$. Thus (CL3) is a literal holomorphic local map with a common fine-cutoff-independent radius. It satisfies

$$\mathcal{B}_U(U) = \mathcal{M}(U).$$

Its exact gauge covariance is

$$h(U^g, V^g)(y) = g(y)h(U,V)(y)g(y)^{-1}. \tag{CL5}$$

Indeed the two endpoint factors $g(x)$ cancel in R_x, and conjugation equivariance of $\mathfrak{m}$ proves the identity. Consequently

$$\mathcal{B}_{U^g}(V^g) = \mathcal{B}_U(V)^{\,Ng}, \tag{CL6}$$

where N is actual retained-node restriction. This is a joint background/variable covariance statement, not a statement that the one-step mean is pointwise independent of the background.

84.1.1 The derivative is the compatible boxed derivative

Let Q_U denote the physical left differential of the actual raw mean. The derivative of $h(U,V)$ at $V=U$ is

$$\rho_U v = -\frac{1}{L^4}\sum_x \delta P_U(y,x) P_U(y,x)^{-1}. \tag{CL7}$$

The minus sign agrees with the convention in (CL3). Therefore

$$D\mathcal{B}_U(U) = A_U = Q_U + d_{\mathcal{M}(U)}\rho_U, \qquad S_U = N + \rho_U d_U, \qquad A_U d_U = d_{\mathcal{M}(U)} S_U. \tag{CL8}$$

These are precisely the endpoint-removed operators of CK, after a consistent choice of the sign of the gauge generator. In particular at the identity

$$A_1 = Q_{\mathrm{bl}}, \qquad S_1 = S_{\mathrm{site}}. \tag{CL9}$$

The definition (CL3) is important: it constructs a genuine local nonlinear gauge from relative path variables. It does not assume that the connection one-form ρ is globally integrable.

84.2 2. The full scalar gauge differential has no thin path term

Keep U fixed and transform only V by a fine gauge g. The exact coarse gauge comparing the two outputs is

$$\gamma_U(V,g) = h(U,V^g)\,(Ng)\,h(U,V)^{-1}, \qquad \mathcal{B}_U(V^g) = \mathcal{B}_U(V)^{\,\gamma_U(V,g)}. \tag{CL10}$$

It has the ordinary cocycle identity under two successive fine gauges.

There is a useful exact cancellation in its derivative. For $g = e^{t\zeta}$,

$$R_x(U,V^g) = e^{t\zeta(y)} R_x(U,V) e^{-t\,\mathrm{Ad}_{P_U(y,x)}\zeta(x)}. \tag{CL11}$$

Left equivariance of $\mathfrak{m}$ extracts the first factor $e^{t\zeta(y)}$. Differentiating its inverse in h cancels the retained-node term in (CL10). Thus the differential $\mathcal{S}_{U,V}$ of $\gamma_U(V,e^{t\zeta})$ has the form

$$\mathcal{S}_{U,V}\zeta(y) = \sum_{x\in B_y} C_x(U,V)\zeta(x). \tag{CL12}$$

The C_x are analytic finite-dimensional coefficients. There is no fine derivative of ζ, no line-supported derivative of it, and no unresolved endpoint term in (CL12). At $V=U$, it is the actual covariant site average S_U. At $U=V=1$, its coefficients are $L^{-4}I$.

Let $\mathcal{Q}_{U,V}$ be the physical differential of (CL4), rather than its logarithmic-coordinate differential. Differentiating (CL10) gives the exact identity

$$\mathcal{Q}_{U,V} d_V = d_{\mathcal{B}_U(V)} \mathcal{S}_{U,V}. \tag{CL13}$$

This holds for arbitrary fields in the local analytic domain. No critical equation or Ward condition is imposed here.

84.3 3. Composition and the full-space fractional adjoints

Set $U_0 = U$, $U_{j+1} = \mathcal{M}(U_j)$. Starting from $V_0 = e^{ab}U$, define recursively

$$V_{j+1} = \mathcal{B}_{U_j}(V_j), \qquad b_j = a_j^{-1}\log(V_j U_j^{-1}), \qquad q_U(b) = \log(V_k U_k^{-1}). \tag{CL14}$$

The retained spacing in the last expression is one. Every determining path of this composition has physical length at most $C_L\sum_j \ell_j \le C_L$. It is a local retained-scale map, even though its internal description has k levels.

For some exactly defined coarse gauge $G_U(V_0)$,

$$V_k = \mathcal{M}^k(V_0)^{\,G_U(V_0)}, \qquad q_U(0) = 0, \qquad Dq_U(0) = A_U^{(k)} = A_{U_{k-1}}\cdots A_{U_0}. \tag{CL15}$$

The corresponding scalar differential at $b = 0$ is $S_U^{(k)} = S_{U_{k-1}}\cdots S_{U_0}$. Hence

$$A_U^{(k)} d_U = d_{U_k} S_U^{(k)}. \tag{CL16}$$

Below A_U, S_U denote these complete products when no superscript is written.

Here is the quantitative full-space estimate. Work in a weak analysis chart in which all raw reference intermediate fields have physical potentials of supremum at most M, and $\|b_j\|_\infty \le M$. The first condition follows from the actual raw coarsening bounds WP on a small $W^{1,p}$ input chart. The second is preserved on a smaller initial supremum ball by the one-step estimate below and $\sum_j \ell_j \le C_L$.

All one-step determining sets have a fixed number of coordinates, depending only on L. In physical tangent normalization their kernels are analytic functions of the dimensionless link coordinates. The value at zero is (CL9), and every other coefficient contains at least one factor of size $O(\ell_j M)$. Thus their physical row and column sums obey

$$\begin{aligned} \|\mathcal{Q}_j - Q_{\mathrm{bl},j}\|_{\ell^p\to\ell^p} + \|\mathcal{Q}_j^{\mathsf{t}} - Q_{\mathrm{bl},j}^{\mathsf{t}}\|_{\ell^p\to\ell^p} &\le C_L \ell_j M, \\ \|\mathcal{S}_j - S_j^0\|_{\ell^p\to\ell^p} + \|\mathcal{S}_j^{\mathsf{t}} - (S_j^0)^{\mathsf{t}}\|_{\ell^p\to\ell^p} &\le C_L \ell_j M. \end{aligned} \tag{CL17}$$

Here $1 \le p \le \infty$; the volume factors on the two lattices are included. These are full-space bounds. In particular the site estimate is obtained from the exact cancellation (CL11), not by assigning a Ward identity to an error field.

The flat maps are normalized positive averages, or their adjoints, and are contractions in these physical L^p norms. Their products therefore have norm at most $\exp(CM\sum \ell_j)$. The actual physical tangent factors at adjacent levels are the same ones, so they telescope when the derivatives are composed. No independent logarithmic chart factor is repeated at each level.

A fine inverse difference on one error of (CL17) costs $C_L \ell_j^{-s}$. A complete flat fine-prefix adjoint is a direct box-hold/line-filter map, uniformly bounded in $W^{s,p}$ for $sp < 1$. For the site case it is just direct block hold. Duhamel expansion, with a whole flat fine-prefix and the remaining nonlinear coarse-suffix, gives

$$\begin{aligned} \|\mathcal{Q}_{U,V}^{\mathsf{t}}\lambda\|_{W_a^{s,p}} &\le C\left(\|\lambda\|_{W_1^{s,p}} + M\sum_j \ell_j^{1-s}\|\lambda\|_{\ell^p}\right), \\ \|\mathcal{S}_{U,V}^{\mathsf{t}}\xi\|_{W_a^{s,p}} &\le C\left(\|\xi\|_{W_1^{s,p}} + M\sum_j \ell_j^{1-s}\|\xi\|_{\ell^p}\right). \end{aligned} \tag{CL18}$$

Since the final mesh is one, both right sides are bounded by the corresponding coarse ℓ^p norm times a fixed constant. The L^∞ adjoint bounds also follow from (CL17). These statements are uniform on ordinary complex weak balls and hence allow ordinary parameter Cauchy estimates.

Only a supremum bound on the potentials is used for the fractional adjoint calculation. To use $Dq_U(b)$ as a logarithmic differential, the physical fine chart factor is multiplied once at the endpoint. For $b \in W^{1,p}$, $p > 8$, that factor is a bounded multiplier on $W^{s,p}$, $s < 1/p$. The coarse endpoint factor is at mesh one.

84.5 5. The physical-gradient nonlinear augmented system

Put

$$W = e^{ab}U, \qquad \omega = \Delta_U \phi.$$

Consider the equations

$$\begin{aligned} \mathcal{E}_a(b)^{\mathsf{t}}\{J_W(W) + d_W\omega\} + Dq_U(b)^{\mathsf{t}}\{\kappa q_U(b) + \mu\} &= 0, \\ \Delta_U\psi + S_U^{\mathsf{t}}\nu &= 0, \\ -\delta_U b + \psi + \Delta_U\phi &= 0, \\ S_U\phi &= 0, \\ q_U(b) &= 0. \end{aligned} \tag{CL22}$$

Here $\mathcal{E}_a$ is the physical logarithmic chart derivative, and J_W is the actual Wilson current. The additional force in the first row is the physical covariant gradient $d_W\omega$. It is not the independent-current extension of the original renormalization maps.

The derivative at $b = \psi = \phi = \nu = \mu = 0$ is (CL19); the inhomogeneous term on the left is $J_W(U)$ in the first row, so the corresponding right-side source is $\mathbf{f}_U = (-J_W(U), 0, 0, 0, 0)$. The nonlinear difference

$$\mathcal{E}_a(b)^{\mathsf{t}} d_W \Delta_U \phi - d_U \Delta_U \phi \tag{CL23}$$

is quadratic in the weak type norm. Indeed $d_W - d_U$ is a zero-order coefficient of size $O(b)$, while $\mathcal{E}_a(b) - I$ carries the explicit fine factor a. Multiplication by a $W^{1,p}$ coefficient is bounded on $W^{-1,p}$ for $p > 4$; the required dual product uses $W^{1,p'} \hookrightarrow L^{p/(p-2)}$. Thus there is no unaccounted fine inverse in (CL23).

The actual Wilson logarithmic gradient is an analytic map $W^{1,p} \to W^{-1,p}$ on this small ball, as quantified in BI. The constraint terms obey the analogous bounds by (CL17). Consequently, writing the equations as $\mathbb{K}_U \mathbf{b} = \mathbf{f}_U - N_U(\mathbf{b})$,

$$\|\mathbf{f}_U\|_{Y_{p,\mathrm{ul}}} \le C\epsilon, \qquad \|N_U(\mathbf{b})\|_{Y_{p,\mathrm{ul}}} \le C\|\mathbf{b}\|^2_{X_{p,\mathrm{ul}}}, \qquad \|DN_U(\mathbf{b})\|_{X_{p,\mathrm{ul}} \to Y_{p,\mathrm{ul}}} \le C\|\mathbf{b}\|_{X_{p,\mathrm{ul}}}. \tag{CL24}$$

Here "ul" means the uniformly local typed norms on the fixed retained-scale cover, with negative components defined by the same localized weak decompositions as in BI. The exponential cell majorants bound the inverse in these norms. Thus neither the smallness premise nor the displayed constants grow with the number of boxes. The corresponding global inequalities hold for globally small data; a merely local ceiling does not imply a volume-independent global L^p norm.

The wall filing is performed **before forming a strong derivative**. File the Wilson reference force and nonlinear terms as their weak divergence/product vertices, or equivalently weight each complete plaquette action and constraint-coupling vertex before taking its fine variation. A wall factor is a scalar coefficient of the whole vertex. It is not a sharp spatial multiplier applied to an already formed $W^{-1,p}$ force: such multiplication need not be bounded.

For (CL23), use its weak product-rule expression, with derivatives on the test field and on the local analytic chart coefficients. Keep every term arising from that expression on the same complete carrier.

In particular, do not multiply a distributional derivative of ω by an unrelated wall hold. The resulting weak norm bound uses the original product estimates and the bounded carrier weights, with no derivative of a sharp wall factor. Locality and the weak variation are then literal, including their endpoint terms.

With this convention, file each vertex on its complete actual determining carrier and use the same carrier wall factors as in AL. Let the resulting weak maps be $N_{U,\mathbf{s}}$ and $\mathbf{f}_{U,\mathbf{s}}$. The equation

$$\mathbf{b}_{\mathbf{s}} = \Gamma_{U,\mathbf{s}}[\,\mathbf{f}_{U,\mathbf{s}} - N_{U,\mathbf{s}}(\mathbf{b}_{\mathbf{s}})\,] \tag{CL25}$$

has one small holomorphic solution on the common wall polydisc. The contraction proof uses (CL24), after the localization geometry and wall radius have been fixed. Repeated nonlinear carrier occurrences are controlled by that contraction; they are not incorrectly counted as independent Gaussian integrals.

All entries in (CL22), the reference force, the nonlinear vertex filing, and the inverse construction are jointly gauge covariant. Uniqueness therefore gives covariance of every wall value $W_{\mathbf{s}}(U)$. On a protected region whose complete entry carriers have weight one, the same-carrier identity proves the literal nonlinear equations (CL22). This is stronger than exponential proximity to a solution.

84.6 6. Exact protected Noether identity

On the protected equations set

$$W_k = e^{q_U(b)}U_k, \qquad \lambda = \mathcal{E}_1(q_U(b))^{-\mathsf{t}}[\,\kappa q_U(b) + \mu\,].$$

Removing the endpoint chart factor in the first row of (CL22) gives

$$J_W(W) + d_W\omega + \mathcal{Q}^{\mathsf{t}}_{U,W}\lambda = 0. \tag{CL26}$$

The exact Wilson gauge identity is $\delta_W J_W(W) = 0$. Applying δ_W and using (CL13), composed over all levels, yields

$$\Delta_W\omega = -\mathcal{S}^{\mathsf{t}}_{U,W}\delta_{W_k}\lambda. \tag{CL27}$$

The right side is a site-adjoint lift, not a derivative of a possibly rough fine force. Equation (CL27) holds only after shrinking the protected region by every differential stencil and every complete averaging carrier used in this calculation. There is no assumption of an independent coarse Ward condition.

At retained spacing one, $\delta_{W_k}\lambda$ has a bounded local coarse ℓ^p norm. Equation (CL18) therefore puts the right side of (CL27) in $W^{s,p}$, $s < 1/p$, uniformly in the fine cutoff. It also gives a supremum bound if the coarse multiplier is bounded.

84.7 7. Protected fractional curvature, without an initial high reference norm

Let $B \Subset B_1 \Subset B_2 \Subset B_3$ be fixed retained-scale collars on which the required protected equations hold, including their complete stencils. The weak solution gives W a small $W^{1,p}$ potential in a local analysis chart, and $\omega \in L^p$. The following bootstrap gives

$$\|\omega\|_{W^{2+s,p}(B)} + \|J_W(W)\|_{W^{s,p}(B)} + \|F_a(W)\|_{W^{1+s,p}(B)} \le C\epsilon. \tag{CL28}$$

The $W^{2+s,p}$ norm of ω in this display is asserted in the final total-field Coulomb analysis gauge constructed below. It is not transported through a merely $W^{2,p}$ gauge change, and it is not identified with a high-order covariant norm relative to the original rough reference U. The curvature and

current bounds, of orders $1+s$ and s, do transfer to the original controlled weak chart: the gauge is $W^{2,p} \subset C^{1,\alpha}$, with $\alpha > s$, so the required product estimates apply. They can equivalently be expressed by the corresponding low-order covariant norms. No arbitrary spatial jet estimate is asserted.

First, (CL27) implies $\omega \in W^{1,p}(B_2)$ and then $W^{2,p}(B_1)$. For completeness, the lower terms of the covariant Laplacian, acting on an L^p scalar, are in $W^{-1,p}$. The product involving Dv and ω is tested using the same embedding $W^{1,p'} \hookrightarrow L^{p/(p-2)}$; all other terms are divergences of L^p quantities or zero-order products. Interior flat elliptic estimates and a small-coefficient perturbation give the first derivative bound. Since $p > 4$, this makes ω bounded. The terms $vD\omega$, $(Dv)\omega$, and $v^2\omega$ are now in L^p, giving the second derivative bound. Smooth collar cutoffs use only the previously obtained lower norms.

Equation (CL26) and the full-space L^p adjoint estimate give $J_W(W) \in L^p$ on the next collar. Put the total connection W, and at the same time its reference U, in an ordinary local Coulomb analysis gauge. The weak gauge construction uses a fixed enclosing box and a small $W^{2,p}$ gauge parameter; it preserves both weak charts. It is used only for estimates, not in defining the wall family or its input gauge covariance.

In that gauge, write $W = e^{av}$, $\delta_a v = 0$. The exact Wilson identity WB15 gives

$$\delta_a d_a v = J_W(v) + f_a(v), \qquad \|f_a(v)\|_{L^p} \le C\epsilon^2. \tag{CL29}$$

Interior elliptic estimates now give $v \in W^{2,p}$. In particular $v \in C^{1,\alpha}$ for any $\alpha < 1 - 4/p$, and $F_a(v) \in W^{1,p} \cap L^\infty$. This is obtained from the actual current, not postulated as regularity of a stationary label.

Here is a direct check that the scalar upgrade does not introduce an inverse-spacing coefficient. In one direction put

$$B_+(v) = a^{-1}(e^{a\,\mathrm{ad}\,v} - I).$$

The covariant Laplacian differs from the flat one by first differences of ω with bounded coefficients $B_+(v)$ and their negative-direction counterparts, and a zero-order coefficient of the form

$$-D^- B_+(v) - a^{-2}\{e^{a\,\mathrm{ad}\,v_-} + e^{-a\,\mathrm{ad}\,v_-} - 2I\}. \tag{CL30}$$

Signs depend on the incidence convention and do not affect the norm statement. The second term is $O(v_-^2)$; it has no a^{-1} loss. The first term is controlled by Dv. Thus these coefficients have a uniformly bounded C^α norm on the $W^{2,p}$ ball. Apply (CL27) again, now with its $W^{s,p}$ right side and $s < 1/p < \alpha$. The ordinary product bounds and interior constant-coefficient elliptic estimate give $\omega \in W^{2+s,p}$. Equation (CL26) then gives $J_W(v) \in W^{s,p}$.

Finally this current bound upgrades the curvature by the exact Wilson expansion, not by a continuum substitution. Write

$$F_a(v) = d_a v + G_a(v), \qquad \|G_a(v)\|_{W^{1+s,p}} \le C\epsilon^2. \tag{CL31}$$

To prove the last inequality, subtract the linear curl from the four-link BCH series. Each word of degree $n \ge 2$ has coefficient a^{n-2}; $v \in W^{2,p} \subset W^{1+s,p}$, and the latter is an algebra for the present p, s. The analytic word series sums uniformly.

The zero-order Wilson incidence error times F is $O(\epsilon^2)$ in $W^{s,p}$. The analytic flux correction $a^4\mathcal{P}(F)$ costs at most one fine inverse difference:

$$a^4\|\delta_W \mathcal{P}(F)\|_{W^{s,p}} \le Ca^3\|F\|_\infty^2\|F\|_{W^{s,p}} \le C\epsilon^3. \tag{CL32}$$

Consequently (CL29) holds in $W^{s,p}$, with $\|f_a(v)\|_{W^{s,p}} \le C\epsilon^2$. The Coulomb condition and a final interior elliptic estimate give $v \in W^{2+s,p}$; (CL31) proves (CL28).

The same argument supplies the original low readout bounds on a smaller collar. The full adjoint L^∞ estimate and $\omega \in W^{2+s,p}$ give $J_W(v)$ bounded. The first derivative of $G_a(v)$ is bounded, and the flux correction still has its factor a^3. Thus

$$\|v\|_{C^{1,\beta}} + \|\Delta_a v\|_\infty + \|J_W(v)\|_\infty + \|F_a(v)\|_\infty \le C_\beta \epsilon \tag{CL33}$$

for fixed $\beta < 1 + s - 4/p$. This identifies an actual protected weak core together with its fractional improvement. Sharp held currents prevent replacing (CL28) by arbitrary high spatial derivatives without further work.

84.8 8. Endpoint gauge elimination

At $\mathbf{s} = 1$, all equations hold globally. Let R_U be the symmetric complex-bilinear projection onto $\Delta_U \ker S_U$, defined by the constrained scalar biharmonic block. BI's scalar estimates imply its boundedness and the bounded invertibility of

$$\zeta \longmapsto (R_U \Delta_U \zeta,\ S_U \zeta) : H^2 \longrightarrow \operatorname{Ran} R_U \oplus \ell^2. \tag{CL34}$$

The scalar rows of (CL22) give $\omega = \Delta_U \phi \in \operatorname{Ran} R_U$.

The corresponding map

$$\zeta \longmapsto (R_U \Delta_W \zeta,\ \mathcal{S}_{U,W} \zeta) \tag{CL35}$$

is a small perturbation of (CL34). Here $\Delta_W - \Delta_U : H^2 \to L^2$ has norm $O(\|b\|_{W^{1,p}})$, and (CL17) controls the second row. For $p > 4$, the coefficient containing Db is legitimate because H^2 embeds into $L^{2p/(p-2)}$ on each fixed cell. The exponentially localized reference estimates retain uniform control of the full map. Thus (CL35) is onto.

Pair (CL26) with a physical gauge variation $d_W \zeta$ having $\mathcal{S}_{U,W} \zeta = 0$. Wilson gauge invariance and (CL13) remove the action and constraint terms, leaving

$$\langle \Delta_W \zeta, \omega \rangle = 0.$$

For any fine scalar f, choose ζ in (CL35) with data $(R_U f, 0)$. Symmetry of R_U and $R_U \omega = \omega$ give $\langle f, \omega \rangle = 0$. Hence $\omega = 0$, also for complex fields; no inference from a complex squared norm is used. The scalar inf-sup estimate then gives $\phi = 0$.

Surjectivity of $\mathcal{S}_{U,W}$ and (CL26) imply $\delta_{W_k} \lambda = 0$. The remaining Euler equation is the true constrained physical one. Since (CL15) differs from the actual raw constraint only by a coarse gauge, its extra vertical differential is killed by this Ward condition. Thus the endpoint is a physical critical orbit for the raw mean, with the prescribed coarse orbit.

84.9 9. Exact matching on the original physical critical graph

If the input U is itself a small regular critical configuration for the actual raw k-step mean, let its actual multiplier be λ_0, with its actual Ward identity. Telescoping the linear products in (CL8) gives

$$A_U - Q_{\mathrm{raw},U} = d_{U_k} \rho_U^{(k)}. \tag{CL36}$$

This identity is used only as a differential identity; it does not integrate $\rho_U^{(k)}$. The Ward identity therefore gives $A_U^{\mathsf{t}} \lambda_0 = Q_{\mathrm{raw},U}^{\mathsf{t}} \lambda_0$. Consequently

$$b = \psi = \phi = \nu = 0, \qquad \mu = -\lambda_0 \tag{CL37}$$

is the small solution at $\mathbf{s} = 1$. Its multiplier bound follows from the reference inf-sup estimate. Uniqueness in the small contraction ball proves

$$W_{\mathbf{1}}(U) = U. \tag{CL38}$$

This is an exact graph identity, not merely equality of a trace after choosing an uncontrolled gauge bridge.

The nesting argument NL53 applies to actual inner reconstructions whose constraint fibers and boundary data are genuinely nested. For a gauge-invariant source readout on that graph, composing with $W_{\mathbf{s}}(U)$ supplies a gauge-invariant local wall extension. The pre-existing Gaussian variables, Jacobian and physical cutoff need not be changed merely to evaluate this identical endpoint function.

This conclusion must not be enlarged. A raw field $e^{gB}U$, an arbitrary independent pair (U, J), or a local minimizer with different boundary constraints need not lie on this graph. The new map agrees with the original physical readout only after the actual appropriate inner reconstruction has been made. Likewise, the presence of both fine and coarse retained constraints in an R determining set is not covered by relabeling that set as one periodic boxed constraint.

85.1 1. The periodic two-scale geometry

Fix odd L>=3. Let the fine spacing be a=1/n, with n odd, on a compatible periodic four-torus. The two retained grids have spacings 1 and L. Every centered L-block is a union of Lˆ4 centered unit

blocks. Let Omega be an arbitrary union of these L-blocks. An L-block in Omega is unrefined; a block outside Omega is refined into all its unit children.

The mixed determining rows are:

- all unit-grid bonds with both endpoints outside Omega;
- an arbitrary chosen subset of unit-grid bonds with one endpoint outside and one endpoint inside Omega;
- all L-grid bonds with at least one endpoint in Omega.

The second class consists of retained inward rows. Its endpoints inside Omega are the junction nodes. No unit bond with both endpoints inside Omega is included. Let I contain all unit-grid sites outside Omega and the distinct junction nodes. Let C contain the L-grid centers in Omega. The scalar node set is I union C. A junction is in the boundary unit shell of an L-block. In particular its central unit child is never a junction when L>=3. This fact will supply a scalar and a field pivot.

Write S_r for the full centered site average from a to r and Q_r for the full box-and-line physical one-form average. Define

$$(S_m\phi)(v) = \begin{cases} S_1\phi(v), & v \in I, \\ S_L\phi(v), & v \in C, \end{cases} \qquad (Q_m A)(c) = Q_{r_c}A(c), \quad r_c \in \{1, L\}. \tag{MB1}$$

Node and row pairings have physical weights r_v^4 and r_c^4. Since the two lengths are 1 and L, changing to counting norms costs only fixed L-dependent factors. The fine pairing has weight a^4. All derivatives and the Laplacian are periodic fine differences. Coarse mass weights are comparable to r_c^{-2}; at these two lengths they have fixed positive lower and upper bounds.

There are exact maps R and P such that

$$S_L = RS_m, \qquad Q_L = PQ_m. \tag{MB2}$$

For R, retain the coarse node value on an unrefined L-block and average the L^4 unit values on a refined L-block. Extra junction values are not needed by R. For P, retain a mixed L-row if it is present. If an L-row has both endpoints outside Omega, its two endpoint L-blocks are both refined. Every unit row used by the exact composition Q_L=Q_{L,1}Q_1 then has both endpoints outside Omega and is a retained mixed row. Thus no interface or inward row is missing in this factorization. Both maps have fixed finite range and bounded physical Lp norms.

There is also a finite mixed derivative D_m with

$$Q_m d = D_m S_m. \tag{MB3}$$

On a unit row it is the ordinary difference of its two node values; both values occur in I, including a junction when needed. On an L-row it is the difference of R c at its endpoints, divided by L. At an outside endpoint R c is the average of the unit children, rather than the value at its unit center. This distinction is essential. In particular D_m is not silently identified with incidence on the actual mixed nodes.

85.2 2. The mixed scalar block

For every 1<p<infinity there is a local scalar right inverse E_m with

$$S_m E_m = I, \qquad \|E_m c\|_{W_a^{2,p}} \le C_{L,p} \|c\|_{\ell_m^p}. \tag{MB4}$$

First choose normalized smooth bumps in the middle of every unit pillar whose center is in I. They realize its S_1 moment, have uniform first and second physical differences, and have disjoint supports. Such a bump at a junction also contributes to S_L on its containing unrefined L-block. After these unit values have been set, correct that L-block's S_L value using a bump in its central unit child. That child has no unit or junction scalar constraint. Its S_L moment is bounded below by a positive L-dependent constant, so the correction is uniformly bounded. Different coarse corrections have disjoint supports. For a=1 the same construction uses point bumps; the finitely many difference factors are then bounded by a fixed constant. This proves MB4, including arbitrary retained inward subsets. No point-sampling estimate on a continuum Sobolev space is being assumed.

The kernel inclusion in MB2 gives a simple scalar coercivity estimate. If S_m phi=0, every full L-block mean vanishes. Block Poincare and periodic summation by parts give

$$\|\phi\|_2 \le C_L \|D\phi\|_2, \qquad \|D\phi\|_2^2 = \langle \phi, \Delta\phi \rangle, \qquad \|\phi\|_{H_a^2} \le C_L \|\Delta\phi\|_2. \tag{MB5}$$

The second-difference part uses the exact periodic flat multiplier identity, with equivalent sums of component differences. The scalar constant is fixed by the averages. Therefore the constrained biharmonic block has the uniform energy inverse obtained by testing against E_m and solving its coercive form on ker S_m.

Let R_m be the L2 orthogonal projection onto Delta ker S_m, and let H_m c be the unique minimizer of $\|\Delta\phi\|_2^2$ subject to S_m phi=c. The preceding proof gives

$$S_m H_m c = c, \qquad R_m \Delta H_m c = 0, \qquad \|H_m c\|_{H_a^2} \le C_L \|c\|_{\ell_m^2}. \tag{MB6}$$

For the last bound compare the minimum with E_m c. Subtract that lift and apply MB5 on the kernel. This avoids an inverse of a small global Fourier momentum.

85.4 4. Mixed slice coercivity by comparison to one full coarse grid

Write R_L for the projection onto Delta ker S_L. MB2 implies

$$\ker S_m \subset \ker S_L, \qquad \ker Q_m \subset \ker Q_L, \qquad \operatorname{Ran} R_m \subset \operatorname{Ran} R_L. \tag{MB13}$$

Suppose Q_m A=0 and R_m delta A=0. There is a unique beta in ker S_L with Delta beta=R_L delta A, by the uniform scalar block. Set B=A-d beta. Then Q_LB=0, R_L delta B=0, and dB=dA. The uniform-scale AL coercivity, rescaled from retained length one to fixed length L, gives

$$\|B\|_{H^1_a} \le C_L \|dA\|_2. \tag{MB14}$$

Moreover B is also in the mixed slice, by the last inclusion in MB13. Consequently R_m Delta beta=0. If c=S_m beta, then

$$\beta = H_m c, \qquad Rc = S_L \beta = 0, \qquad D_m c = Q_m d\beta = -Q_m B. \tag{MB15}$$

The node vector c has the following finite-block Poincare estimate:

$$Rc = 0 \quad \Longrightarrow \quad \|c\|_{\ell^2_m} \le C_L \|D_m c\|_{\ell^2_m}. \tag{MB16}$$

Indeed c vanishes on every unrefined coarse node. On each refined L-block its L^4 unit values have zero arithmetic mean. All unit edges inside that block are retained determining rows, so ordinary finite grid Poincare controls its unit values by those differences. Finally each junction value is its outside endpoint value plus the difference on one chosen retained inward row. Every outside value is used a bounded number of times. This proves MB16 with no missing interface edge and no connectivity assumption on Omega.

The averaging map Q_m:H_a^1 to mixed ell2 is uniformly bounded, since its two retained lengths have positive lower bound one and its physical kernels have bounded local L2 norm. Combining MB6 and MB14–16 gives

$$\|\beta\|_{H^2_a} \le C_L \|c\|_{\ell^2_m} \le C_L \|Q_m B\|_{\ell^2_m} \le C_L \|B\|_{H^1_a}, \qquad \|A\|_{H^1_a} \le C_L \|dA\|_2. \tag{MB17}$$

This is coercivity on the actual mixed slice and constraint kernel. For general A in ker Q_m, subtract d phi_0 with S_m phi_0=0 and Delta phi_0=R_m delta A. MB3 preserves its constraint, and MB5 controls that subtraction. For arbitrary A, first subtract the row lift E_row Q_m A and then apply this kernel estimate. The result is

$$\|A\|^2_{H^1_a} \le C_L \left(\|dA\|_2^2 + \|R_m \delta A\|_2^2 + \|Q_m A\|^2_{\ell^2_m} \right). \tag{MB18}$$

There is no comparison of a nonlocal projected gauge with a full Feynman gauge by dropping its subtraction. The nested scalar kernels and the finite block estimate MB16 are what control that subtraction.

85.7 7. Exact flat raw target recovery and minimizing orbits

Let A_red be the physical curl-energy minimizer for Q_red,m A=B in the mixed slice. It is the one-form output of MB19 with f=g=h=k=0, using Q_red,m. Its gauge auxiliary vanishes: apply delta to its symmetric first row, use MB24, and pair the resulting equation Deltaˆ2 phi+S_mˆ*D_node*ˆ(kappa B+mu)=0 with phi in ker S_m. Thus Delta phi=0 and then phi=0. Its remaining equation is the physical constrained Euler equation, and convexity proves minimality. The bound from MB20 is $\|A_{\mathrm{red}}\|_{W^{1,p}} \le C\|B\|_{\ell^p}$.

The mixed nodes are a subset of the unit grid, so they are separated by distance at least one. Smooth node bumps give an exact lift E_N with N_node E_N=I and ellp to W2p norm C_{L,p}. The

connector estimate FW23, applied at the two fixed retained lengths and restricted to the actual nodes, gives

$$\|K_{\mathrm{node}}A\|_{\ell^p_m} \le C_{L,p}\|A\|_{W^{1,p}_a}, \qquad p > 2. \tag{MB25}$$

Set phi_N=E_N K_node A_red and A_raw=A_red+d phi_N. Then

$$Q_{\mathrm{raw},m}A_{\mathrm{raw}} = B, \qquad dA_{\mathrm{raw}} = dA_{\mathrm{red}}, \qquad \|A_{\mathrm{raw}}\|_{W^{1,p}_a} \le C_{L,p}\|B\|_{\ell^p_m}. \tag{MB26}$$

Conversely, if Q_raw,m A=B, choose phi with S_m phi=-K_node A, using MB4. MB24 then gives Q_red,m(A+d phi)=B, without changing its curl. Thus the two constraint fibers have the same minimum curl energy, and MB26 represents an actual raw constrained minimizing orbit. No assertion that the final raw representative remains in the earlier mixed slice is needed. This is a flat exact-target statement. It does not restore the false unrestricted raw hidden inverse of FI: adding a gradient preserves the curl action, but it does not preserve an arbitrary external one-form forcing.

85.8 8. What this local theorem does and does not identify

The two-scale models are the natural candidates for the adjacent-layer localization cubes in the original variational geometry. CMP 102 (1985), pp. 277–279, states both a separation of successive regions in their own units and the use of enlarged cubes contained in Bˆj(A_j) union Bˆ{j+1}(A_{j+1}). The inward-row option above is explicit because different source formulations of determining bond sets do not all use the same inward-bond convention.

MB20 supplies a uniform periodic local inverse for either convention, and MB24–26 supplies its actual flat raw target bridge. To use these models in a global many-scale parametrix one must still construct a partition whose complete row and scalar stencils meet at most the two adjacent scales, match each actual mixed row on its inner carrier, keep physical row masses r_cˆ{-2}, and verify the commutators in the scaled X/Y types. A smooth cutoff cannot turn an unmatched boundary action into the periodic component action. Those boundary rows must be retained or separately compared.

For nonlinear relative constraints, MB22 is only their derivative at the identity. The required outer node factor must be computed from the prescribed finer internal datum, with its actual noncommutative order, before claiming exact raw target preservation. The newly defined CL relative endpoint means are suitable objects to examine, but their nonlinear mixed seam identity is not assumed by the flat proof above.

85.9 9. An adaptive cover for a separated hierarchy

We now give a global inverse under explicit determining-set hypotheses. This uses the actual nodal relative map CL39–48, whose local flat derivative is Q_red,m. It does not assume a globally invertible seam transformation or take a product of the local inverse constants.

Let xi_j=Lˆj a and let Omega_0 contain Omega_1 contain ... contain Omega_k be an aligned nested hierarchy. Its primary pillars partition the active region, and have scale xi_j in layer j. Retain the actual interior/outward determining rows and any specified inward rows with their junction nodes. Assume

$$\operatorname{dist}(\Omega_j^c, \Omega_{j+1}) \ge H\xi_j. \tag{MB27}$$

Every complete averaging or prefix-site carrier at level j has diameter at most c_L xi_j and is contained in its prescribed endpoint blocks and their fixed adjacent-layer collar. The latter is the actual finite path locality, not a collapse of the nonlinear composition. The aligned block geometry makes an outside endpoint of a level-j outward row a level-(j-1) node; all children of its outside level-j block are at that finer level. Extra inward nodes use their actual finer prefix. These are precisely the row incidences matched in MB1 and CL40.

Fix a localization radius R_loc, eventually large enough for the commutator estimate. Choose H>=C_L R_loc, with the numerical constant also large enough for all complete-stencil enlargements below. This is an explicit quantitative requirement on the geometric separation. It is compatible with choosing the original fixed large geometry constants before the field ceiling. A merely unquantified adjacent- layer label would not establish it.

Here is one construction of the cover. Let sigma(x) be the scale of the primary pillar containing x. Define a Lipschitz minorant

$$s(x) = \inf_y \left\{ \sigma(y) + \frac{\operatorname{dist}(x,y)}{C_0 R_{\mathrm{loc}}} \right\}, \qquad L^{-1}\sigma(x) \le s(x) \le \sigma(x). \tag{MB28}$$

The lower bound follows from MB27. If x is in layer j, a point in layer at most j-2 is outside Omega_(j-1), hence is at distance at least H xi_(j-1)=H sigma(x)/L. Such a point cannot lower the infimum when H>=C_0 L R_loc. A point in the immediately finer layer has scale sigma(x)/L, and points at coarser scales cannot lower that number. Fine corner conventions change this by fixed L factors, which can equivalently be absorbed by a one-fine-cell enlargement of the pillars.

Select a maximal disjoint family of balls of radius a fixed small multiple of R_loc s(z). Since s has Lipschitz constant (C_0 R_loc)^(-1), intersecting enlarged balls have comparable radii. Euclidean packing gives a bounded overlap independent of the depth. Fixed smooth radial cutoffs, normalized by the square root of their sum of squares, give a sampled smooth partition

$$\sum_z \chi_z^2 = 1, \qquad |D^m \chi_z| \le \frac{C_{m,L}}{(R_{\mathrm{loc}}\ell_z)^m}, \quad m = 1,2,3,4. \tag{MB29}$$

Here ell_z is the smallest retained length on the fully enlarged tile. It is comparable to s(z), within a fixed L factor. Each enlarged tile meets at most two adjacent levels: to meet both a layer below j and a layer above j it would have to cross the gap H xi_j/L, whereas its diameter is O_L(R_loc xi_j). Increasing the fixed constant in MB27 rules this out, including the complete carrier collar of diameter c_L xi_j. On an interior tile the exact local pattern is consequently one MB two-scale pattern after rescaling by ell_z. Extend that pattern periodically on a larger compatible tile, agreeing on all inner stencils. Arbitrary geometry outside the inner carrier is not being identified with the actual hierarchy.

85.11 11. The literal bare action boundary

There is a separate exact local model at the outer action boundary. The original separation includes j=0, so a sufficiently small tile meeting that boundary lies in the bare collar before Omega_1. Every dynamic bond and node there has its level-zero identity constraint: Q_0=S_0=I. Fixed exterior or ghost coordinates are excluded from the unknowns, and their gauge variations are zero. Keep the actual local action, its actual plaquette convention, and its actual dynamic bond/site graph. Do not replace it by a periodic Maxwell action.

For any Wilson Hessian H on that graph, the symmetric five-type system has the explicit inverse

$$
\begin{aligned}
&A = B, \qquad \phi = k, \qquad \psi = h + \delta_U B - \Delta_U k,\\
&\nu = g - \Delta_U \psi,\\
&\mu = f - (H + \kappa) B - d_U \Delta_U k.
\end{aligned}
\tag{MB32}
$$

This is direct substitution into its five rows. In bare units ell=a the fine mesh equals one, all graph degrees and difference norms are bounded, and W^{-m,p}, Lp, and W^{m,p} on this mesh are uniformly equivalent for each fixed m. Thus MB32 has a uniform typed bound and finite range, independently of the boundary shape. For real compact references its matrix coefficients are bounded; their admitted small complex perturbations have the same bound. No positivity or gauge kernel of an unconstrained boundary Hessian is needed.

The boundary criterion is explicit: all unknowns in every relevant boundary stencil must really be pinned level-zero variables. The collar in CMP 102 (1985), equation (1), includes j=0 and provides the required distance from the outer action boundary to Omega_1. If an alternative model leaves unpinned boundary variables, MB32 does not apply to that model. Its boundary inverse would remain an additional problem.

86.3 3. Weak equations and nonlinear estimates

Let E_a(b) denote the physical left chart derivative. With W=eˆ{ab}U and omega=Delta_U phi, the unweakened equations are

$$\begin{aligned} E_a(b)^{\mathsf{t}}\{J_{\Omega_0}(W)+d_W\omega\}+Dq_U(b)^{\mathsf{t}}\{\kappa q_U(b)+\mu\}&=0,\\ \Delta_U\psi+S_U^{\mathsf{t}}\nu&=0,\\ -\delta_U b+\psi+\Delta_U\phi&=0,\\ S_U\phi&=0,\\ q_U(b)&=0. \end{aligned} \tag{MC9}$$

The Wilson current and Hessian are those of the literal action on its actual graph, with its actual plaquette boundary convention. Every transpose is the invariant complex-bilinear transpose. Hermitian absolute norms are used only in estimates. The extra force d_W omega is a physical covariant gradient, not a flat-logarithmic gradient and not the independent current of an older off-graph map.

The derivative at all unknowns zero is the exact global matrix K_U of MB, including the relative-chart Wilson Hessian and its current term. The inhomogeneous source is f_U=(-J_(Omega_0)(U),0,0,0,0). After subtracting that derivative write

$$\begin{aligned} K_U x&=f_U-N_U(x), & \|f_U\|_{Y,\mathrm{uloc}}&\le C\epsilon,\\ \|N_U(x)\|_{Y,\mathrm{uloc}}&\le C\|x\|^2_{X,\mathrm{uloc}}, & \|DN_U(x)\|_{X\to Y}&\le C\|x\|_{X,\mathrm{uloc}}. \end{aligned} \tag{MC10}$$

Here are the needed uniformity checks. Rescale an adaptive tile by its constant base length ell, using MC3. The reference is small W1p in its CE chart, and p>4 gives the local algebra and multiplier bounds. The actual Wilson logarithmic gradient is W1p to Wˆ{-1,p} analytic, as proved with its full chart term in BI23. The physical auxiliary difference is

$$E_a(b)^{\mathsf{t}}d_W\Delta_U\phi-d_U\Delta_U\phi. \tag{MC11}$$

The connection difference is zero order and O(b); E_a(b)-I has the explicit factor a. Its weak product-rule expression is bounded quadratically from W1p times W2p to Wˆ{-1,p}. The dual differentiated product uses Wˆ{1,p'} into Lˆ{p/(p-2)}. No fine inverse is inserted. The actual relative constraint has the full-space operator estimates of CL17 and CL46: one whole fine-to-unit prefix followed by one mixed top map. The top has fixed L complexity, and the prefix error sum is geometric. This bounds q, Dq and their analytic remainders in the coarse row types with no depth factor. Bounded carrier overlap then gives MC10 in the uniformly local adapted norm.

The force bound uses the divergence of the locally Lp-small Wilson flux plus bounded lower-order terms in the analysis chart. At the bare boundary the scaled mesh is one and it is the direct finite plaquette estimate. It does not suppose that the physical current is initially uniformly bounded at the microscopic scale.

86.4 4. One common wall contraction

Use the common typed carrier expansion of MB35–36 to obtain Gamma_(U,s). Define f_(U,s) and N_(U,s) by weak vertex filing: Wilson plaquette and coarse constraint-coupling vertices are weighted

before variation; the auxiliary term MC11 retains all terms of its weak product rule on the same complete carrier. This preserves the weak bounds and exact covariance. It is not multiplication by a sharp spatial step function on a preformed W^{-1,p} distribution.

An inverse box meets at most C_L adaptive wall cells. The weighted linear remainder is bounded by R_w^(C_L) C/R_loc. This was made less than one half before fixing the field ceiling. The local nonlinear vertices have the same finite carrier bound, so MC10 holds uniformly on the whole wall polydisc, with a fixed enlarged constant. Solve

$$x_{\mathbf{s}} = \Gamma_{U,\mathbf{s}}[f_{U,\mathbf{s}} - N_{U,\mathbf{s}}(x_{\mathbf{s}})]. \tag{MC12}$$

On the ball of radius 2 C_Gamma C_f epsilon, its derivative norm is at most C_Gamma C_N times that radius. Decrease epsilon_* once so this is below one half and the ball is invariant. Banach contraction proves existence, uniqueness, and MC4. Uniformly local estimates use the exponentially summable local cell majorants; a global Lp bound on a large tile is not substituted for them.

All operations are holomorphic in the fixed tube MC2, all logarithm branches remain common, and the iteration converges uniformly on smaller complex domains. Therefore x_s is holomorphic jointly in the field parameters and wall parameters. The operator and weak vertex definitions depend on U and the fixed geometry; the chosen real bar U serves only to fix a normed analytic domain. Joint gauge covariance of those definitions and uniqueness imply covariance of every W_s. On overlaps of admitted tubes the same small branch is obtained by uniqueness in the equivalent covariant norms.

On a protected region, every typed seed and nonlinear vertex whose complete carrier is needed there has wall weight one. The exact same-carrier parametrix identity then gives the literal equations MC9 on that region. These are local equations, not a bound on the distance to some separate stationary field.

86.5 5. Protected Noether equation in the actual own scale

Put V_m=B_(m,U)(W) and remove the coarse logarithmic chart factor from kappa q+mu, obtaining its physical multiplier lambda. The first protected row becomes

$$J_{\Omega_0}(W) + d_W\omega + \mathcal{Q}^{\mathrm{t}}_{U,W}\lambda = 0. \tag{MC13}$$

Wilson gauge invariance on the actual allowed dynamic gauge space gives delta_W J=0. Using MC8 yields the exact scalar equation

$$\Delta_W\omega = -\mathcal{S}^{\mathrm{t}}_{U,W}\delta_{V_m,\mathrm{node}}\lambda. \tag{MC14}$$

To use it locally, protect all mixed rows and nodes whose complete carriers enter the chosen outer collar, including one further incidence stencil. This is stronger than merely declaring the fine plaquettes in the inner set unweighted. After the corresponding fixed shrinkage, MC14 is a literal identity.

In units ell of a two-layer protected tile, lambda is multiplied by ell^3 and its coarse divergence by ell^4. The physical coarse pairings are r_c^4 and r_y^4, and r/ell lies in {1,L}. Thus its mixed incidence is bounded with a fixed L-dependent constant. The weak solution controls the local coarse values, so

$$\|\ell^4\delta_{V_m,\mathrm{node}}\lambda\|_{\ell^p} \le C\epsilon, \qquad \|\ell^4\mathcal{S}^{\mathrm{t}}_{U,W}\delta_{V_m,\mathrm{node}}\lambda\|_{W^{s,p}} \le C\epsilon. \tag{MC15}$$

The last bound is CL46 on this exact two-scale tile. It comes from the site-adjoint lift, not from differentiating a rough fine current. The full L-infinity adjoint bound also holds. No coarse Ward equation has yet been imposed or inferred.

86.6 6. Protected fractional curvature and the bare case

We detail the bootstrap to specify its assumptions. Work in units ell and omit tildes. Initially W has a small W1p potential and omega is Lp. The lower terms of Delta_W-Delta acting on an Lp scalar belong to W^{-1,p}: rewrite the first-order term in divergence form and test (Dv)omega using W^{1,p'} into L^{p/(p-2)}. Equation MC14 and interior constant-coefficient estimates first give omega in W1p, then in W2p, since p>4 makes omega bounded. Smooth nested cutoffs cost only lower norms already obtained. Equation MC13 and the full-space adjoint bound then give J(W) in Lp.

Put W in an ordinary local Coulomb analysis gauge, acting on its reference at the same time. A small W2p gauge parameter preserves both weak charts. This gauge is used only for the estimate; it does not redefine a wall value. Write W=e^{a'v}, delta v=0. The exact lattice Wilson expansion gives

$$\delta dv = J(W) + f_{a'}(v), \qquad \|f_{a'}(v)\|_p \le C\epsilon^2. \tag{MC16}$$

The curl BCH remainder has one derivative in Lp; incidence errors are zero order; the flux correction retains its positive power of a'. The estimates are WB/CL's actual lattice identities, not a continuum substitution. Interior ellipticity gives v in W2p and F in W1p intersect L-infinity. Thus the coefficients of Delta_W have C^alpha regularity for every alpha<1-4/p. Choose s<alpha. The W^{s,p} right side in MC15 now gives omega in W^{2+s,p}; MC13 then gives J in W^{s,p}.

For completeness the last Wilson upgrade uses

$$F_{a'}(v) = dv + G_{a'}(v), \qquad \|G_{a'}(v)\|_{W^{1+s,p}} \le C\epsilon^2, \qquad (a')^4\|\delta_W P(F)\|_{W^{s,p}} \le C(a')^3\|F\|_\infty^2\|F\|_{W^{s,p}}. \tag{MC17}$$

Each degree-n nonlinear BCH word has coefficient (a')^{n-2}. The W^{1+s,p} algebra bound sums these words. In the analytic flux correction the one fine inverse costs only (a')^{-1}, leaving (a')^3. The remaining incidence terms are bounded multipliers. Hence MC16 also holds in W^{s,p}, and its final elliptic estimate gives v in W^{2+s,p}, proving MC5 on the inner collar. The same proof yields

$$\|v\|_{C^{1,\beta}} + \|\Delta v\|_\infty + \|J(W)\|_\infty + \|F(W)\|_\infty \le C_\beta\epsilon, \qquad \beta < 1 + s - 4/p, \tag{MC18}$$

in that final interior analysis frame. No arbitrary higher spatial jet is claimed; HJ's sharp held-current example remains compatible with this ceiling.

At a literal bare boundary tile ell=a, so a'=1. The protected identity constraint is q(b)=b=0. The scalar identity row gives phi=0 on its complete stencil, hence omega=0 on the inner collar. Thus W=U there. All remaining field and plaquette differences on the actual readout graph cost a fixed unit-mesh constant. The assumed curvature bound and the finite actual Wilson force formula give the F/J bounds in MC5 directly, with their exact original scale factors. No periodic action or unproved boundary Coulomb potential is introduced. Statements about differences at that boundary use the actual defined readout stencils; a missing outside plaquette is not supplied by an arbitrary extension.

86.8 8. Exact endpoint elimination and raw constraint

The scalar rows give omega=Delta_U phi in Ran R_U. Pair MC13 with d_W zeta where S_(U,W) zeta=0. The actual Wilson gauge identity removes its action term, and MC8 removes the constraint term. One obtains

$$\langle \Delta_W \zeta, \omega \rangle = 0. \tag{MC21}$$

Given any fine scalar f, choose zeta in MC20 with data (R_U f,0). Using symmetry of R_U and R_U omega=omega, MC21 gives <f,omega>=0. Thus omega=0 even for complex fields. No inference from a complex squared norm is made. The scalar inverse on ker S_U then gives phi=0. Surjectivity of S_(U,W), together with MC13, also gives the actual mixed nodal Ward identity

$$\delta_{V_m,\mathrm{node}} \lambda = 0. \tag{MC22}$$

The relative mixed map MC7 is one nodal gauge of the actual raw connection. Its differential differs from the gauge-conjugated raw differential by a vertical term. MC22 kills the transpose of that vertical term. After transporting lambda back by G_m, the remaining Euler equation is precisely the physical raw constrained equation. Moreover q=0 says

$$M_m(W)^{G_m(U,W)} = M_m(U). \tag{MC23}$$

This is an exact target-orbit identity, with the actual junction node factors and their actual noncommutative order retained.

One can restore the literal raw target without changing that critical orbit. The complete read set of a node prefix of length r_y lies in its actual pillar and a fixed adjacent-layer carrier. Separation excludes a third level there, so sigma is comparable to r_y within fixed L factors, including at an inward junction. In one CE frame on that whole carrier, every intermediate relative physical potential is O(epsilon/r_y), by the same whole-prefix weak estimates as CL. The successive node gauge increments therefore have size at most C ell_j epsilon/r_y. Since sum_(j<j(y)) ell_j<=C_L r_y, their ordered product has $\| \log G_m(y) \| \le C_L \epsilon$. This is one geometric sum on the complete prefix, with no count of its depth.

The bumps can be chosen literally disjoint. Distinct same-level nodes are separated by r_y. Nodes at adjacent levels lie on the finer grid and are separated by at least max(r_y,r_z)/L. A coarse center is not also a finer junction: the central unit child of its L-block has no inward endpoint. For more widely separated levels, MB27 gives an even larger distance; a junction changes that distance by only one finer edge. Consequently choosing, for example, c_L<1/(8L) makes the balls of radius c_L r_y disjoint. The fixed finite coarse-grid cases use the center-point bump with the same L-dependent difference bounds. In a CE analysis frame on the node's complete carrier, extend log G_m(y) as a constant color times that bump, and transform back. Its covariant W2p bound follows from the W1p connection in that frame and two cutoff differences. The point value at y is exact.

Summing these bumps and exponentiating gives a fine gauge g_N with N_node g_N=G_m and uniformly small own-scale W2p parameter. Bare node factors are G_0=I and no nonbare bump reaches a bare node; the separation collar also keeps its support away from fixed exterior sites. Thus the gauge is the identity on every pinned bare endpoint and on the fixed exterior data. The frame construction can be chosen covariantly by CE's canonical rooted convention; existence of the bounded gauge already suffices here. Raw covariance then proves

$$M_m(W^{g_N}) = M_m(U). \tag{MC24}$$

The actual action and critical equation transform covariantly under this allowed fine gauge. Absolute curvature/current bounds change only by its bounded adjoint factors. This operation is made at the unweakened endpoint. Independently gauge-fixing already localized wall terms is not being used as a replacement for the common wall construction.

86.9 9. Exact identity on the actual critical graph

Suppose the input U is itself a regular critical configuration of the actual action on the actual raw mixed fiber and boundary class. Let lambda_0 be its actual physical multiplier, with the actual nodal Ward identity. Here their existence can be checked without assuming an inverse for arbitrary raw forcing. The mixed inverse makes A_U onto. If a row covector alpha annihilates D M_m(U), the exact raw gauge chain D M_m(U)d_U=d_target N_node implies d_targetˆt alpha=0: N_node is algebraically onto the actual allowed mixed node values. Equation MC25 then gives A_Uˆt alpha=0, hence alpha=0. Thus D M_m(U) is onto at each finite cutoff. The usual finite-dimensional tangent-space theorem and criticality give a multiplier. Wilson gauge invariance and the same raw gauge chain give its nodal Ward identity. This argument proves qualification and does not assert a uniform unrestricted raw hidden inverse.

Differentiating the exact nodal gauge formula at W=U gives

$$A_U = DM_m(U) + d_{M_m(U),\mathrm{node}}\rho_U \tag{MC25}$$

in physical tangents. The Ward identity therefore implies A_U^t lambda_0=DM_m(U)^t lambda_0. Thus the five unknowns

$$b = \psi = \phi = \nu = 0, \qquad \mu = -\lambda_0 \tag{MC26}$$

solve MC9 exactly. Their smallness is part of the proof: apply the global mixed inverse to (A_U^t lambda_0,0,0,0,0). The candidate (0,0,0,0,lambda_0) is its exact output, so the adapted multiplier norm is bounded by the weak physical current norm C epsilon. This uses the full mixed inverse and its correct negative types, rather than dualizing a right inverse at an unrelated exponent.

After the fixed enlargement of the contraction ball, MC26 lies in its uniqueness class. Uniqueness proves MC6. All its node gauge factors are then the identity, so the optional target restoration MC24 also does nothing on this graph. The identity includes the actual input gauge and the actual action boundary.

For an inner readout it can be used only after that readout's actual reconstruction is completed. If two reconstructions have genuinely nested fibers and the same boundary class, restriction of a minimum (or critical-orbit uniqueness in the admitted class) gives the usual nesting identity. A raw fluctuation field e^{gB}U is not thereby declared a critical input. Nor are two different determining boundary problems made equal by giving them the same scale label.

87.2 2. The actual current scales exactly as r^3

Let $\mathfrak{J}(U)$ be the unscaled left derivative of the plaquette Wilson action, including its actual orientations and endpoint transports. In physical units write $J_a = a^{-3}\mathfrak{J}$, up to the fixed action normalization. The link configuration itself is unchanged by a rescaling of coordinates, so

$$F_\xi(U^g) = r^2\ \mathrm{Ad}\, g\, F_a(U), \qquad J_\xi(U^g) = r^3\ \mathrm{Ad}\, g\, J_a(U). \tag{DA6}$$

The adjoint action is at the actual base or tail of the corresponding plaquette or current bond. These are exact lattice identities: $\xi^{-2} = r^2 a^{-2}$ for the plaquette logarithm, and $\xi^{-3} = r^3 a^{-3}$ for the Wilson derivative.

In particular

$$\|F_\xi(U^g)\|_\infty \le C_D r^2 \epsilon, \qquad \|J_\xi(U^g)\|_\infty \le C_D r^3 \epsilon. \tag{DA7}$$

This current is the physical current of this same field. It is not an independent outer variable, and (DA7) is not deduced from the C^1 potential bound alone. A generic low potential norm would not control its fine second difference. Here the protected actual-current bound in (DA1) is essential.

87.3 3. Admission to the original paired core

Bałaban I, CMP 109 (1987), (1.11)–(1.16), defines the local paired domain as a union of complex gauge orbits. Its first three clauses control a real regular reference, a small complex first-order deformation, and the curvature/current. The remark immediately following (1.16), printed p. 263, proves that sufficiently smaller first-three-clause constants imply the additional nested-minimizer clause by the published Proposition 9 estimates.

On the local carrier just considered choose the real reference to be the identity, and choose the deformation potential to be $\widetilde{A}$. The real-reference conditions then hold trivially. Equations (DA5)–(DA7) give all the remaining low-core conditions. Thus, whenever the smaller-core inclusion constants for the chosen original domain are $c_{\mathrm{ad}}\alpha_{0,j}$, $c_{\mathrm{ad}}\alpha_{1,j}$, and $c_{\mathrm{ad}}\gamma_j$, admission follows from the explicit inequalities

$$\begin{aligned} C_D r^2 \epsilon_k &< c_{\mathrm{ad}} \min(\alpha_{0,j}, \alpha_{1,j}), \\ C_D r^3 \epsilon_k &< c_{\mathrm{ad}} \gamma_j. \end{aligned} \tag{DA8}$$

The constant c_{ad} is the actual smaller-core margin, not an extra analyticity radius created by the present argument. For the fixed clean core its scale independence is the published statement. For an R-born shrinking core the corresponding original preparation inclusion, with its genuine radii, is required.

The current component of the argument is the actual current $J_\xi(U^g)$. If a proposed extension instead substitutes an arbitrary independent J, it must separately verify (DA8); there is no r^3 gain for an unconstrained independent variable.

The utility for source-only extensions is direct. An old admitted function $E_j(U, J)$ can be read on $(U_{\mathrm{current}}, J_{\mathrm{physical}}(U_{\mathrm{current}}))$, expressed in its own units. At a genuine nested critical endpoint this has the same gauge-invariant value as its original inner minimizer readout. Off that graph it is a different extension; the original Gaussian, coarea factors, and tests are not altered by this statement.

87.5 Source and dependency

The domain identification uses:

T. Bałaban, Renormalization group approach to lattice gauge field theories I, CMP 109 (1987), pp. 262–263, (1.11)–(1.16) and the following smaller-core inclusion.

The current and first-order input is CL33 and the precisely protected own-scale MC variant. The holomorphic gauge construction and exact rescaling identities (DA2)–(DA7) are proved in this note. The physical nested-graph equality is the scoped PS1–4 statement; arbitrary off-shell or cold stored configurations are not included.

88.1 1. The nested variational identity

Let the actual averaging maps be composed in their prescribed order, with their actual intermediate meshes. Denote the composition from the fine lattice to level j by M_j. For j<k write

$$M_k = M_{k:j} \circ M_j. \tag{PS1}$$

This is an identity of compositions, not an associativity assertion for the group mean. Fix the same Wilson action and boundary data in both variational problems. Let U_k(v) be a regular critical minimum on the fiber M_k(U)=v. On an admitted regular branch put w=M_j(U_k(v)). The fiber M_j(U)=w is contained in the fiber M_k(U)=v. Therefore U_k(v) is a minimum on the smaller fiber. Regular-orbit uniqueness for that smaller problem gives

$$U_j\big(M_j(U_k(v))\big) \sim U_k(v), \tag{PS2}$$

where the equivalence uses the allowed gauge transformations of the same boundary problem. For a gauge-invariant readout O,

$$O\big(U_j\big(M_j(U_k(v))\big)\big) = O(U_k(v)). \tag{PS3}$$

The real statement follows directly from minimization and uniqueness. For two analytic choices of representative, gauge-invariant readouts are holomorphic functions of the finite collection of coarse group coordinates. Equality on an open real coordinate set implies equality on the connected admitted complex neighborhood by the identity theorem, applied successively to those coordinates. Thus PS3 also identifies the complex physical germs. A common uniform complex neighborhood must be proved separately; the identity theorem does not provide its radius.

These statements require actual nesting. If either problem changes boundary data, removes an old determining constraint, or replaces a composition by a direct mean, the fiber inclusion must be checked again. They do not identify unrelated mixed-boundary variational problems.

88.2 2. A new extension may be used only in the source slots

For a uniform regular periodic problem, let W_s(U) be the covariant wall family CL; for the specified separated regular mixed hierarchy, use its MC extension with the actual pinned bare boundary. Its unweakened value satisfies

$$W_1(U) = U \tag{PS4}$$

whenever U lies on the genuine regular critical graph for the specified constraint. This is the exact CL38 or MC6 identity in the input gauge. Define the extended readout on its common analytic domain by

$$O_s(U,J) = O(W_s(U)). \tag{PS5}$$

The independent current J is deliberately unused in this readout. It may still occur in the unchanged source-free integration kernel. On the physical graph, PS4 implies O_1(U,J_W(U))=O(U). If the inserted field is an actual inner minimum obtained from a nested constraint, PS2–3 justify the same equality at that insertion. For a fixed finite product of readouts, the equality holds for their product and for any fixed polynomial in them.

There is no change of Gaussian coordinates in PS5. In particular, if the original integral is

$$I_O(x) = \int_{D_x} dB\; j_x(B)\, e^{-S_x(B)} O\big(\Phi_x(B)\big), \tag{PS6}$$

and every physical Phi_x(B) in the relevant source slot lies on the verified critical graph, its value is exactly

$$I_O(x) = \int_{D_x} dB\; j_x(B)\, e^{-S_x(B)} O_1\big(\Phi_x(B), J_W(\Phi_x(B))\big). \tag{PS7}$$

The domain D_x, the density j_x, the action, all tests, and all other stored variables in this identity are literally the original ones. This point matters: an arbitrary fluctuating connection exp(gB)U is not generally a constrained minimum. PS7 applies when Phi_x(B) is the actual minimum used by the reconstruction, not merely because the notation contains a background field.

Ordinary source exponentials also obey PS7 at every finite cutoff where the integrals exist. Finite source jets obey it algebraically without requiring an ordinary exponential domain for a high-degree operator. The cutoff-uniform estimates for those jets remain a separate task.

88.3 3. An induction principle for preserving the same physical measure

Here is a precise form that also includes stored histories. Let P_j be the set of actual physical states at step j; a state includes the current gauge field, its physical current, every still-stored old variable, and the geometric tests. Suppose the frozen unmarked integration at the next step has the form

$$(\mathcal{T}_j F)(x) = \int K_j(x, d\xi)\, F\big(\Phi_j(x,\xi)\big), \qquad x \in P_{j+1}. \tag{PS8}$$

No normalization of K_j to a probability kernel is needed. Require that its actual input state Phi_j(x,xi) belongs to P_j wherever its tests admit xi. A source-preserving R replacement may be included in this notation by retaining the **old marked numerator** and its old variables, and using the matched unmarked conditional denominator. That is the fiber identity of S53–58; an arbitrary multiplication by a source-dependent normalized density is not PS8.

Let F_j and F'_j be two analytic extensions which agree on P_j. Then

$$(\mathcal{T}_j F_j)(x) = (\mathcal{T}_j F'_j)(x), \qquad x \in P_{j+1}. \tag{PS9}$$

This follows by equality of their integrands on the same support. One may choose any analytic output extension that agrees with the result on P_{j+1}. Induction proves equality of the terminal marked integrals for every finite sequence of frozen transformations. In particular the final generating functional or finite source jet is that of the original Wilson measure. It is not the generating functional of a newly chosen background measure.

Exact finite-cutoff rearrangements, insertion of a partition of unity, and convergent connected expansions do not alter PS9. If a logarithm of an activity is used, its branch is fixed at the same source-zero value before the rearrangement. This argument does not assert convergence of a proposed connected expansion; it applies after that convergence or an exact finite rearrangement is known.

The principle explains the permissible change in analytic extension. Equality on the physical graph is sufficient for preservation of the physical integral. It is insufficient for its bounds unless the chosen extension is also controlled on every complex domain actually used by the iteration. CL supplies such a local regular wall family for its specified uniform constraint. MC extends that classical statement to its separated mixed hierarchy, with the specified fixed complex tube and actual bare boundary. It does not by itself show that every inherited sourced activity stays in its original analytic core.

88.4 4. Nonwrapping coefficients agree when their completed constructions agree

Consider two compatible periodic lattices with the same fine and retained spacings. A completed carrier C consists of all sites, links, plaquettes, constraint paths, tree paths, local extension boxes, and integration variables used by a term. Suppose C has an injective common lift to the infinite lattice, so that no geometric identification within it wraps around either torus. Require the following literal rules.

1. Every geometric choice in the carrier is a fixed function of its translated local coordinates. The same rule is used on both tori.
2. A local periodic inverse is built on the same artificial local box and the same locally extended connection on that box. Its period is part of C, independent of the ambient torus.
3. Every local nonlinear vertex, characteristic function, coarea density, and finite integration is the same function of the identified variables.
4. The term's complete carrier includes every factor created by a derivative, a source mark, a denominator expansion, or an old history. It is not just the support of the original bare mark.

Under these conditions the two coefficients are exactly equal. For a walk term, each local inverse in rule 2 is the inverse of the same finite matrix, so its kernel is identical. Every matrix product and its internal sums are consequently identical. This remains true after inserting the same local nonlinear vertex. For an analytic fixed point, compare its successive approximants: the initial values and each subsequent local operation coincide. Uniform convergence then gives identical analytic coefficients. Finally, each finite integration, finite source differentiation, and connected combinatorial operation is the same. This proves the claim, including all numerical and gauge-volume constants.

The tree extension CE obeys rules 1–2: its root, coordinate order, cutoff and artificial periodic box are fixed; it uses only the connection inside that box. Its possible rough outer gauge is part of that local construction, not a gauge chosen using the ambient torus. The AL/CL common-carrier rule includes the whole box and every tree path. Therefore these new local inverse and nonlinear readout coefficients have exact nonwrapping compatibility.

A uniform exponentially summable carrier bound then controls the remaining wrapping terms by its usual large-carrier tail. Exact equality of nonwrapping terms alone does not prove that tail bound. Nor does it automatically prove compatibility of an original R term whose stored history or normalized

reference has not yet been filed on its full determining carrier. In that case rules 3–4 are the explicit checks still required.

88.5 5. What this interface establishes

For uniform periodic regular critical readouts, CL and the exact nesting identity justify use of the new gauge-covariant wall extension while keeping the original physical integration kernel. MC gives the same classical graph identity on its specified separated mixed regular hierarchy and fixed analytic tube. The local construction also has literal equality across compatible tori for every completed nonwrapping carrier. These are finite-cutoff identities and do not require a new coarea theorem.

For the full construction the outstanding quantitative questions remain visible: identification of the actual completed readout in each source slot, preservation of inherited activity cores, comparison of its complete stored carriers, and renormalized estimates for arbitrary operator insertions. The identities above supply no uniform bound on derivatives of a bare high-dimensional operator. In particular they do not evade the genuine-minimizer counterexample HJ.

The source-preserving R fiber itself is now identified by S13.7 and its final Section 20. PS8 uses that source-independent carrying kernel, retaining every old source in its old numerator slot. What remains for a new CL-based source induction is different: its C-times-input local estimates give a common smaller analytic tube, but do not by themselves preserve every inherited member's original core radius through arbitrarily many iterations. One must use an actual preparation inclusion and the original strict age readout, or prove invariance of a new source norm. Primitive trace polynomials do not have that radius issue; reuse of already constructed analytic activities does.

89.1 1. The finite-cutoff object and the maps that must remain fixed

Let a be the bare spacing and let T be a compatible finite torus. With one fixed choice of oriented plaquettes, put

$$a_p(U) = 1 - \frac{1}{N}\operatorname{Re}\operatorname{Tr} U(\partial p), \qquad S_x(U) = \sum_p x_p a_p(U). \tag{S1}$$

On real compact-group fields, $a_p \geq 0$. The physical source functional under discussion is

$$Z_a[w] = \int \exp\{-S_x(U) + \sum_p w_p a_p(U)\}\, dU. \tag{S2}$$

Normalization or running field factors can be incorporated into w_p; whenever this is done their size must be included in the source budget. All tests, coordinate maps, gauge conventions, determining sets, reference replacement densities, and their denominators are frozen at the real source-zero construction. Thus an analytic continuation changes the carrying function in an existing integral. It does not move a characteristic-function boundary.

The elementary estimate

$$\sup_{z\in D} |e^{-S_x(U)+\sum_p w_p(z)a_p(U)}| \leq e^{-S_x(U)+\sum_p d_p a_p(U)}, \qquad d_p = \sup_{z\in D}|w_p(z)|, \tag{S3}$$

is an exact real-field inequality. It alone gives no useful cutoff-uniform partition bound: the normalizer and the large number of integrated variables still have to be treated. Its proper use is to consume a specified fraction of the action that remains in a particular stored integral.

We use a commutative unital Banach algebra $\mathcal{A}$ with $\|1\| = 1$. The examples needed here are $H^\infty(D)$ on a finite source polydisc and finite coefficient algebras with a submultiplicative coefficient norm. The constants below do not depend on the dimension of D or on the number of source labels.

Here is the concrete ordinary profile domain used in the final theorem. Fix $0 < \alpha < 1$, one physical length convention, and finitely many real scalar profiles $f_i \in C^\alpha(T)$. The same scalar profile is evaluated at the center of each plaquette, with the prescribed lattice sampling. Choose positive radii r_i and put

$$D_{\boldsymbol{r}} = \{z : |z_i| < r_i\}, \qquad w(z,x) = \sum_i z_i f_i(x), \qquad \sum_i r_i(\|f_i\|_\infty + [f_i]_{C^\alpha}) \leq \rho. \tag{S2a}$$

Thus $\sup_{z\in D_{\boldsymbol{r}}} \|w(z)\|_{C^\alpha} \leq \rho$, and the source oscillation on a scale-ℓ carrier is at most $C\rho\ell^\alpha$ times its fixed carrier-diameter power. Every bare or running normalization multiplying an insertion must be included in this same budget. The number of profiles may be arbitrary and finite; their total weighted norm, not that number, enters the estimate. This is not an independent complex disc for every microscopic plaquette with no spatial regularity cost.

After fixing the original geometry and spare analytic/locality margins, choose the weak floor and then a positive ρ, independent of the bare cutoff, so that

$$C_{\mathrm{in}}\rho < a_*/4, \qquad \rho < r_{\mathrm{hom}}/8, \qquad C_M\rho < \min(\theta,\tau)/4. \tag{S2b}$$

Here a_* is the formation allowance in (S4), r_{hom} is the uniform scalar disc constructed in Section 11.5, and θ, τ are the unused positive reserves in (S86). If the optional sourced reference normalization

of Section 9 is used, impose also $C_M\rho < h_M$, its fixed component allowance. The preferred frozen-reference construction does not need that optional condition. The fixed constant C_{in} includes the ordinary scalar packet and heterogeneous propagation constants. Choose the fluctuation-to-field margin so that $1 + C \sup_k \theta_k < L^{\alpha}$, as in (S43)–(S44).

89.3 3. The actual completed large-field operator already accepts carrying functions

There is a useful primary-source point stronger than a scalar mass estimate. In Large Field Renormalization II, equations (1.71)–(1.75), the operator $T_k(X)$ acts on an arbitrary function (F) of its integrated variables. On a real reference background its density is positive. The complex paired-core comparison puts a small additional expression in that same exponential. In particular, the argument for (1.73) is a triangle inequality before integration, not an inference from a bound on (T_k(X)1).

At this level one may write its exact consequence schematically but unambiguously as

$$\|T_{k,b}F\|_{\mathcal{A}} \le \int \|F(\xi)\|_{\mathcal{A}} e^{|\sigma_b(\xi)|}\, d\mu_{k,\bar{b}}(\xi), \tag{S12}$$

where $\bar{b}$ is the real comparison background, $\mu_{k,\bar{b}}$ is the positive reference measure of that very stored operation, and σ_b is its actual complex comparison term. Constants in its definition can be included in $|\sigma_b|$. For a composition of stored operations, all shared variables remain in one joint measure until the corresponding integration has been performed.

The normalized estimate (1.75) is unnecessary here. Its small source-free denominator is not a reason to introduce a sourced denominator. The first inequality (S12) is the appropriate one for ordinary sources and for unresolved live components.

89.6 6. One vacuum convention and a valid denominator comparison

89.6.1 6.1 Exact scalar completion

Let $v_b(w)$ be the universal regular vacuum births, with their logarithmic density signs, indexed by birth scale, complete carrier, and type. Include both clean and regular R births. Set

$$V_a = \sum_b v_b(0), \qquad E_a(w) = \sum_b [v_b(w) - v_b(0)]. \tag{S19}$$

For a history whose displayed regular births have index multiset I_ω,

$$e^{\sum_{b\in I_\omega}[v_b(w)-v_b(0)]} = e^{E_a(w)} e^{-\sum_{b\notin I_\omega}[v_b(w)-v_b(0)]}. \tag{S20}$$

If the rooted ordinary norm of the source excess vacuum is at most $C\rho$, assigning each absent term to its first intersected birth-scale record gives

$$\sum_{b\notin I_\omega} \|v_b(w) - v_b(0)\|_{\mathcal{A}} \le C\rho N_\omega. \tag{S21}$$

This is exactly a term allowed in (S14). The assignment uses complete carriers, and each term is charged once. In particular, a fine-scale R vacuum is not bounded by summing its small activity over physical volume without subtracting it: the fine root count would invalidate that argument.

The clean part of the rooted vacuum norm follows from (S5)–(S7) by evaluation at the trivial paired field. The completed R part follows from (S12)–(S17), followed by its convergent connected resummation, provided the actual incoming norm and reserve conditions hold. Evaluation of a vacuum uses no extra shrinking field-radius derivative.

89.6.2 6.2 The source-zero lower bound can use the published convention

The original full-density description already uses the required kind of vacuum convention. In III, (2.23) separates the effective action and vacuum constant, while (2.30) explicitly subtracts the flat values of the regular R members. The paragraph before Theorem 1 defines the initial normalization constant from full-lattice one-step vacuum contributions. Corollary 3 is a stability bound for that normalized density, conditional on the stated weak coupling trajectory.

Consequently the source-zero denominator should be obtained from that source-free normalized density, after checking that the deterministic vacuum convention equals (V_a) in (S19), including signs and Gaussian count constants. If the conventions differ by D_a, the exact relation is

$$e^{-V_a} Z_a(0) = e^{D_a}\, e^{-V_a^{\text{published}}} Z_a(0), \qquad D_a = V_a^{\text{published}} - V_a. \tag{S22}$$

A terminal-volume estimate on (D_a), or exact equality, is required. An unspecified ultraviolet-extensive difference cannot be absorbed into a terminal constant.

A direct lower-sector proof is also possible when the literal positive density representation is retained. The term with no surviving terminal live component has $T_k(\varnothing) = \mathrm{Id}$. It contains the completed regular R family, including the integrated healed histories. On a small terminal link ball the action

and vacuum-subtracted regular terms are bounded by CN_k; its positive Haar volume is at least $\exp[-C(1+\log x_k)N_k]$. Hence

$$Z_a(0) \geq \exp\{V_a - C(1+\log x_k)N_k\}. \tag{S23}$$

The hypotheses needed for this last argument are explicit: positivity of that exact sector, admission of the displayed terminal ball by its current tests, and the source-free regular action bound on the ball. Positivity of an arbitrarily selected Mayer term would not suffice. The sector is not the event that no fine-scale bad test ever occurred; that event would remove the healed R vacuum family.

Given the upper estimate with the same (V_a), (S23) yields the usual ordinary normalized source bound after the local source vacuum functional (E_a(w)) is removed. It supplies no global complex zero-free theorem and does not identify the reference-field law with the physical coarse-field law.

89.8 8. The literal reference quotient: a precise normalization obstruction

89.8.1 8.1 What is and is not visible in R-II (1.1)

The quotient at printed p.356 is subsequently analyzed in (1.10) using the relaxed reference $V_\Lambda(V)$, rather than an arbitrary copy of the current interior field. Schematically its reference denominator is

$$D(V) = \int dV'_\Lambda \, \delta_{G_0}(V') \, \chi(V') \, r_b(V' V_\Lambda(V)), \qquad r_b(Y) = \exp\{-g_k^{-2} A(\zeta_0, U_{k,Z}(Y; b))\}. \tag{S24}$$

It occurs outside the old numerator integral, with the reference density in the outside factor. The remaining old operators and integrations have been suppressed in the displayed formula, not discarded. The distinction between $V_\Lambda(V)$ and V is essential: if its dependence on the interior of V is only a gauge choice of the exterior-determined minimum, the exact gauge-fiber lemma of Section 13 makes (S24) exterior-measurable. One must then match the output measure to that same fiber. The later removal of a cutoff on a completed regular component, by itself, establishes neither fact for an earlier live component.

This is a limitation of the presently verified identification, not a counterexample to the complete R operation. Additional coordinate identities, the omitted integrations, or the full sum over preparation pieces may establish the required identity. They must actually be used.

89.8.2 8.2 An exact operator test

Let (Ω, μ) be a finite measure space, let K be a positive integral operator, and let $r > 0$ with $Kr > 0$. Consider the displayed-pattern replacement

$$\mathscr{R}_r q = \frac{r}{Kr} \, Kq. \tag{S25}$$

Fubini gives the exact condition

$$\int \mathscr{R}_r q \, d\mu = \int q \, K^* \left(\frac{r}{Kr} \right) d\mu. \tag{S26}$$

Thus mass is preserved for every integrable q if and only if $K^*(r/(Kr)) = 1$ almost everywhere. On the other hand, $\mathscr{R}_r r = r$ holds without that condition. A correct source-zero output at one density is therefore insufficient even for first-source matching.

For an explicit example take the compact group $\mathbb{Z}_2$ with normalized counting measure and

$$K = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}, \qquad r = (1, 2), \qquad f = (1, 0). \tag{S27}$$

Then $Kr = (5/4, 7/4)$, the source-zero density r is fixed, but

$$\int rf\,d\mu = \frac{1}{2}, \qquad \int \mathscr{R}_r(rf)\,d\mu = \frac{1}{2}\left(\frac{3}{5} + \frac{2}{7}\right) = \frac{31}{70}. \tag{S28}$$

The obstruction is not peculiar to a disconnected group. Let K be a symmetric heat-convolution operator on a compact connected semisimple group, and let f be a bounded real nonconstant class eigenfunction with mean zero and $Kf = cf$, $0 < c < 1$. For sufficiently small real ε, put $r = 1 + \varepsilon f > 0$. The coefficient of a source coupled to f changes by

$$\int \mathscr{R}_r(rf)\,d\mu - \int rf\,d\mu = \varepsilon c(1-c)\int f^2\,d\mu + O(\varepsilon^2). \tag{S29}$$

Indeed, expand $r/(Kr) = 1 + \varepsilon(1-c)f + O(\varepsilon^2)$ and $K(rf) = cf + \varepsilon K(f^2)$. On $SU(N)$ one may take $f(U) = N^{-1}\operatorname{Re}\operatorname{Tr} U$; then $1 - f$ is a nonnegative Wilson-form class observable, and its mismatch has the opposite sign because the constant-source part is preserved at $q = r$. This example only tests the inference from a quotient's shape; it does not assert that the original Wilson preparation is this heat-convolution operator.

If K is a genuine conditional expectation onto the exterior sigma-algebra, then Kr is exterior-measurable and $K(r/(Kr)) = 1$. That is one exact sufficient repair. Another is to derive a joint old/new kernel with the correct conditional denominator inside its old-variable integral. A third is to prove the integrated identity only after all the actual preparation pieces and omitted operations have been restored. No one of these alternatives follows from (S24) alone.

89.9 9. A sourced reference denominator on a bounded component

89.9.1 9.1 The regular reference component, rather than a live-island integral

R-II (1.2)–(1.11) analyzes (S24) on a component Λ contained in a cube of side $100M$. After the physical rescaling $V' = \exp(ig_k B')$, its axial gauge leaves a bounded number n_M of real coordinates. The reference minimizer is regular, the local saddle current vanishes, and the quadratic form has a positive lower bound depending on M. The remaining linear tail, nonlinear action, and Haar factors are controlled on the fixed original cutoff. These are properties of the regular reference integral; they do not replace the joint large-field operator estimate of Section 4.

The following lemma makes exactly the analytic consequence needed for an added source. It also states explicitly which quantitative source-free bounds are used.

89.9.2 9.2 Proposition: a cutoff-independent local ordinary source disc

Let b range over a source-free complex reference-field domain. After removing the explicit saddle-action and coordinate prefactors, suppose the reference denominator is

$$J_b(0) = \int_{\mathcal{C}} j_b(B)\,dB, \qquad B \in \mathbb{R}^n, \quad n \le n_M, \tag{S30}$$

where $\mathcal{C}$ is the frozen real cutoff set. Assume constants $c_M, C_M, t_M > 0$, independent of the microscopic cutoff, such that

$$|J_b(0)| \ge c_M, \qquad \int_{\mathcal{C}} |j_b(B)| e^{t_M(1+|B|^2)}\, dB \le C_M. \tag{S31}$$

Let $H_b(B)$ be an analytic $\mathcal{A}$-valued source potential satisfying

$$\|H_b(B)\|_{\mathcal{A}} \le h(1+|B|^2). \tag{S32}$$

There is $h_M > 0$, depending only on the constants in (S31), for which

$$\frac{J_b(H)}{J_b(0)} = \frac{\int_{\mathcal{C}} j_b(B) e^{H_b(B)}\, dB}{J_b(0)} = e^{\ell_b(H)}, \quad \|\ell_b(H)\|_{\mathcal{A}} \le C'_M h, \tag{S33}$$

whenever $h \le h_M$. The logarithm is the branch equal to zero at $H = 0$, and the ratio is invertible in $\mathcal{A}$. In particular, with $\mathcal{A} = H^\infty(D)$, it has no zeros on D. Its first derivative is bounded by C'_M in the quadratic-growth norm, and its Taylor remainder after the linear term is bounded by $C''_M h^2$.

Proof. Put $Q(B) = 1 + |B|^2$. For $0 \le h \le t_M/2$,

$$\begin{aligned} \left\| \frac{J_b(H)}{J_b(0)} - 1 \right\|_{\mathcal{A}} &\le c_M^{-1} h \int_{\mathcal{C}} |j_b(B)| Q(B) e^{hQ(B)} dB \\ &\le C_M c_M^{-1} h \sup_{q \ge 1} q e^{-t_M q/2}. \end{aligned} \tag{S34}$$

Choose h_M so that the last bound is at most $1/2$. The Banach-algebra logarithm series $\log(1+z)$ converges absolutely, proving (S33). Analyticity and a fixed smaller ball give the derivative and quadratic remainder, either directly from this series or by the same one-dimensional Cauchy argument as in (S11). All bounds are independent of the number of source labels. □

89.9.3 9.3 Why the hypothesis has no hidden cutoff-volume cost

For the reference block, (S31) follows from a uniformly positive real quadratic form, a small complex quadratic perturbation, an integrable small nonlinear remainder, and the source-zero analytic lower bound, after the explicit powers of g_k and saddle action have been factored out. The number of remaining coordinates is at most n_M. For example, a bound

$$|j_b(B)| \le C_M e^{-c_M|B|^2}, \qquad |J_b(0)| \ge c'_M \tag{S35}$$

immediately implies (S31), on any subset $\mathcal{C}$. The cutoff is not moved, and its radius may tend to infinity in the canonical variable B. The moment bound, rather than the supremum of B on that cutoff, is what keeps the source radius independent of the coupling and the microscopic spacing.

The displayed R-II regular denominator analysis supplies this structure on each bounded component, on a slightly reduced original complex field domain. Its Gaussian lower bound and analytic representation (1.10)–(1.11) control $J_b(0)$; the same quadratic comparison gives (S35), with constants depending on the fixed geometric parameter M. The source disc may be very small as a function of M; it is nevertheless a fixed positive disc once the source-free geometry has been chosen.

The denominator counterterm is therefore an analytic local member with a proved bounded first source difference, provided its incoming source potential satisfies (S32) on this actual reference chart. This

result is stronger than a phase estimate on an unbounded component, and weaker than a claim that all full-history denominators have already been factorized into these blocks.

89.9.4 9.4 The additive coupling disc is compatible with this local lemma

A source-independent fluctuation constant need not be small. Write $H_b(B) = H_b(0) + \widehat{H}_b(B)$. The exact identity

$$\log \frac{J_b(H)}{J_b(0)} = H_b(0) + \ell_b(\widehat{H}), \tag{S35a}$$

holds whenever the fluctuation part is in the ball of Section 9.2. The factor $e^{H_b(0)}$ is invertible regardless of its norm. It must be retained in the same classical-action/normalization convention as the outside reference factor.

For a homogeneous additive Wilson coefficient w, the local analytic action expansion on the original regular reference chart gives, with all fixed powers from that chart included in q,

$$|w\,[A(U_b(gB)) - A(U_b(0))]| \le C_M|w|(g\alpha_k + g^2)(1+|B|^2), \qquad \alpha_k \le Cg(\log x_k)^q, \quad x_k = g^{-2}. \tag{S35b}$$

This is a quadratic-growth estimate, not a supremum over the expanding canonical cutoff. It follows by applying the analytic second-order action bound to the reference response; a residual linear term is bounded by $C_M\alpha_k g|B|$. The extra local critical-current cancellation in (1.4)–(1.6) improves that term but is not needed here. The nonlinear remainder is absorbed because the physical fluctuation ceiling is a fixed small fraction of the admitted chart.

Consequently the ordinary additive allowance

$$|w| \le c\,\frac{x_k}{(\log x_k)^{M_0}}, \qquad M_0 > q, \tag{S35c}$$

puts $\widehat{H}$ inside the fixed ball of (S32), after c is reduced once. If it is useful to bound the constant part as well, take $M_0 > 2q$. Thus the small fixed local source radius in Section 9.2 is compatible with the growing additive coefficient discs used for the homogeneous running-coupling derivative. It does not force their radius to be a cutoff-dependent constant. Inherited regular source differences are handled by (S40)–(S41), and the allowed total in (S32) is shared between them and (S35b).

89.10 10. What the source-dependent-denominator route does and does not repair

Suppose one defines the sourced construction by adding the primitive physical source to the initial local potential and rerunning the exact algebraic RG formulas, without freezing any denominator that those formulas genuinely change. For a regular reference factor of the form (S30), Section 9 supplies an analytic logarithmic ratio. That logarithm must be included among the new boundary/regular source members. It is not an optional normalization that can be dropped after differentiating.

For a collection of exactly factorized bounded reference blocks, the logarithm is the sum of their local logarithms (S33), and its rooted norm is bounded by C_M times the incoming rooted source norm. Arbitrarily many blocks cause no decrease of the source radius. The same conclusion holds for a joint prepared Gaussian reference integral if its actual conditioned covariance, walls, and carrier covering

satisfy Section 2: apply the connected expansion to the added source and subtract the zero-source connected expansion. This gives

$$\log \frac{J_\Omega(H)}{J_\Omega(0)} = \sum_{X \subset \Omega} \ell_X(H), \qquad \sup_q \sum_{X \ni q} e^{\kappa d(X)} \|\ell_X(H)\| \le C\|H\|_\mu. \tag{S36}$$

The identity first holds near zero by the convergent connected expansion and therefore holds throughout its common domain. Its exponential is the actual normalized integral. This is a genuinely local zero-free theorem; it is not obtained by dividing a global upper estimate by a global lower estimate.

For the literal R-II formula, the inspected pages establish the bounded-component calculation but do not explicitly display either a product factorization of the complete denominator or the complete joint covariance/wall statement needed for (S36). The exterior-dependent reconstructed saddle means that product factorization should not be assumed from the words "one of its components." If the original R-I definition performs ordered single-component normalizations, (S33) applies successively with all other variables treated as admitted parameters; the exact ordering and parameter-domain inclusion are then the facts to verify. If it performs one joint normalization, (S36) is the corresponding precise target.

There is a separate exactness question. Proving that a chosen sourced quotient is analytic and zero-free does not prove that its integral equals the physical sourced Wilson partition function. The complete original algebraic R identity must hold for the perturbed incoming density, or a carrying extension must be derived with the correct old/new coordinates and normalizer placement. Thus the source-dependent-denominator route removes a concrete analytic obstacle, while (S26) identifies the exact algebraic check that cannot be replaced by that obstacle's removal.

89.13 13. Exact normalization on a matched gauge fiber

89.13.1 13.1 The finite graph disintegration, including its Jacobian

Fix all exterior and retained physical data b of one component. Here b may include older stored variables that will be integrated later; it is not restricted to geometric exterior links. In particular the older operations explicitly omitted before R-II (1.1) must remain outside this conditional calculation. Let the free current-scale links form a finite graph. Choose a gauge-fixing forest G_0 with one fixed boundary root in each tree and one tree edge for each unfixed gauge vertex. Let $R(b)$ be a fixed reference connection on that graph with the prescribed exterior data. Write $h = 1$ at the fixed vertices.

Every free link configuration has the unique coordinates

$$V_e = h_{e_-}\, C_e R_e(b)\, h_{e_+}^{-1}, \qquad C_e = 1 \quad (e \in G_0), \tag{S53}$$

where C_e on the other free edges and h_x on the unfixed vertices vary independently over the compact group. With normalized Haar measures,

$$\prod_{e\ mfree} dV_e = \left(\prod_{x\ munfixed} dh_x \right) \left(\prod_{e\ mfree\setminus G_0} dC_e \right). \tag{S54}$$

Proof. Solve for h successively from the root along the tree edges. At each step the relevant old edge variable is a fixed left/right translate of the newly introduced h_x, possibly followed by inversion; Haar measure is unchanged. Conditional on these tree variables, each remaining V_e is a left/right translate of C_e. Fubini proves (S54), with Jacobian exactly one. This also proves the formula when the reference depends on b, since b is held fixed. It does not justify an arbitrary field-dependent replacement of the reference without these coordinates. □

The usual delta notation $\delta_{G_0}(C)\,dC$ is a shorthand for the second product in (S54). It must not be interpreted as inserting a new delta function of an identity that is already enforced by the coordinate definition.

If the component has a residual global gauge, one fixes a root while proving (S54) and then integrates that one global group variable. Its normalized volume is one for a jointly gauge-invariant integrand. Boundary gauges not allowed by the retained datum cannot be included among the free h_x.

89.13.2 13.2 The relaxed reference removes its apparent interior dependence

Suppose the regular variational problem defining the relaxed reference has only the retained constraints b, and has a unique admitted solution modulo the supported gauge group. Let $V_\Lambda(V)$ be that solution represented in the tree gauge selected by V. Then

$$V_\Lambda(V) = h \cdot R(b), \qquad V_e V_{\Lambda,e}(V)^{-1} = h_{e_-} C_e h_{e_-}^{-1}, \tag{S55}$$

with the same h as in (S53). Let χ be an invariant norm cutoff in the relative variables, and let r_b be gauge invariant. Then the denominator in (S24) equals

$$D(b) = \int_{C|_{G_0}=1} \chi(C)\, r_b(CR(b))\, dC. \tag{S56}$$

Indeed use the change $V_e' = h_{e_-} C_e' h_{e_-}^{-1}$. Its Haar measure, tree condition, and cutoff are unchanged. Moreover $V' V_\Lambda(V) = h \cdot (C' R(b))$, so gauge invariance removes h. This proves exterior measurability. In the real calculation the transformations are unitary; analytic extensions are subsequently determined by the identity theorem on the admitted complex chart. No noncompact Haar integral or complex conjugation Jacobian is asserted.

89.13.3 13.3 Exact transport of the physical old source

Let $q_0(\xi, b)$ be the old numerator density, including its actual old tests and integrations, and let $M(\xi, b; z)$ be any jointly gauge-invariant physical carrying source. Define

$$Q_M(b; z) = \int q_0(\xi, b) M(\xi, b; z)\, d\mu_b(\xi). \tag{S57}$$

On the new gauge fiber the replacement is

$$d\nu_b(C;z) = \frac{\chi(C) r_b(CR(b))}{D(b)} Q_M(b;z)\, dC. \tag{S58}$$

Integration over C, and then over b, gives exactly the original marked integral. The old support need not equal the reference support; its full measure is retained in (S57). If the gauge transformation h is also applied to stored variables ξ, the joint change $(V,\xi) \mapsto (V, h(V)\cdot\xi)$ has triangular derivative and unit Haar Jacobian. Joint gauge invariance preserves the carrying function. There is therefore no unaccounted Jacobian from a gauge transformation selected by the new tree coordinates.

A source-dependent reference works equally well: replace r_b and $D(b)$ by $r_b e^{H_b(\cdot;z)}$ and its integral. Section 9 makes that integral invertible on the local source domain. The exact compensation $-\log[D_b(z)/D_b(0)]$ must then be retained. This is a finite-component coarea/normalization identity, not a claim about a reference-field marginal of the original Wilson measure.

89.13.4 13.4 The two literal matching checks in the original notation

The exterior-only variational problem in R-II (1.3), its uniqueness argument following (1.12)–(1.13), and the relaxed extension in (1.38)–(1.43) supply the variational structure used in (S55). The statement that $V_\Lambda = V_k$ outside Λ, together with (1.33)–(1.35), identifies which current variables are being replaced. To apply (S56)–(S58) literally, however, two coordinate statements must be matched:

1. The retained mixed constraint **data** on $Z \setminus \Lambda$ must stay fixed under the displayed integration over the interior current variables. Dependence on the older variables held in b is harmless: the denominator remains inside their later integration. Nonlocality of the averaging operation on the unknown fine field is also harmless; hidden dependence of its prescribed right-hand side on the relaxed current variables would not be. An exact test is $\mathcal{C}_{\mathrm{ext}}(U_{k,Z}(CR(b))) = b$ for every admitted C. A tangent identity only at the relaxed saddle is insufficient. The usual actual right-inverse constraint identity throughout the admitted chart would establish the test by integration along a path.
2. The output reference measure must be exactly the same tree/cutoff fiber as (S56), or the same measure after the explicit Haar/gauge substitution (S53)–(S54). The late integrated reverse-Faddeev–Popov step at p.378 is consistent with this interpretation, but is not a replacement for checking it at the reference insertion.

Thus Section 13 proves the candidate repair, including the field-dependent tree-gauge Jacobian, rather than merely observing that the denominator might be gauge invariant. It also identifies precisely the remaining notation-to-coordinate checks. The quotient counterexample in Section 8 does not apply once these matched-fiber statements hold.

89.13.5 13.5 The primary construction fixes the reference before the output variable

There is a useful further resolution of the first concern. In its proof of Proposition 1, printed p.359, R-II first uses the exterior data to choose a regular extension from $Z \cap \Lambda^c$ onto Z, fixes that extension, and only then solves the variational problem for $V' \mapsto A(U_{k,Z}(V'V_k))$ in the relative axial gauge. The V_k in that auxiliary construction is the fixed extension. It is not the later free output integration variable.

Consequently, at fixed full datum b, choose the extension once as a function of b. The uniqueness proved in (1.12)–(1.13) then defines $V_\Lambda(b)$ in that fixed relative gauge. The current output can be written independently as

$$V_{\mathrm{out}} = C\,V_\Lambda(b). \tag{S58a}$$

This is exactly the new-relative-variable structure in (1.34). The later formula on printed p.378 uses $V_k' = V_k V_\Lambda^{-1}$, with the same relative axial gauge and relative cutoff. Thus the suspected arbitrary dependence of the reference on the free current output does not arise in this construction. The tree condition on C is a genuine restriction of an independent output coordinate; it is not a delta function of an identity manufactured by a reference that copies that output.

The conditional denominator is therefore (S56) once the relative output measure is identified. If another fixed representative of the reference orbit is chosen at the same b, a supported unitary gauge substitution gives the same integrated answer for every jointly invariant carrying function. Equations (S53)–(S54) provide that substitution with its exact Jacobian. This uses no norm bound for a complex gauge transformation.

The remaining literal issue is now the placement of the **output** $\chi(C)\delta_{G_0}(C)$ factor and the old numerator in the same conditional integral, not the existence, uniqueness, or output independence of $V_\Lambda(b)$. The original tests in the old numerator are retained there; they need not equal the output reference cutoff. The matched-fiber statement (S57)–(S58) is already proved and applies as soon as that placement is fixed.

89.13.6 13.6 Output selectors need normalize only after their labels are summed

The small/large output decomposition on printed p.378 is a partition of unity. Let its labels be α, and write its nonnegative real selector functions as $\pi_\alpha(C,\xi,b)$, with $\sum_\alpha \pi_\alpha = 1$. The exact selected carrying kernel is

$$d\nu_{b,\alpha}(C;z) = \frac{\chi_{\mathrm{ref}}(C) r_b(CR(b))}{D(b)} \int q_0(\xi,b) M(\xi,b;z) \pi_\alpha(C,\xi,b)\, d\mu_b(\xi)\, dC. \tag{S58b}$$

Summing α and integrating C gives $Q_M(b;z)$. An individual selected output history need not have unit conditional mass. In particular a new small-field selector depending on C belongs in π_α; it is not an extra restriction to insert silently into the reference denominator.

All variables on which the reference depends are held fixed in this conditional normalization. If they are older integration variables, their integration remains outside the displayed C integral. Equivalently one may write (S58b) with a reference probability kernel conditional on the entire old state, integrate C first, and then perform the original old integrations. There is no assertion that D factors out of an older stored operation.

89.13.7 13.7 The collective primary identities identify this kernel

The same-fiber placement can now be checked without reconstructing erased symbols in (1.1). The relevant readable identities establish the following chain.

1. The relaxed reference uses the exterior constraints in (1.3), with a fixed extension chosen before the relative critical equation on p.359. Thus $R(b) = V_\Lambda(b)$ is fixed before the current output.
2. Formula (1.34) introduces the new relative current-scale output $C = V_k'$ through $V = CV_\Lambda$. Formula (1.50), in contrast, writes the **old** reconstruction using the old V'', the fluctuation B, and $U_0(b)$. That old readout does not acquire a dependence on the independent output C.

3. The operations omitted immediately before (1.1) are explicitly left unchanged. Keep their old variables and old field slots under their original integrals. They do not become kernels evaluated at the new background merely because the effective action is later completed. Any rewriting of such a slot carries its full difference term. Consequently it is the old marked numerator in (S57), not a source evaluated on the replacement field, that enters the conditional move.
4. Printed p.378 retains the same relative cutoff $\chi(C)$ and the relative axial gauge factor $\delta_{G_0}(C)$. The reference denominator (1.10) integrates precisely that relative tree/cutoff fiber. The same forest is fixed on both sides; averaging over its allowed choices can be performed afterwards. Haar translation by the fixed $V_\Lambda(b)$ and the tree disintegration (S54) give exactly the output measure used in that denominator.
5. The additional small/large component labels on p.378 are the partition in (S58b). Removing the relative gauge delta on a completed small component is expressly permitted only under the corresponding current-variable integral. The same integrated gauge substitution preserves any jointly gauge-invariant carrying function, including the original plaquette source.

Thus (S57)–(S58b) is the canonical source-preserving extension of the **published source-zero R' identity**, with its old integrations left in place. It is not inferred from a quotient merely having a denominator, from a bound on $T1$, or from the later optional normalization $T/T1$. The published source-zero preparation identities themselves are the imported source-free input; the marked extension and its Haar/selector algebra have been proved here.

The completion terms subsequently introduced can depend on C individually, but their addition and subtraction is an exact identity. They therefore do not change the preceding conditional mass calculation. The original component source-free cancellation and range estimates apply to the same old/new fields, tests and relative measure. No independently redefined replacement operation is needed for this identification.

89.16 16. Complex action domination from the original accretivity estimate

There is a second useful route to the primitive bound. It uses a pointwise complex action estimate, rather than reconstructing a covariant first-difference bound from an unread coefficient formula. Its exact weighting requirement matters.

89.16.1 16.1 A dimension-independent positive-real-part lemma

Let $\mathcal{B}$ be the unit ball of a complex Banach space, and let H be holomorphic on $\mathcal{B}$. Suppose

$$H(0) = E \geq 0, \qquad \operatorname{Re} H(z) \geq cE - B \quad (z \in \mathcal{B}), \qquad 0 \leq c \leq 1, \quad B \geq 0. \tag{S63}$$

For every fixed $0 < t < 1$,

$$\sup_{\|z\| \leq t} |H(z)| \leq E + \frac{2t}{1-t}\big[(1-c)E + B\big] \leq C_t(E + B). \tag{S64}$$

Proof. Restrict to a complex line through the origin and put $f = H - cE + B$. Its real part is nonnegative and $f(0) = (1-c)E + B \geq 0$. When $f(0) > 0$, the function $(f - f(0))/(f + f(0))$ maps the disc into the closed unit disc and vanishes at the origin. Schwarz's lemma gives $|f(z) - f(0)| \leq 2tf(0)/(1-t)$. When $f(0) = 0$, apply the preceding argument to $f + \epsilon$ and let $\epsilon \downarrow 0$. Finally $H(z) - E = f(z) - f(0)$. The bound is independent of the dimension of the deformation space. □

The base point and the energy E may depend on every still-unintegrated real history variable. The proof is pointwise in those variables. Thus if an original cold action satisfies (S63) with $B \leq CN_\omega$, a constant complex source coefficient of magnitude at most ρ obeys

$$|wH(z)| \leq C_t\rho(E + N_\omega). \tag{S65}$$

This is exactly the form required by the one-joint-integral reserve. A single fixed reduction of the original complex deformation radius suffices; there is no cutoff-dependent phase estimate and no division by a live conditional expectation.

89.16.2 16.2 The precise extension to ordinary spatial sources

Let $a_p(z)$ be the holomorphic primitive plaquette functions, with $a_p(0) \geq 0$. Assume that the same pointwise comparison is available, with constants independent of the selector, for every real vector ϑ satisfying $0 \leq \vartheta_p \leq 1$:

$$\operatorname{Re} \sum_p \vartheta_p a_p(z) \geq c \sum_p \vartheta_p a_p(0) - B_\omega, \qquad B_\omega \leq C N_\omega. \tag{S66}$$

Then every complex vector w, with $\sup_p |w_p| \leq \rho$, satisfies

$$\left| \sum_p w_p a_p(z) \right| \leq C_t \rho \left(\sum_p a_p(0) + N_\omega \right) \qquad (\|z\| \leq t). \tag{S67}$$

To prove this, decompose w/ρ into the positive and negative parts of its real and imaginary components. Each is a real vector in $[0,1]^P$. Apply (S64) to those four weighted subactions and sum. At most four copies of B_ω occur, independently of the number of plaquettes. The same argument applies pointwise in an ordinary complex source polydisc, and hence in its supremum Banach algebra.

A stronger, directly usable sufficient condition is the absolute Taylor estimate

$$\sum_p |a_p(z) - a_p(0)| \leq \epsilon \sum_p a_p(0) + C_\epsilon N_\omega. \tag{S68}$$

It implies (S66) and (S67) immediately. The plaquette estimate (S49) provides a version of this argument after the linear term is bounded by Young's inequality. More generally, any original action comparison obtained by summing absolute values of plaquette Taylor remainders retains arbitrary multipliers $0 \leq \vartheta_p \leq 1$. A cancellation that holds only after summing the unweighted action does not have this property.

89.16.3 16.3 What the published comparison establishes, and what still has to be checked

R-II (1.46)–(1.48) explicitly writes the complexified cold Wilson action as a real positive action plus a small analytic remainder. Its discussion explains that exponential separation suppresses the large historical coupling weights; (1.47) is the estimate of the potentially largest linear term. That equality directly gives the constant-source conclusion (S65), with the displayed positive action as E. In that application the positive comparison action need not equal $H(0)$: the modulus bound on the analytic remainder is already sufficient. The lemma (S63)–(S64) is useful when one retains only an accretivity version of the comparison.

For a heterogeneous primitive source, one must check that the bounds of the three Taylor contributions in that calculation are absolute plaquette sums, or prove the selector-uniform version (S66) by repeating that estimate with ϑ. The field maps, their characteristic functions, and their cold variables are kept frozen. In particular, no saddle is recomputed for ϑ. Formula (1.47)'s exponential-separation argument is compatible with such a repetition, but a bound for the total unweighted action alone is not its proof. Section 12 gives an independent sufficient covariant estimate when that repetition is inconvenient.

There is a simple sharp warning. Let T_n denote the Chebyshev polynomial and set

$$a_{1,n}(z) = \frac{1 + T_{4n}(z)}{2}, \qquad a_{2,n}(z) = \frac{1 - T_{4n}(z)}{2}. \tag{S69}$$

Both functions are nonnegative on the real interval $[-1, 1]$, and their sum is identically 1 on the whole complex plane. Thus the unweighted action has perfect accretivity and no remainder. At a fixed $z = it$, however,

$$T_{4n}(it) = \cosh\bigl(4n \operatorname{arsinh} t\bigr), \tag{S70}$$

so the individual weighted action $a_{1,n}(it)$ is unbounded as $n \to \infty$. The point is logical: unweighted accretivity and positivity on real inputs do not supply a uniform spatial-source bound. This is not an example of the actual Wilson reconstruction.

89.18 18. Exact inheritance of the published source-zero normalizer

When the literal original transformations are used, the source-zero vacuum is not an additional quantity to be guessed or matched after the calculation. It can be fixed at the start. This observation removes a possible independent normalization hypothesis from Section 15.

89.18.1 18.1 The source-zero inheritance lemma

Fix a finite bare lattice, its real source-zero tests, all reference factors, all gauge-volume conventions, and the exact order of the original Gaussian, coordinate, completion, and connected-expansion operations. Denote the published source-zero identity by

$$Z_a[0] = e^{V_a^{\mathrm{pub}}} \int \rho_{K,a}^{\mathrm{pub}}(V)\, dV. \tag{S74}$$

Carry a physical source through those same operations, and suppose their marked integral identities have been verified as in Sections 5 and 13. Use precisely the same scalar normalizations at source zero. For every universal regular birth b, including clean and healed R births, let $v_b(z)$ be the scalar term produced by the same local expansion. Define only its excess

$$\Delta v_b(z) = v_b(z) - v_b(0), \qquad \mathcal{E}_a(z) = \sum_b \Delta v_b(z), \qquad \mathcal{E}_a(0) = 0. \tag{S75}$$

All sums are finite at finite cutoff and volume, or absolutely convergent finite-volume connected sums on their proved domain.

Then the exact marked representation has the form

$$Z_a[z] = e^{V_a^{\mathrm{pub}} + \mathcal{E}_a(z)}\, W_a(z), \qquad W_a(0) = \int \rho_{K,a}^{\mathrm{pub}}(V)\, dV. \tag{S76}$$

Proof. At source zero, each Banach-valued integrand, coordinate factor, and selected connected coefficient reduces to its original scalar value. Prove this successively in the actual order of the operations. The source-independent selectors and reference denominators do not change. A source-dependent analytic Gaussian reorganization, if used merely as an identity inside the same integral, has the same source-zero determinant and coordinate factors; its excess logarithm is fixed to vanish at zero. The convergent local connected logarithm has its prescribed source-zero branch, so it introduces no arbitrary scalar constant.

At each completion of the regular action, factor the source-excess scalar using the exact identity (S20), and retain every missing copy under its original history integral. The source-zero scalar stays in the already fixed V_a^{pub}. Section 14 bounds the missing excess copies but does not alter this equality. Induction proves (S76). In particular, all source-zero R vacua, Gaussian normalizations, Haar factors, and gauge volumes agree because they are the same factors, not because their difference was estimated. □

It is enough to define V_a as V_a^{pub} and never redefine it. Replacing it by a sum of selected pure-clean vacua would invalidate this argument. A newly defined operation such as (S58), if it has not yet been

identified with the original source-zero transformation, is a different case: its exact mass identity is established, but its source-zero local coefficients and normalization must still be compared before importing (S74).

89.18.2 18.2 The terminal lower bound uses the full original density

Suppose the published conditional stability theorem, on the tuned weak trajectory, gives

$$\rho_{K,a}^{\mathrm{pub}}(V) \geq \chi_{K,a}(V)\, \exp\{-x_K A(U_{K,a}(V)) - C_- N_K\}. \tag{S77}$$

Here $\chi_{K,a}$ is the theorem's actual regular-core indicator. If a fixed retained-scale product Haar ball $\mathcal{O}_K$ satisfies

$$\chi_{K,a} = 1 \text{ on } \mathcal{O}_K, \qquad x_K A(U_{K,a}(V)) \leq C_0 N_K \text{ on } \mathcal{O}_K, \qquad \mathrm{Haar}(\mathcal{O}_K) \geq e^{-C_1 N_K}, \tag{S78}$$

uniformly in the bare cutoff, then

$$W_a(0) \geq e^{-(C_- + C_0 + C_1) N_K}. \tag{S79}$$

This is an integral of the full positive source-zero density. It is not a lower estimate obtained by selecting one signed Mayer summand, or by deleting every history in which a fine-scale bad test occurred.

The standard small-core check is quantitative. A retained weak-chart lift of size δ has finer link and plaquette deviations at depth t bounded by $C\delta L^{-t}$ and $C\delta L^{-2t}$. The tuned running tube has $x_{K-t} \leq A_K + Bt$. Consequently

$$\sup_{t \geq 0} L^{-t} \sqrt{A_K + Bt} < \infty. \tag{S80}$$

Choose δ below the fixed reciprocal of this supremum, including any specified polynomial or logarithmic threshold factors. The original small-core readout bounds then put this ball inside every retained background test. Fluctuation tests centered on the critical background vanish there. The uniform small lift has action $O(\delta^2 N_K)$, and its constrained minimum has no larger action. A product of fixed small Haar balls costs only $e^{-C_1 N_K}$. This is the retained-resolution core argument of the earlier continuum-bridge module; it does not require identifying the terminal reference field with a physical Wilson pushforward. If a different operation changes these tests, its version of (S78) has to be checked anew.

Finally the proved absolute marked-history majorant, with the common excess vacuum removed, gives $|W_a(z)| \leq e^{C_+ N_K}$. Together with (S76) and (S79),

$$\left| e^{-\mathcal{E}_a(z)} \frac{Z_a[z]}{Z_a[0]} \right| \leq e^{C N_K}. \tag{S81}$$

Thus no separate comparison of arbitrary vacuum conventions is necessary on the literal frozen-map route. The unresolved task is the marked same-fiber/local-domain construction that licenses that route, rather than the elementary inference (S74)–(S81).

89.21 21. A strictly local physical normalizer follows on each compatible torus

This section closes the local-counterterm step mentioned in Section 20.2. It uses the complete-carrier scalar source construction just proved, rather than postulating a separate ordinary observable-source theorem. Exact equality between two ambient tori is distinguished at the end.

89.21.1 21.1 The source ball and the complete vacuum norm

At finite cutoff, every primitive source is a continuous linear function of a smooth profile sampled on finitely many plaquettes. All frozen operations therefore define holomorphic functions on the ball $\mathcal{B}_\rho^\alpha = \{J : \|J\|_{C^\alpha} < \rho\}$. The estimates above also apply with the Banach algebra of bounded holomorphic functions on this ball: they use only its supremum norm, multiplication, and scalar directional Cauchy estimates. Alternatively, restrict to any finite-dimensional profile subspace and then use compatibility under enlarging that subspace. No number of fine plaquettes enters the source norm.

Let $\nu_{j,X}(J)$ be the source-excess scalar birth, with every clean, healed R, and explicit Gaussian type included. After spending fixed locality and source-radius margins, Sections 2, 11, 14, and 19 give

$$\sup_{j,Q} \sum_{X \ni Q} e^{\kappa d_j(X)} \sup_{J \in \mathcal{B}_\rho^\alpha} |\nu_{j,X}(J)| \le C\rho, \qquad \nu_{j,X}(0) = 0. \tag{S87}$$

Source-dependent quadratic Gaussian terms can either remain among the small quadratic-growth incoming potentials or be expanded in the same current-step connected determinant series. Neither representation takes an operator-norm bound and traces it over all fine variables in a deep background lift. The current fluctuation has the fixed number of degrees of freedom per current cell used in Section 2.

The carrier X in (S87) contains all source sites used by that coefficient, together with its fixed determining collars and internal records. The exact source-free localization identities are unchanged by adding a carrying function. Inductively, a selected potential, local integration, or connected product reads only the union of its input carriers. A reference denominator is frozen and carries its original determining variables; an unresolved piece remains an operator until that whole piece is included. Consequently $\nu_{j,X}(J)$ depends only on the values of J on this completed carrier. Equality of profiles there gives equality of the coefficient. This is a property of the constructed coefficient, stronger than its small norm.

89.21.2 21.2 Fourth-order source jets leave a summable remainder

Fix a physical radius $a_0 > 0$ smaller than one quarter of the minimum torus period under consideration. Choose a smooth real cutoff χ_0, invariant under coordinate reflections, equal to one in the a_0 ball and

supported in the $2a_0$ ball. Give each completed scalar carrier a fixed root y, with any pointing factors paid from spare locality. Put

$$(P_yJ)(x) = \chi_0(x-y)\sum_{|\gamma|\le 4}\frac{\partial^\gamma J(y)}{\gamma!}(x-y)^\gamma, \qquad \mathcal{L}_{a,T}(J) = \sum_{j,X}\nu_{j,X}(P_{y(j,X)}J). \tag{S88}$$

Choose a fixed C^5 radius $r_0 > 0$ small enough that all these profiles lie in $\mathcal{B}^\alpha_{\rho/4}$. The constants of this embedding depend on a_0, not on the fine cutoff or the ambient volume. Every summand in $\mathcal{L}$ is a holomorphic function of the jet through order four at one point, and vanishes at the zero jet. Thus $\mathcal{L}$ is a strictly local contact functional, even though its coefficients need not converge individually as the cutoff is removed.

Let the carrier and its collars fit in a ball of radius $r_X = C\ell_j(1+\operatorname{diam}_j X)$ about the root. If $r_X < a_0/4$, take a smooth cutoff χ_X, equal to one on that ball and supported in the ball of radius $2r_X$, and set

$$h_X = \chi_X(J - T^4_y J), \qquad \|h_X\|_{C^\alpha} \le C r_X^{5-\alpha}\|J\|_{C^5}. \tag{S89}$$

Taylor's integral remainder gives the supremum estimate $Cr_X^5\|J\|_{C^5}$ and the first-derivative estimate $Cr_X^4\|J\|_{C^5}$. At separations below r_X use the derivative estimate, and at larger separations use the supremum estimate; this proves (S89), including derivatives of the cutoff. Since $P_yJ + h_X = J$ on the complete carrier, source locality identifies their coefficients. Scalar Cauchy in the genuine global direction h_X, using the spare ball radius and (S87), gives

$$|\nu_{j,X}(J) - \nu_{j,X}(P_yJ)| \le Ce^{-\kappa' d_j(X)} r_X^{5-\alpha}\|J\|_{C^5}. \tag{S90}$$

For $r_X \ge a_0/4$, use the original bound and zero-source Schwarz estimate, retaining the tail e^{-c/ℓ_j}; wrapping carriers are included in this case. Root counting supplies $C\operatorname{Vol}(T)\ell_j^{-4}$. Summing the diameter polynomial with spare exponential decay and then summing the geometric scales proves

$$|\mathcal{E}_a(J) - \mathcal{L}_{a,T}(J)| \le C\operatorname{Vol}(T)\|J\|_{C^5}\sum_{\ell_j\le 1}\left(\ell_j^{1-\alpha} + \ell_j^{-4}e^{-c/\ell_j}\right) \le C'\operatorname{Vol}(T)\|J\|_{C^5}. \tag{S91}$$

The finitely many current scales above the chosen physical unit only change the fixed constant. This is the actual Taylor/carrier argument of the earlier local-map Section 16, now applied to the source coefficients whose incoming and live estimates have been checked in this note.

89.21.3 21.3 Physical normalized bound and reflection positivity

Combining (S81) and (S91), and absorbing the fixed terminal-cell volume convention, gives

$$\left|e^{-\mathcal{L}_{a,T}(J)}\frac{Z_{a,T}[J]}{Z_{a,T}[0]}\right| \le e^{C\operatorname{Vol}(T)}, \qquad \|J\|_{C^5} < r_0. \tag{S92}$$

The constants are uniform in the bare cutoff and in the compatible torus volume with the fixed lower period bound. The functional in this inequality is the original Wilson source functional, by the matched-fiber carrying identity. It is not a functional of the replacement coordinate.

If a fixed blocking or pointing convention breaks a lattice symmetry, average the resulting local normalizers over the finite torus translation/reflection group. For each transformed convention, physical covariance gives the same inequality for $\log|Z[J]/Z[0]|$; averaging its right side preserves (S92). This averages counterterms, not measures. It preserves strict locality and gives $\mathcal{L}(\theta\overline{J}) = \overline{\mathcal{L}(J)}$.

For positive-side sources with a nonzero margin from the reflection plane, the source and its reflected partner have disjoint jets. Hence

$$\mathcal{L}(\theta\overline{J}_1 + J_2) = \overline{\mathcal{L}(J_1)} + \mathcal{L}(J_2). \tag{S93}$$

The renormalized reflection matrix is the original physical positive matrix multiplied by the diagonal factors $e^{-\overline{\mathcal{L}(J_1)}}$ and $e^{-\mathcal{L}(J_2)}$. Positivity therefore survives exactly. Every mixed derivative of $\mathcal{L}$ involving two disjoint source supports is zero. These conclusions need no equality of normalizers on different tori and no uniform zero-free disc for the whole partition function.

89.21.4 21.4 The separate exact intertorus locality statement

The inspected primary formulas give exact component localization: II (1.6)–(1.10), the interior whole-lattice identification following III (3.47), and the component/bridge unions in R-II (1.71)–(1.72), (1.90)–(1.98). Those facts, with the canonical source-independent carrying extension, prove exact source support and the universal regular birth family used above. Unsummed exterior history stays in a boundary operator and does not enter that regular scalar sum.

For literal equality of a nonwrapping coefficient on two different tori, one also fixes the same volume-independent coupling extraction and the same elementary local-factor construction on isomorphic lifted domains. Under that convention the complete term consists of identical finite-domain kernels, local fixed-point equations, relative Haar/tree measures, selector supports, old field slots, and frozen reference denominators. Induction through their products, convergent local series, and integrations proves equality before root counting. This is PS's naturality theorem applied to the complete carrier, not an inference from (S87).

The source-independent carrying extension introduces no new ambient dependence: it keeps exactly the original denominator and old determining data. However, a statement merely saying that an elementary typed factor depends on $U|_X$ does not by itself exclude an explicit ambient-volume constant. The literal local-factor definition in the original propagator construction, or its fixed universal local-choice convention, is the source-definition input for that last intertorus equality. The readable II/III/R-II component formulas support it but should not be presented as a proof from a norm bound alone. None of this qualification weakens the within-torus theorem (S92)–(S93), which is already sufficient for the strict-local scalar chessboard step.

89.22 22. The volume-transfer interface only needs source-zero naturality

89.22.1 22.1 The common source normalizer lives on one reference torus

Let T_* be one fixed compatible reference torus and let $T = qT_*$ be a larger compatible torus. Write $V_{a,T}$ for the original source-zero vacuum extracted in Section 18. The only additional cross-volume numerical estimate needed below is

$$\frac{V_{a,T_*}}{\mathrm{Vol}(T_*)} - \frac{V_{a,T}}{\mathrm{Vol}(T)} \le C_0, \tag{S94}$$

uniformly in the fine cutoff and q. The absolute nonwrapping comparison of marked-bridges (18.4) implies (S94), but the one-sided form is sufficient.

Fix a cell of side l_c in T_*, and place a scalar source J inside it, at a fixed positive distance from every face. Reflect and conjugate it into all cells to obtain W_J. The physical transfer identity and the

positive A^*A slab pair give

$$|Z_{a,T}[J]/Z_{a,T}[0]| \le \left(\frac{Z_{a,T}[W_J]}{Z_{a,T}[0]}\right)^{1/n_T^4}, \qquad Z_{a,T}[W_J] \le Z_{a,T_*}[W_J]^{q^4}, \tag{S95}$$

where n_T is the number of cells in one direction. These are inequalities for the original Wilson functional. Applying the proved reference-torus ordinary-source upper bound, the source-zero lower bound on T, and (S94) gives

$$|Z_{a,T}[J]/Z_{a,T}[0]| \le \exp\left\{n_*^{-4}\,\mathrm{Re}\,\mathcal{E}_{a,T_*}(W_J) + Cl_c^4\right\}. \tag{S96}$$

Use only the source-excess births on T_* to define $\mathcal{E}_a^\circ(J)$: keep carriers contained inside one cell, undo the complex conjugation in odd cells, and average the resulting holomorphic functions over the reference cells. Every omitted source-excess carrier both meets the source support and reaches a cell face. By exact source support and (S87), its total contribution per cell is at most

$$C\sum_{\ell_j\le 1}\ell_j^{-4}e^{-c/\ell_j} < \infty. \tag{S97}$$

Consequently

$$\left|e^{-\mathcal{E}_a^\circ(J)}\frac{Z_{a,T}[J]}{Z_{a,T}[0]}\right| \le Q, \qquad \|J\|_{C^\alpha} < \rho_*, \qquad \mathcal{E}_a^\circ(0) = 0, \tag{S98}$$

with the same $\mathcal{E}_a^\circ$, radius, and Q for every compatible T, including T_*. No source-dependent coefficient on T was compared with a coefficient on T_*. All sourced coefficients used in this argument belong to the one reference torus. The only cross-volume renormalization input is the source-zero estimate (S94).

The bounded functions in (S98) equal one at zero. A smaller fixed source disc therefore gives bounded local logarithms, without a global zero-free theorem. Their common $\mathcal{E}_a^\circ$ cancels when logarithmic derivatives for two volumes are subtracted. For scale-d bumps, packing cd^{-4} equally translated pairs in the cell and using one independent sign per pair yields

$$|\,\mathrm{Cov}_T(O(f),O(h)) - \mathrm{Cov}_{T_*}(O(f),O(h))| \le Cd^{4-2\alpha}\|f\|_\infty\|h\|_\infty. \tag{S99}$$

The source norms of the packed sums are $O(d^{-\alpha})$, with no packing-count factor. Sign averaging removes unequal translated pairs. This proves the stated scalar volume-transfer consequence conditional only on (S94), in addition to the already established source theorem. The strict-local C^5 theorem remains a separate unconditional within-torus consequence.

89.22.2 22.2 Which original local factor still needs identification

The exact objects are the initial and typed factors in the generalized random-walk expansion (3.107) of Bałaban's *Propagators for lattice gauge theories in a background field*, CMP 99 (1985), 389–434, as imported in II (1.6). This includes the local inverses used to form the gauge projector and the coarse Schur inverses, not just the final vector Green kernel. Copying a completed typed domain must copy its padded domains, artificial local boundary rule, background extension, gauge conventions, finite scalar/coarse inverses, cutoffs, and normalization measures. If these factors are defined by the same finite-domain matrices on the copied data, equality is exact: conjugation by the finite-coordinate identification commutes with matrix inversion, products, local convergent fixed points, and the retained integrations. The same argument then applies inductively to the source-zero vacuum family and gives (S94) after the wrapping tail sum.

There is a concrete primary verification of this mechanism in the flat/Abelian construction. Dimock, *Multiscale block averaging for QED in d=3*, arXiv:1712.10029, Section 3.7.1, equation (278), defines the local gauge inverse on its enlarged cube with that cube's own periodic boundary and local projector. Section 3.7.4, equations (310)–(312), expands the remaining projector pieces into short typed factors and cites the background-field propagator construction. These are genuine autonomous local inverses, not restrictions of the ambient inverse. Thus that flat local construction has the required exact naturality.

The inspected II/III/R-II statements prove component support, component fixed-point splitting, interior whole-lattice coefficients, and complete history carriers. They do not display the non-Abelian definition of every typed inverse in (3.107). Establishing that the same autonomous finite-domain convention is used for those background-dependent factors would finish this narrow source-zero matching step. A new canonical parametrix could also finish it if it is shown to reconstruct the same unweakened original propagators with the published formation bounds. The boxed/orbit inverses constructed elsewhere in this work cannot simply be substituted without that identification.

89.22.3 22.3 Flat values and finitely many jets do not suffice

Even a source-zero scalar birth integrates nonzero fluctuation fields. Inherited members are evaluated on nonlinear reconstructed backgrounds inside that integral. Thus the vacuum is not determined merely by the typed factors at $U = 1$, nor by any fixed finite number of their field derivatives there.

A one-variable diagnostic makes the issue quantitative. Fix a small $b > 0$ and define

$$I_c(g) = \frac{1}{\sqrt{2\pi}} \int_{|gB|<b} \exp\left[-\frac{B^2}{2} + c(gB)^{2m}\right] dB. \tag{S100}$$

For bounded small real c, the added analytic potential has zero jets through order $2m-1$, a fixed-core bound independent of g, and a quadratic-growth Gaussian majorant on the fixed physical cutoff. Direct Gaussian Taylor expansion gives

$$\log I_c(g) - \log I_0(g) = c\,\mathbb{E}[B^{2m}]g^{2m} + O(g^{4m}) + O(e^{-c_b/g^2}). \tag{S101}$$

An unmatched error of this size per scale cell would be multiplied by ℓ^{-4}. Along $g^2 \asymp 1/\log(1/\ell)$, no fixed m makes that root sum ultraviolet finite. This example is not a counterexample to Bałaban's coefficients or the Wilson measure. It shows precisely why flat equality plus finite-jet matching and ordinary analytic norm bounds cannot replace equality, or an exponentially small boundary comparison, for the full background-dependent local factors.

89.22.4 22.4 An alternative only requires a second source-zero stability construction

There is an exact comparison lemma which avoids identifying individual coefficients of two decompositions. Let both $V_{a,T}$ and $\widetilde{V}_{a,T}$ normalize the SAME positive Wilson partition function, with

$$|\log Z_{a,T}[0] - V_{a,T}| \le C\,\mathrm{Vol}(T), \qquad |\log Z_{a,T}[0] - \widetilde{V}_{a,T}| \le \widetilde{C}\,\mathrm{Vol}(T). \tag{S102}$$

The constants are independent of cutoff and compatible volume. Subtracting these two inequalities gives

$$|V_{a,T} - \widetilde{V}_{a,T}| \le (C + \widetilde{C})\,\mathrm{Vol}(T). \tag{S103}$$

If the second vacuum has a canonical local construction giving $\widetilde{V}_{a,T_*}/\operatorname{Vol}(T_*) - \widetilde{V}_{a,T}/\operatorname{Vol}(T) \le \widetilde{C}_0$, then

$$\frac{V_{a,T_*}}{\operatorname{Vol}(T_*)} - \frac{V_{a,T}}{\operatorname{Vol}(T)} \le \widetilde{C}_0 + 2(C + \widetilde{C}). \tag{S104}$$

This is exactly (S94). Neither a sourced theorem for the second decomposition nor equality with the original vacuum coefficients is needed.

The lemma has not supplied that second construction. In particular the new boxed/relative augmented maps cannot supply (S102) merely because they agree with the genuine physical critical graph. They would need their own complete source-zero formation, positive integration, and terminal estimates for the same Wilson partition function. Thus (S102)–(S104) records a precise alternative to the literal original-factor verification; it does not remove that remaining interface by renaming it.

92.2 2. A literal same-variable torus gluing identity

There is no need to change the Haar space or fix a gauge. Label the positive links by

$$(b, x, \mu), \qquad b \in (\mathbb{Z}/q\mathbb{Z})^4, \quad x \in \{0, \dots, n-1\}^4, \quad \mu = 1, \dots, 4.$$

For the product of M small periodic tori, an outgoing boundary link wraps inside its own block. For the large torus, it connects the next block. The independent group variables and their product Haar measure are identical in these two readings.

Let S_{per} be the sum of the M small-torus plaquette actions without the coefficient β_a, and let S_{big} be the corresponding large-torus sum on this same variable space. Put

$$\Delta S = S_{\mathrm{big}} - S_{\mathrm{per}}, \qquad \mu_0 = \mu_{a,T_*}^{\otimes M}.$$

Then, exactly,

$$\frac{Z_{a,T}}{Z_{a,T_*}^{M}} = \mathbb{E}_{\mu_0} e^{-\beta_a \Delta S}, \qquad G_{a,q} := M \log Z_{a,T_*} - \log Z_{a,T} = -\log \mathbb{E}_{\mu_0} e^{-\beta_a \Delta S}. \tag{SV5}$$

In particular $G_{a,q}/(M\ell_*^4) = p_{a,T_*} - p_{a,T}$. Normalized Haar measure contributes no gauge-volume factor.

For a fixed orientation $\mu < \nu$, only plaquettes whose starting site satisfies $x_\mu = n-1$ or $x_\nu = n-1$ can change. Their number in each block is $2n^3 - n^2$. Consequently

$$N_{\mathrm{seam}} = M(12n^3 - 6n^2), \qquad |\Delta S| \le s_{\max} N_{\mathrm{seam}}. \tag{SV6}$$

The displayed count is an upper bound also for $q = 1$, when the actual difference is zero. For $q \ge 2$, it counts the rewired plaquettes. The factor is $s_{\max}$, not $2s_{\max}$, since both compared plaquette actions lie in the same interval $[0, s_{\max}]$.

Reflection positivity, or S95 at zero source, supplies $G_{a,q} \ge 0$. Combining it with (SV5)–(SV6) proves

$$0 \le p_{a,T_*} - p_{a,T} \le \beta_a s_{\max} \left(\frac{12}{\ell_* a^3} - \frac{6}{\ell_*^2 a^2} \right). \tag{SV7}$$

This estimate is uniform in q but not in a. Even replacing its factor β_a by a bounded constant would leave an a^{-3} divergence. A bound per lattice site with an error $O(1/n)$ has precisely this problem when converted to physical volume.

92.3 3. Cutting and local weak-field energy do not repair (SV7)

Delete the same set of seam plaquettes from both actions. The remaining action is identical and factorizes into M open boxes. Boundary link variables that are no longer used integrate to one. Therefore, with Z_{open} denoting this exact open-box Wilson integral,

$$e^{-\beta_a s_{\max} N_{\mathrm{seam}}} Z_{\mathrm{open}}^{M} \le Z_{a,T},\ Z_{a,T_*}^{M} \le Z_{\mathrm{open}}^{M}. \tag{SV8}$$

This is the explicit cut-boundary estimate. Comparing each periodic system separately to the open one loses the cancellation between their large interface free energies.

One cannot improve Jensen's bound merely by invoking small plaquette energy in the small-torus measure. For $q \ge 2$ and $n \ge 2$, every rewired plaquette contains a single open bond in a different small block. In the product measure that block is separately gauge invariant. The expectation of this open bond, in the nontrivial fundamental representation, is zero: Haar averaging a gauge transformation at one endpoint annihilates it. If both relevant coordinates lie on a seam, the two neighboring blocks each supply such a single bond. Hence

$$\mathbb{E}_{\mu_0} s(U_p^{\mathrm{big}}) = 1 \quad (p \text{ rewired}).$$

Translation and coordinate symmetry on a small torus give, with $\bar{s}_{a,*} = \mathbb{E}_{\mu_{a,T_*}} s(U_p)$,

$$\mathbb{E}_{\mu_0} \Delta S = N_{\mathrm{seam}}(1 - \bar{s}_{a,*}), \qquad G_{a,q} \le \beta_a N_{\mathrm{seam}}(1 - \bar{s}_{a,*}). \tag{SV9}$$

Even if one grants the favorable estimate $\bar{s}_{a,*} = O(\beta_a^{-1})$, the right side of (SV9) remains of order $\beta_a N_{\mathrm{seam}}$. The weak plaquette estimate concerns the old closed plaquettes; it does not control the newly glued plaquettes under an independently gauged product of blocks.

The finite interaction range also gives conditional independence of cell interiors after the separating links are fixed. It leaves an interface density obtained from those conditioned cell partition functions. Bounding that density by its supremum and infimum gives the same surface cost as (SV8). Finite range does not provide a cutoff-uniform bound on this interface density.

92.4 4. The exact missing signed boundary estimate

Define a finite positive interpolating integral on the same Haar space:

$$S_t = S_{\mathrm{per}} + t\Delta S, \qquad d\mu_t = Z_t^{-1} e^{-\beta_a S_t} dU, \qquad 0 \le t \le 1.$$

Both endpoint actions are nonnegative, so S_t is nonnegative. Differentiation under this finite compact integral gives

$$\frac{d}{dt} \log Z_t = -\beta_a \mathbb{E}_{\mu_t} \Delta S, \qquad G_{a,q} = \beta_a \int_0^1 \mathbb{E}_{\mu_t} \Delta S \, dt. \tag{SV10}$$

The following is a sufficient and, given (SV3), quantitatively equivalent replacement for S94:

$$\boxed{-\log \mathbb{E}_{\mu_{a,T_*}^{\otimes q^4}} e^{-\beta_a \Delta S} = \beta_a \int_0^1 \mathbb{E}_{\mu_t} \Delta S \, dt \le C q^4 \ell_*^4} \quad \text{uniformly in } a, q. \tag{SV11}$$

Only this signed integral is required. A sufficient bound on the integral of the absolute expectation would be stronger. The absolute pointwise estimate instead gives (SV7), so (SV11) is additional mathematical content.

There is an exact entropy formulation of the cancellation. Let μ_1 be the large-torus measure on the same labeled variables. Equation (SV5) gives

$$\log \frac{d\mu_1}{d\mu_0} = -\beta_a \Delta S + G_{a,q}, \qquad G_{a,q} = \beta_a \mathbb{E}_{\mu_0} \Delta S - D(\mu_0 \| \mu_1). \tag{SV12}$$

Thus positivity of relative entropy reproduces Jensen, while a uniform volume comparison needs the appropriate cancellation between the seam energy and the relative entropy. No such cancellation follows from their separate nonnegativity.

In a multiscale attempt, an uncancelled surface cost at scale r has root count $O(\mathrm{Vol}(T)/(\ell_* r^3))$. Multiplication by any fixed inverse power of $\log(1/r)$ is still not summable in the ultraviolet. Exact equality of local nonwrapping pieces, or another estimate which explicitly cancels their interface terms, is needed before these roots are summed. The original wrapping mechanism would instead produce a summable bound of the form

$$C \, \mathrm{Vol}(T) \sum_j r_j^{-4} e^{-c\ell_*/r_j}. \tag{SV13}$$

This observation explains what the missing naturality input accomplishes; it does not prove that input from finite range.

92.5 5. What reflection positivity and transfer operators do give

The source-zero consequence of S95 is

$$Z_{a,T} \le Z_{a,T_*}^{q^4}, \qquad p_{a,T_*} - p_{a,T} \ge 0. \tag{SV14}$$

This is the lower side of the difference in S94. Reversing it is not an application of the same trace inequality.

There is a precise alternative sufficient estimate. Fix a transverse three-torus Σ of physical volume A. Let $\mathsf{T}_{a,\Sigma}$ be the positive Wilson transfer operator on the gauge-invariant Hilbert space, including the spatial half-action factors. The projected periodic identity in marked-bridges (18.1)–(18.2) is an identity for the normalized Haar integral, retaining temporal holonomy integration. With $n = \ell_*/a$, put

$$K = \mathsf{T}_{a,\Sigma}^{n}, \qquad \lambda = \|K\| > 0, \qquad \widehat{K} = K/\lambda.$$

Then

$$Z_{a,\ell_*\times\Sigma} = \operatorname{Tr} K, \qquad Z_{a,q\ell_*\times\Sigma} = \operatorname{Tr} K^q,$$

and the scalar normalization λ cancels exactly:

$$\begin{aligned} p_{a,\ell_*\times\Sigma} - p_{a,q\ell_*\times\Sigma} &= \frac{1}{\ell_* A}\left[\log \operatorname{Tr}\widehat{K} - \frac{1}{q}\log \operatorname{Tr}\widehat{K}^{\,q}\right], \\ 0 \le p_{a,\ell_*\times\Sigma} - p_{a,q\ell_*\times\Sigma} &\le \frac{\log \operatorname{Tr}\widehat{K}}{\ell_* A}. \end{aligned} \tag{SV15}$$

The lower bound uses the positive-eigenvalue trace inequality. The upper bound uses that a compact positive nonzero operator attains its largest eigenvalue, so $\operatorname{Tr}\widehat{K}^{\,q} \ge 1$.

It would therefore suffice to prove

$$\boxed{\log \operatorname{Tr}\left(\frac{\mathsf{T}_{a,\Sigma}^{\ell_*/a}}{\|\mathsf{T}_{a,\Sigma}^{\ell_*/a}\|}\right) \le C_{\ell_*} A,} \tag{SV16}$$

uniformly in the cutoff and all compatible transverse tori whose side lengths are ℓ_* or $q\ell_*$. Enlarge the four sides in turn. At each step (SV15)–(SV16) bounds the pressure change by C_{ℓ_*}/ℓ_*; summing the four changes proves (SV4).

Equation (SV16) is a normalized spectral counting estimate at a fixed physical slab width. It does not require a mass gap, but it is not a consequence of positivity or trace class. For example, $K = R^{-1} I_R$ is positive with trace one and has $\operatorname{Tr}(K/\|K\|) = R$; its bracket in (SV15) is $(1 - 1/q)\log R$. This operator example is only a diagnostic of the abstract inference, not a counterexample for a Wilson transfer operator.

Equivalently, defining the nonnegative spectral operator

$$H_{a,\Sigma} = -a^{-1}\log(\mathsf{T}_{a,\Sigma}/\|\mathsf{T}_{a,\Sigma}\|)$$

on its spectral support, (SV16) asks for $\log \operatorname{Tr} e^{-\ell_* H_{a,\Sigma}} \le C_{\ell_*} A$. This is the exact finite-spacing transfer operator, not a replacement by an approximate Hamiltonian. All common scalar vacuum shifts cancel in its definition. The present source-free inputs have not established this uniform normalized trace bound.

92.6 6. Monotonicity in the coupling and the cost of changing the trajectory

At a fixed cutoff,

$$\frac{d}{d\beta}p_{a,R}(\beta) = -\frac{1}{\mathrm{Vol}(R)}\mathbb{E}_{R,\beta}S_R. \tag{SV17}$$

Each pressure is decreasing. This does not order the two energy densities. With $\epsilon_{a,R}(\beta) = \mathrm{Vol}(R)^{-1}\mathbb{E}_{R,\beta}S_R$, normalized Haar gives $p_{a,R}(0) = 0$, and hence the exact endpoint identity

$$p_{a,T_*}(\beta_a) - p_{a,T}(\beta_a) = \int_0^{\beta_a} [\epsilon_{a,T}(u) - \epsilon_{a,T_*}(u)]\, du. \tag{SV18}$$

The elementary bound $0 \le \epsilon_{a,R} \le 6s_{\max}a^{-4}$ does not make (SV18) uniform. Reflection positivity of the pressure gap for every u does not imply a useful bound on its derivative. No non-Abelian correlation-ordering theorem is being assumed here.

All comparisons above have used the same exact endpoint β_a. If separate volume-dependent bare values were substituted, their cost would have to be included:

$$|p_{a,R}(\beta) - p_{a,R}(\beta')| \le 6s_{\max}a^{-4}|\beta - \beta'|. \tag{SV19}$$

Even a uniform weak-coupling estimate $\epsilon_{a,R}(u) \le Ca^{-4}/u$, if independently proved on the whole intervening interval, would give only

$$|p_{a,R}(\beta) - p_{a,R}(\beta')| \le Ca^{-4}|\log(\beta/\beta')|. \tag{SV20}$$

Finite-order asymptotic agreement of two coupling runs is not enough for these errors. The existing common trajectory avoids this additional tuning issue exactly; it does not itself bound (SV11).

92.7 7. An accessible elementary stability theorem does not fill this slot

Faria da Veiga and O'Carroll, *On Yang–Mills Stability Bounds and Plaquette Field Generating Function*, arXiv:2205.07376v2, define volume as lattice site count in Section I. Their Theorems 1–2, (71)–(78), bound the periodic integral between products of one-plaquette factors, with an extra boundary bond count in the lower bound. The scaled free energy in (79)–(81) is normalized per retained lattice bond; the thermodynamic limit precedes the continuum limit. Primary PDF.

In physical units, an $O(1)$ error per retained lattice bond becomes $O(a^{-4})$. The extra $O(n^3)$ boundary count can also leave $O(a^{-3}\log\beta_a/\ell_*)$. These estimates therefore do not provide S102 with an $O(\mathrm{Vol}_{\mathrm{physical}})$ remainder or the diagonal finite-physical-volume comparison (SV4). This is a normalization distinction, not a challenge to the stated lattice-volume bounds.

93.1 1. Conventions and globally smooth observables

For definiteness take SU(N) in its defining representation on a finite periodic four-dimensional lattice of spacing a. The arguments involving Haar adjoints also work for any compact Lie group. Let T^A be an orthonormal anti-Hermitian traceless basis for

$$\langle X, Y\rangle = -\operatorname{tr}(XY), \qquad \sum_A T^A T^A = -C_F I. \tag{AI1}$$

All pairings on real Lie-algebra fields are real. Set

$$s(P) = N - \operatorname{Re}\operatorname{tr} P, \qquad S_0(U) = \sum_{x,\mu<\nu} s(P_{\mu\nu}(x)), \qquad d\mu_a = Z_a^{-1} e^{-\beta_a S_0(U)} \prod_b dU_b. \tag{AI2}$$

There is no source-dependent denominator in the identities below. A bond derivative is defined, without an invariant-vector-field naming convention, by

$$X_b^A F(U) = \frac{d}{dt} F(\dots, e^{tT^A} U_b, \dots)\Big|_{t=0}, \qquad J_b^A = a^{-3} X_b^A S_0, \qquad \kappa_a = (\beta_a a^3)^{-1}. \tag{AI3}$$

Here beta_a is positive. A globally smooth lattice observable means a smooth function on the entire finite product of compact groups. In particular, polynomials in plaquette fluxes, their finitely many covariant differences, and compact parallel transports are allowed. A polynomial in a local logarithm chart is not automatically allowed. If it is multiplied by a smooth chart cutoff, every derivative of that cutoff in the formulas below is retained. A sharp chart cutoff instead has boundary distributions and does not satisfy the displayed smooth formulas without those terms.

The exact based flux is

$$X_{\mu\nu}(x) = a^{-2} \pi_{\mathfrak{su}(N)} P_{\mu\nu}(x) = \pi_{\mathfrak{su}(N)} \frac{P_{\mu\nu}(x) - P_{\mu\nu}(x)^{-1}}{2a^2}. \tag{AI4}$$

The projection is essential for N>2. On real matrices it denotes the orthogonal anti-Hermitian traceless projection. We use oriented plaquettes, so X_nu,mu=-X_mu,nu at their common basepoint. With

$$T_\mu Y(x) = U_\mu(x) Y(x + a\hat{\mu}) U_\mu(x)^{-1}, \qquad D_\mu = (T_\mu - I)/a, \qquad D_\mu^* = (T_\mu^{-1} - I)/a, \tag{AI5}$$

the star is the exact adjoint in the physical sum a^4 sum_x. Direct differentiation of the two plaquettes in each mu,nu plane meeting a nu bond gives

$$J_\nu(x) = \sum_{\mu \neq \nu} D_\mu^* X_{\mu\nu}(x). \tag{AI6}$$

For example, the plaquette based at x contributes -a^{-1}X_mu,nu(x), and the plaquette based at x-a mu contributes a^{-1}T_mu{-1}X_mu,nu(x). This verifies both the physical power of a and the sign. J is the variation of S_0; with this convention its continuum leading term is minus the usual D_mu F_mu,nu.

93.4 4. A complementary dimension-six sector really remains

For a nu bond split S_0 into its three incident plane families and put

$$J_{b;\mu} = a^{-3} X_b S_0^{(\mu\nu)}, \qquad Q_{b;\mu} = J_{b;\mu} - \tfrac{1}{3} J_b, \qquad \sum_{\mu\neq\nu} Q_{b;\mu} = 0. \tag{AI33}$$

Then, exactly,

$$\sum_{\mu\neq\nu} |J_{b;\mu}|^2 = \tfrac{1}{3}|J_b|^2 + \sum_{\mu\neq\nu} |Q_{b;\mu}|^2. \tag{AI34}$$

Subtracting one third of the Casimir counterterm AI15 removes the first term in separated correlations. It leaves the explicitly defined, nonnegative real complementary insertion sum|Q|^2. Its classical symbol is

$$\sum_{\mu\neq\nu} |D_\mu F_{\mu\nu}|^2 - \frac{1}{3}\sum_{\nu} \left| \sum_{\mu} D_\mu F_{\mu\nu} \right|^2 . \tag{AI35}$$

It is not legitimate to remove this hypercubic tensor by the scalar O(4) identity. A direct symbol check makes the distinction sharp. In a Cartan color take complex momentum p=(1,i,0,0), polarization A_3=H, and all other polarizations zero. Use two formal waves of momenta p and -p. Then p^2=p dot A=0, so both linear EOMs vanish. The total-momentum-zero coefficient of the first sum in AI35 is proportional to p_1^{4+p_-2}4=2, hence is nonzero. A total derivative has zero coefficient at that total momentum, and a cubic curvature polynomial has no quadratic jet. Consequently no real polynomial identity can reduce AI35 to EOMs, total derivatives, and cubic curvature alone: its complexification would contradict this test. This is an algebraic symbol test, not a claim that these complex waves define a real periodic Wilson configuration.

93.4.1 4.1 The smallest scalar center justified by the symmetry

At the continuum engineering-jet level, consider reflection-even hypercubic scalars of dimension six for this simple gauge group. A safe small scalar list contains C_sharp, the complementary H=sum|Q_mu|ˆ2 from AI34, EOM/contact representatives, and descendants of the full dimension-four quadratic family. Vacuum and dimension-four action mixing must be retained as well. The derivative scalar A itself is eliminated by AI32d. Before a continuum O(4) tensor law is proved, H cannot be removed.

For exact **pointed lattice** hypercubic covariance, average the entire identity AI32d, with its tests and all its terms, over the finite signed coordinate-permutation group. Do the same for AI34. This defines fixed symmetrized completions; the average preserves locality, gauge invariance, field valuation, the EOM identities, and every displayed compact bound. The forward based plaquette at a fixed root is not by itself declared an exact same-root two-form under reflections. If the average shifts a plaquette base, its shifted test and transported field are retained. The notation for the candidate center below may use these symmetrized completions. Its continuum jet classification is unchanged.

The need for exactly one extra non-EOM quadratic scalar at this jet order can be checked directly. A translation-invariant quadratic hypercubic scalar in the potential has a symmetric degree-four momentum matrix K(p) satisfying K(p)p=0. Reflections and coordinate permutations give its general entries, with S_mu=sum_{rho different from mu}p_rhoˆ2,

$$\begin{aligned} K_{\mu\mu} &= Ap_\mu^4 + Bp_\mu^2 S_\mu + C\sum_{\rho\neq\mu} p_\rho^4 + D\sum_{\rho<\sigma;\ \rho,\sigma\neq\mu} p_\rho^2 p_\sigma^2, \\ K_{\mu\nu} &= p_\mu p_\nu \left[E(p_\mu^2 + p_\nu^2) + F\sum_{\rho\neq\mu,\nu} p_\rho^2 \right] \quad (\mu\neq\nu). \end{aligned} \tag{AI35a}$$

Transversality imposes A=0, B=C=-E, D=-2F. This is a two-dimensional space. The full EOM square has (E,F)=(-1,-1), up to a common convention; the separate directional-current square has (E,F)=(-1,0). Their independence, and the non-EOM test AI35, leave precisely one quadratic hypercubic direction after the EOM quotient.

A reflection-even scalar made from three curvature factors has each coordinate index occurring an even number of times. Its three antisymmetric index pairs must therefore form a triangle. Permutation symmetry gives one scalar contraction, and the nonzero color contraction is the antisymmetric invariant tr(F[F,F]); the symmetric-color contraction cancels under reversal of that triangle. This is the single C-even cubic scalar direction above. For quadratic operators expressed with covariant derivatives, integration by parts and commuting derivatives leave the two flat symbols AI35a, curvature cubics from the commutators, and total derivatives. This is a statement about the local engineering jets, not an equality between arbitrary lattice completions.

For nonconstant tests, retain the pointed total derivatives explicitly. They can be chosen among derivatives through order two of the already retained full quadratic tensor family, together with EOM descendants. One cannot drop them by a constant-source classification. A tensor-valued problem must retain the full corresponding quadratic momentum jets and C-even cubic tensors, rather than just these two scalar directions.

The exact lower-dimensional counterterm in AI32d is also transparent. With

$$I_a(f) = a^4\sum_x f(x), \qquad W_a(f) = a^4\sum_x f(x)a^{-4}\sum_{\mu<\nu} s(P_{\mu\nu}(x)), \tag{AI35b}$$

AI29 reads

$$\mathcal{C}_a(f) = 48NC_F\beta_a^{-1}a^{-6}I_a(f) - 8C_F\beta_a^{-1}a^{-2}W_a(f). \tag{AI35c}$$

W_a is the literal Wilson action density. If Q_a uses a different compact quadratic completion, that distinction is retained; they are not silently identified after multiplication by a^{-2}. Their common dimension-four normalizations and the associated finite-cutoff completion differences belong to the lower-sector bookkeeping.

Accordingly a minimal **candidate scalar transfer system** has the two new non-EOM directions

$$\mathbf{O}_{6,a} = (C_a^\sharp, H_a), \tag{AI35d}$$

along with the lower vacuum/quadratic sectors, their retained descendants, the EOM/contact module, and exact remainder kernels. The one-source transfer must prove a bounded, uniformly invertible two-dimensional center evolution after the physical engineering rescaling, plus a strict stable-remainder bound. Lower-dimensional entries are triangular by engineering degree, with their actual scale factors. This is a finite system to be analyzed; finite dimensionality alone proves none of those norm estimates.

There is a further exactness issue when using AI32d inside an RG step. The EOM in AI7 belongs to the full original Wilson action. A partially integrated effective measure generally has Wilson action **plus** local interaction terms, so its EOM also contains the derivatives of those terms. The correct pushforward is AI13, with its boundary/contact contributions. Discarding the Wilson-current completion pointwise in an effective field would reintroduce precisely the missing near-insertion interaction. Thus AI32d is an exact triangular bare-basis reduction and AI32c proves its primitive moment norm; the invariance and strict remainder estimate for the coupled system AI35d remain the analytic work, rather than following from the reduction.

93.5 5. A normalized local Wilson Taylor theorem

This theorem supplies a finite-jet statement without supposing that exp(z O) has a complex source disc for a high-degree insertion.

Fix a finite set Lambda of integrated bonds, of cardinality and lattice diameter bounded by a constant depending only on the chosen insertion. Let B(Lambda) be the unintegrated boundary bonds entering plaquettes which meet Lambda. Fix any real outside configuration omega. Its conditional Wilson law is

$$d\mu_\Lambda^\omega(U) = Z_\Lambda(\omega)^{-1}e^{-S_\Lambda(U,\omega)}\prod_{b\in\Lambda}dU_b, \qquad S_\Lambda = \beta_a\sum_{p:p\cap\Lambda\neq\varnothing}s(P_p). \tag{AI36}$$

Only terms depending on the integrated links need appear. Let A_a be any bounded measurable observable depending only on those integrated links, with |A_a|<=C_D a^{-D}. In particular it can be a globally smooth fixed-D curvature/difference polynomial whose determining carrier is contained in Lambda. On each boundary bond perturb

$$\omega_b(Z) = e^{aZ_b}\omega_b, \qquad Z_b \in \mathfrak{sl}(N,\mathbb{C}). \tag{AI37}$$

The holomorphic Wilson action uses N-tr(P+P^{-1})/2 in place of N-Re tr P. Continue the conditional density, not the possibly nonsmooth observable. Define

$$F_a^\omega(Z) = \frac{\int A_a(U) e^{-S_\Lambda(U,\omega(Z))} \prod_\Lambda dU_b}{\int e^{-S_\Lambda(U,\omega(Z))} \prod_\Lambda dU_b}. \tag{AI38}$$

Theorem. If beta_a>=1, there are constants c_D,C_D independent of a, the outside field, the total lattice volume, and beta_a, such that AI38 is holomorphic on the boundary sup ball

$$\|Z\|_\infty < R_a := \frac{c_D}{a\beta_a}, \qquad |F_a^\omega(Z)| \le C_D a^{-D}. \tag{AI39}$$

For $\|Z\|_\infty \le R_a/2$ and every K>=0,

$$\left| F_a^\omega(Z) - \sum_{j=0}^{K} \frac{1}{j!} D^j F_a^\omega(0)[Z,\dots,Z] \right| \le C_D a^{-D} (C_D a \beta_a \|Z\|_\infty)^{K+1}. \tag{AI40}$$

There are only binomial(n_B+K,K) scalar Taylor monomials, with n_B=(N^2-1)|B(Lambda)| independent of a. Mixed differential norms have the bound C_D^{j+1} j! a^{j-D} beta_a^j; changing from diagonal to mixed norms costs at most the usual exponential polarization constant.

Proof. Every affected plaquette has four factors, and there are a bounded number of them. Unitary real interior and exterior factors have operator norm one. Matrix exponential telescoping and the corresponding inverse estimate therefore give, uniformly in the integrated links,

$$|S_\Lambda(U,\omega(Z)) - S_\Lambda(U,\omega)| \le C_D \beta_a a \|Z\|_\infty e^{C_D a \|Z\|_\infty}. \tag{AI41}$$

Choose c_D so that this is at most 1/4 on the stated ball. Divide the complex denominator by the positive real conditional partition function. The resulting ratio is E_{mu_Lambda^omega} exp(-Delta S), whose distance from 1 is at most exp(1/4)-1<1/2. It is nonzero uniformly, without any lower bound for the unnormalized partition function. The numerator divided by that same positive partition function is bounded by $e^{1/4}\|A_a\|_\infty$. This proves AI39. Apply one-variable Cauchy to t -> F_a^omega(tZ) on $|t| < R_a/\|Z\|$, and sum the geometric tail, to get AI40. The multivariate rank statement is elementary Taylor counting.

If the observable is a finite Wilson-word polynomial also involving the boundary links, its own holomorphic continuation obeys the same compact bound times $\exp(C_D a\|Z\|)$, and the argument still applies. A local log observable with a nonanalytic cutoff does not fall under that extension; the version with observable supported inside Lambda continues only the density and has no such problem.

For a bounded physical perturbation $\|Z\| \le M$, AI40 has remainder O(a^{K+1-D} beta_a^{K+1} M^{K+1}). In particular it gains a strict power when K>=D and beta_a grows at most polylogarithmically. The last growth assumption is explicit; the theorem itself is the bound AI40 for arbitrary beta_a>=1. Integrated root insertions receive their additional a^4 factor.

93.5.1 5.1 Scope of the theorem

AI36–41 are exact statements about the original Wilson conditional measure. They work around arbitrarily rough **real** outside configurations and for insertions of arbitrary fixed polynomial growth. No ordinary composite-source analyticity or interacting source majorant was assumed. They concern the response to a bounded physical boundary variation, not the assertion that the actual boundary potential is bounded uniformly.

The coefficients D^j F_a^omega(0) remain functions of omega. The theorem does not turn them into coefficients of a fixed physical operator basis, nor impose any normalization on them. If a block of fixed physical radius is used instead, its boundary dimension grows as a decreases, and the constant in AI41 involves an unbounded number of plaquettes. The stated finite-rank and radius bounds then no longer hold uniformly. These are the precise interfaces that prevent direct iteration into the missing all-scale, dimension-filtered mixing theorem.

Also, perturbing integrated variables and the observable together would merely perform a change of Haar variables. AI37 deliberately perturbs the unintegrated boundary and is a nontrivial conditional response.

93.5.2 5.2 Exact local EOM reduction to the conditional mean

There is also a direct connection between the local conditional theorem and the contact module. For a **smooth** A_a supported in Lambda, set F(omega)=E_{mu_Lambda^omega} A_a. Let X_i^dagger be the weighted adjoint in this conditional measure, and let

$$\mathcal{L}_\Lambda^\omega = \sum_{i\in\Lambda} X_i^\dagger X_i. \tag{AI41a}$$

This is an elliptic operator on a fixed compact product of groups. Its kernel consists of constants. If lambda_H>0 is the product-Haar Poincare gap, a density comparison proves the uniform-in-omega lower bound

$$\operatorname{gap}(\mathcal{L}_\Lambda^\omega) \geq \lambda_H \exp[-\operatorname{osc} S_\Lambda] \geq c_\Lambda e^{-C_\Lambda \beta_a}. \tag{AI41b}$$

Indeed, bound the conditional variance by the maximum density times the Haar variance, use the Haar Poincare inequality, and bound the Haar Dirichlet form by the inverse minimum density times the conditional one. Their ratio is exp(osc S_Lambda). The number of affected plaquettes is fixed, and each has bounded action, which gives the last inequality.

The unique conditional-mean-zero solution of L_Lambda^omega psi=A_a-F(omega) is smooth in the integrated links and the real boundary links. This follows from the positive gap and ordinary elliptic regularity on the fixed compact product; differentiating the equation in a boundary parameter and its zero-mean condition proves smooth parameter dependence. Define the globally smooth vector field, supported on the integrated bonds, by

$$V_i = (a^4\kappa_a)^{-1} X_i\psi, \qquad i\in\Lambda, \qquad V_i = 0 \quad \text{otherwise.} \tag{AI41c}$$

Then the following is an exact identity of functions on the full compact configuration space:

$$A_a - F(\omega) = \mathscr{E}_a^\circ[V]. \tag{AI41d}$$

The Wilson derivative on an integrated bond differentiates only S_Lambda, so its weighted adjoint is the same as in the full measure. Equations AI7 and AI41a now prove AI41d immediately. Uniqueness of the zero-mean solution and bi-invariance of the bond Laplacians also show that a gauge-invariant A_a produces the corresponding equivariant V.

Thus local conditional integration is exactly a quotient by a smooth local EOM divergence, with its contact rule AI9. This does not reduce arbitrary curvature polynomials to a **finite polynomial**

EOM ideal: the solution psi is generally not a polynomial of bounded engineering dimension, and the field F(omega) remains on the surrounding boundary. Even its elementary inverse bound costs exp(C_Lambda beta_a). No cutoff-uniform completed insertion norm follows from AI41b. This exact decomposition makes the remaining mixing question concrete rather than removing it: normalize and estimate the boundary function F across scales, without allowing unrelated choices for each marked configuration.

93.5.3 5.3 A same-Wilson-measure obstruction to a stronger inference

AI40 cannot be promoted to a pointwise finite-dimensional approximation of the conditional boundary function on **all** real outside configurations. There is a one-link SU(2) example inside the original Wilson measure.

Take a periodic lattice with at least three sites in every direction, and condition on every link except a single link U. Its six incident staple matrices can be prescribed independently by their six distinct middle links, with the other staple edges fixed. Its conditional density has the form

$$Z(C)^{-1}\exp\{\beta\,\mathrm{Re}\,\mathrm{tr}(UC)\}\,dU, \qquad C=\sum_{j=1}^{6}V_j. \tag{AI41e}$$

Orientations can be absorbed into V_j. Choose V_3=I,V_4=-I,V_5=I,V_6=-I, and V_1=diag(exp(i theta),exp(-i theta)), V_2=V_1^{-1}, with theta=arccos(c/2) and -1<=c<=1. Then C=cI. The observable chi(U)=Re tr U is the trace of the plaquette closed by the fixed third staple, hence is a genuine gauge-invariant Wilson-loop observable in the unconditioned configuration. The conditional expectation is

$$m_\beta(c)=m(\beta c), \qquad m(t)=\frac{\int_{SU(2)}\chi(U)e^{t\chi(U)}dU}{\int_{SU(2)}e^{t\chi(U)}dU}. \tag{AI41f}$$

It has |m(t)|<=2, m(-t)=-m(t) by the central change U to -U, and m(s)>0 for each fixed s>0. The last assertion follows from m(0)=0 and m'(t)=Var_t(chi)>0.

Let V be any fixed finite-dimensional subspace of C^1([-1,1]). There is a finite constant C_V such that $\|v'\|_\infty \le C_V\|v\|_\infty$ for every v in V, by finite-dimensional norm equivalence. Suppose v_beta in V approximated m_beta uniformly with error epsilon_beta. Evaluating at c=plus or minus s/beta gives

$$2m(s)\le 2\epsilon_\beta+\frac{2s}{\beta}C_V(2+\epsilon_\beta). \tag{AI41g}$$

Thus epsilon_beta cannot tend to zero. A fixed-degree polynomial approximation is the special case C_V<=K^2 from the Markov inequality, but no polynomial-specific assertion is needed.

On this analytic one-parameter family, any fixed finite list of local curvature/difference Wilson-word polynomials spans such a space V; their powers of a change coefficients but not that span. Therefore the uniform local Taylor radii in AI39 do not imply a finite physical mixing basis with a vanishing pointwise remainder on all rough boundary data. The transition in AI41f occurs on boundary link scale beta^{-1}, which corresponds to the physical perturbation size (a beta)^{-1} appearing in AI39. There is no contradiction with AI40 for bounded physical Z.

These exterior configurations are far from the regular flat-field core. The example does not disprove a completed, integrated estimate using their true large-field rarity, nor a mixing theorem restricted

to a regular chart and then matched to the large-field measure. It identifies exactly why that measure/normalization step cannot be omitted. Allowing the proposed finite basis itself to contain arbitrary beta-dependent functions such as m_beta would evade this example only by abandoning the fixed polynomial mixing assertion being tested.

93.6 6. Fixed finite jets and one common moment reserve

Let O_{D,a} be a monomial in n flux factors and a total of m covariant differences, so D=2n+m. Expanding each difference into its finitely many transported shifts, using unitary transport and $\|P-I\|^2 \le Cs(P)$, proves the global bound

$$|O_{D,a}(x)| \le C_D a^{-D} \left(\sum_{p \in C_D(x)} s(P_p) \right)^{n/2}. \tag{AI42}$$

The carrier C_D has a bounded number of bonds. Linear combinations are handled termwise; derivative differences do not create an extra infinite family at fixed D. This is the derivative version of the compact cubic birth estimate, not a normalized operator estimate.

Suppose a common integration representation for a **finite** marked coefficient has kernel K and nonnegative energy E, and suppose a positive majorant M is actually proved such that

$$|K| \le e^{-\delta E} M, \qquad \int M \le M_0, \qquad |O_i| \le A_i E^{n_i/2}. \tag{AI43}$$

Spatial carrier/tree weights may be included in the assumed bound on the single integral of M. Setting n=sum n_i gives the rigorous finite-jet bound

$$\left| \int K \prod_i O_i \right| \le M_0 \prod_i A_i \left(\frac{n}{2e\delta} \right)^{n/2}, \tag{AI44}$$

with the n=0 expression interpreted as 1. The proof is simply the maximum of E^{n/2}exp(-delta E) on E>=0. For a fixed engineering cutoff D and k insertions this is bounded by $C_D^{k(k!)}\{q_D\}$ times the primitive engineering factors in the A_i. There is no convergence assertion for the infinite series exp(z O), and none is needed for finite or nilpotent source jets.

The one common majorant in AI43 is substantive. One may not replace it by a product of separately integrated live source coefficients. Its cold and tree reserves must remain after any other original scale estimates have spent their portions. Even when AI43 follows from an original full-core map, AI44 still retains the factors a^{4-D} of integrated primitive insertions. It does not perform their dimension-filtered mixing.

93.6.1 6.1 High-degree finite jets of an actual bounded reference normalizer

There is a useful normalization consequence stronger than AI44. Suppose an actual fixed reference component has a density j_b(B), with its real cutoff set frozen, and satisfies the explicit source-free bounds

$$J_b(0) = \int j_b(B) dB, \qquad |J_b(0)| \ge c > 0, \qquad \int |j_b(B)| e^{t(1+|B|^2)} dB \le C, \tag{AI44a}$$

uniformly on its admitted background domain. These are precisely the sort of bounds S31 verifies for a bounded regular reference component; they are hypotheses here for the component being used, not an assertion about every still-live normalization.

Let eta_1,...,eta_k be commuting squarefree variables, eta_iˆ2=0, and let the globally defined inserted functions obey

$$|O_i(B)| \le A_i(1+|B|^2)^{d_i/2}, \qquad d_i \le d. \tag{AI44b}$$

Define J_b(eta) by integrating the **finite polynomial** exp(sum_i eta_i O_i) against j_b. This is a formal finite jet; no ordinary complex source domain for exp(z O_i) is postulated. Its normalized moments M_I=[eta_I]J_b(eta)/J_b(0), for |I|=m, satisfy

$$|M_I| \le C_d^m (m!)^{d/2} \prod_{i\in I} A_i. \tag{AI44c}$$

To prove this, use the single moment reserve in AI44a on Qˆ{sum d_i/2}, Q=1+|B|ˆ2. Its supremum after multiplication by exp(-tQ) is bounded by (C_d m)ˆ{dm/2}; the elementary bound m!>=(m/e)ˆm gives AI44c after the fixed ratio C/c is absorbed into C_dˆm.

The body of J_b(eta)/J_b(0) is one. Its reciprocal and formal logarithm therefore exist automatically in this finite algebra. With q_d=max(1,d/2), both satisfy

$$\left|[\eta_I]\log\frac{J_b(\eta)}{J_b(0)}\right| + \left|[\eta_I]\left(\frac{J_b(\eta)}{J_b(0)}\right)^{-1}\right| \le C_d^m (m!)^{q_d}\prod_{i\in I} A_i, \qquad I\neq\varnothing. \tag{AI44d}$$

For the logarithm use its exact partition formula sum_pi (-1)ˆ{|pi|-1}(|pi|-1)! product_{B in pi}M_B. Bound the moments by AI44c with factorial exponent q_d. Writing an ordered partition by its positive block sizes b_1+...+b_l=m bounds the partition sum by

$$m!\sum_{l=1}^{m}\frac{1}{l}\sum_{b_1+\cdots+b_l=m}\prod_j (b_j!)^{q_d-1} \le 2^{m-1}(m!)^{q_d}. \tag{AI44e}$$

Here product b_j!<=m! and q_d>=1. The reciprocal has l! instead of (l-1)! in the unordered partition formula, which removes 1/l and gives the same bound. This proves AI44d. Uniform background analyticity of the coefficients follows by dominated integration whenever that of j_b and the O_i is part of the admitted component data.

This proves the local finite-order normalization and contact-cumulant algebra for arbitrary fixed polynomial growth. The engineering factors A_i remain exactly as supplied. It does not prove their incoming normalized old-scale bounds, their spatial derivative filtration, or the physical exactness of a replacement operation. In particular an exact formal quotient here cannot repair an independently misplaced old-variable denominator in a purported carrying identity.

93.7 7. Exact remaining estimate

The all-degree EOM/contact sector is now explicit, and AI30 reduces one important scalar derivative operator to a completely written complement. AI40 supplies a genuine local normalized jet bound;

AI44 supplies the correct finite-source moment mechanism under its stated joint reserve. The missing estimate can therefore be formulated without requiring an unnecessary ordinary source disc:

For each fixed D, choose a finite physical gauge-invariant local basis of dimension at most D, a common normalization independent of other marked points, and an actual local projection P_{<=D}. For the exact near-mark Wilson integration map I_j, prove that

$$(I - P_{\le D})I_j(O) \quad \text{has a completed connected norm bounded by} \quad C_D L^{-\epsilon_D}\|O\|_D \tag{AI45}$$

with all original tree, physical engineering, age, and cold reserves accounted for, and with factorial control of every fixed finite source jet. The projection coefficients must agree with the fixed physical normalization; a different field-dependent Taylor coefficient for each outside configuration is not enough. For the scalar AI30 this includes the explicit R_a and the point-split cubic comparison. For the hypercubic complement AI35 and for higher-spin/higher-derivative operators it remains a genuinely new near-mark mixing problem.

No conditional source bound, classical smooth-field estimate, or finite Gaussian completion has been treated as a proof of AI45.

94 An actual regular Wilson minimizer with an unbounded higher curvature jet

This is a counterexample to a proposed **pointwise conditional-background regularity estimate**, not to a quantum Yang–Mills construction. It uses an exact Cartan Wilson solution with the actual raw constraint. Its potential has uniformly small W^{1,p} and C^{1,beta} norms and its logarithmic curvature has uniformly small supremum, but the square of a second covariant difference of its globally defined plaquette flux has integral proportional to a^{-1}. Full fluctuation completion can cancel conditional singularities; this example does not estimate that completed object.

94.1 1. Lattice, multiplier, and exact box adjoint

Take SU(2), a four-torus of side T, with T a fixed positive even integer. The retained spacing is r=1. Let

$$a = L^{-k} = n^{-1}, \qquad n = L^k \text{ odd}, \qquad H = \frac{i}{\sqrt{2}}\,\mathrm{diag}(1,-1), \qquad -\operatorname{tr} H^2 = 1. \tag{HJ1}$$

Use directions mu=1 and nu=2. On coarse bonds put

$$\lambda_2(y) = \varepsilon(-1)^{y_1} H, \qquad \lambda_\rho(y) = 0 \quad (\rho \neq 2). \tag{HJ2}$$

This has zero mean and is flat co-closed: the only component varies in direction 1, whereas its divergence differentiates direction 2.

Let Q=Q_bl,a,1 be the actual centered box-and-line average. Its physical adjoint first holds a coarse field on centered n^4-site blocks, then applies the normalized own-component line average. Since lambda_2 is constant in direction 2, that line average changes nothing. Hence

$$(Q^*\lambda)_2(x) = j_a(x_1)H, \qquad j_a(x_1) = \varepsilon(-1)^{y_1} \quad \text{for } x_1 \text{ in the centered block at } y_1. \tag{HJ3}$$

The other components vanish. The scalar j_a is a literal step function, not an approximation. It has T jump planes, with jumps of size 2 epsilon, and satisfies

$$j_a(x_1+1) = -j_a(x_1), \qquad \overline{j_a} = 0. \tag{HJ4}$$

Odd n makes the centered boxes an exact fine-site partition. An even number of retained cells makes the half-wave relation periodic, so it is compatible with every allowed odd blocking factor.

94.2 2. Solve the exact scalar Wilson flux equation

Let D=(tau_a-I)/a and D^*=(tau_{-a}-I)/a on the one-dimensional periodic mesh. There is a unique mean-zero scalar Y_a satisfying

$$D^* Y_a = j_a. \tag{HJ5}$$

One may construct it by the cumulative-sum equation Y_a(x)-Y_a(x-a)=-a j_a(x). Periodicity follows from the zero sum of j_a. Alternatively invert the nonzero discrete Fourier modes. Uniqueness and HJ4 imply

$$Y_a(x+1) = -Y_a(x), \qquad \|Y_a\|_\infty \le C\varepsilon, \|DY_a\|_\infty = \varepsilon. \tag{HJ6}$$

The bound follows directly by summing at most one unit block before using half-wave symmetry; its constant is independent of a.

For the SU(2) generator H, the projected Wilson flux of a logarithmic plaquette tH is the scalar odd function

$$\Phi_a(tH) = \phi_a(t)H, \quad \phi_a(t) = \frac{\sqrt{2}}{a^2}\sin\left(\frac{a^2 t}{\sqrt{2}}\right). \tag{HJ7}$$

Choose epsilon small once and define its real analytic inverse

$$F_a(x) = \frac{\sqrt{2}}{a^2}\arcsin\left(\frac{a^2 Y_a(x)}{\sqrt{2}}\right). \tag{HJ8}$$

The argument stays in one fixed neighborhood of zero for every a<=1. The inverse is odd, so F_a(x+1)=-F_a(x) and F_a has mean zero. Therefore there is a unique mean-zero periodic scalar potential v_a satisfying

$$Dv_a = F_a. \tag{HJ9}$$

Again a cumulative sum proves existence and gives $\|v_a\|_\infty \le C$ epsilon. The derivative of the inverse in HJ8 is uniformly bounded, so $\|DF_a\|_\infty \le C\epsilon$. These assertions include exact periodic solvability; an arbitrary inversion of the flux followed by an unjustified integration of a nonzero mean is not being used.

Define the actual fine links by

$$U_2(x) = e^{av_a(x_1)H}, \qquad U_\rho(x) = I \quad (\rho \neq 2). \tag{HJ10}$$

The only nontrivial oriented plaquettes are P_12(x)=exp(a^2 F_a(x_1)H) and P_21=P_12^{-1}. Their logarithmic curvature is F_a H and their globally smooth projected flux X_12 is **exactly** Y_a H. Since the direction-1 transports are identity and the direction-2 transports commute with H, the Wilson current is

$$J_2(U) = D^* Y_a\, H = j_a H, \qquad J_\rho(U) = 0 \quad (\rho \neq 2). \tag{HJ11}$$

Here J is the geometrically normalized physical derivative a^{-3} X_b S_0, as in AI3–6. Multiplying the Wilson action by a positive bare coupling coefficient only multiplies the KKT multiplier by that same coefficient and changes neither the minimizer nor the construction.

94.3 3. Exact raw constraint and full non-Abelian KKT

All links and all contour transports in this configuration lie in the same small Cartan subgroup. Constant Cartan conjugation and uniqueness of each regular mean keep the averaged endpoint transports in that subgroup as well. The actual **outer reference-log** operation CM1 is exactly arithmetic on commuting angles: its reference angle cancels. Each contour angle is an incoming endpoint angle, plus the middle-line angle, minus the translated outgoing endpoint angle. Averaging those endpoint angles gives a coarse scalar kappa_j. Thus one step, and then the exact composition using Q_j d=d R_j, have the form

$$q_{\mathrm{raw}}(z) = Qz - d_r\kappa(z) \quad \text{on Cartan fields}, \qquad Q_{\mathrm{raw}} = Q - d_r K_{a,1}, \quad K_{a,1} = D\kappa(0). \tag{HJ12}$$

The inner endpoint mean is not presumed to be an arithmetic mean; any of its nonlinearity remains in kappa. This distinction does not affect the following Ward pairing. The finite-step endpoint support is the actual one used by the raw averaging map.

Set the prescribed target to B_a=q_raw(v_aH) and V_a=exp(B_a). This is the literal actual raw target. It has uniformly small W_1^{1,p} norm by the actual nonlinear raw bound WP. There is no requirement that its displayed matrix values be independent of a; the counterexample already disproves a uniform estimate on the whole fixed small data ball.

The coarse physical Ward equation is exact. All V_a lie in the Cartan subgroup, lambda is Cartan valued, and the component lambda_2 is constant in direction 2. Thus d_{V_a}^t lambda=delta_r lambda=0.

The physical adjoint of the full, non-Abelian raw differential applied to lambda is also exact. The background and lambda are invariant under every constant Cartan conjugation. Covariance therefore forces the adjoint output to lie in the Cartan centralizer, which in su(2) is the one-dimensional span of H. Its pairing with Cartan variations is given by HJ12; all off-Cartan components vanish. Since Cartan exponential differentials are identity in physical connection units,

$$D\mathcal{M}_r(U)^{\mathsf{t}}\lambda = Q^*\lambda - D\kappa(u)^{\mathsf{t}}\delta_r\lambda = Q^*\lambda = Q^*_{\mathrm{raw}}\lambda = J(U). \tag{HJ13}$$

The penultimate equality also follows directly from Q_raw$^{lambda=Q^}$lambda-K_{a,1}*delta_r lambda. Both the actual and flat endpoint terms vanish because the same flat Ward residual is zero.

This verifies the full physical KKT equation, rather than only its restriction to Abelian variations. The potential u_2=v_aH, u_rho=0 is flat co-closed, so it already lies in the exact R_0 delta slice.

94.4 4. The solution is on the real regular minimizing orbit

For every fixed finite p and every beta in (0,1), HJ6–10 give

$$\|u_a\|_{W_a^{1,p}} + \|u_a\|_{C_a^{1,\beta}} + \|\Delta_a u_a\|_\infty + \|F_a(u_a)\|_\infty + \|\lambda\|_{W_1^{-1,p}} \le C_{p,T,\beta}\varepsilon. \tag{HJ14}$$

For example, Delta u is a shifted first difference of F, which is bounded by HJ8; the discrete first derivative of u is uniformly Lipschitz. The source input A=u_a itself therefore satisfies the real WC weak-plus-curvature hypotheses once epsilon is chosen sufficiently small, uniformly in a.

Minimality can now be checked using the already established real-orbit identification, rather than merely inferred from stationarity. The explicit triple (u_a,chi=0,lambda) is in the CK uniqueness ball, is co-closed, obeys the actual raw target, and satisfies its modified equations because HJ13 and Ward hold. It is therefore the CK branch for that target. Applying CK26–30 with the real input A=u_a identifies this branch with the original genuine regular constrained minimizing orbit. This step uses exactly the published regular-orbit variational theorem already imported in CK, together with its proved uniform weak identification; it does not assert global convexity of the Wilson action on all compact links. Thus the example belongs to the intended small regular minimizing orbit, not an extraneous stationary saddle.

One can also see the corresponding flat statement directly: u=GQˆ* lambda is the unique quadratic curl minimizer with B=Qu in its slice. The Cartan construction above is its exact Wilson flux replacement and the CK argument verifies that it remains on the regular minimum branch.

94.5 5. An exact divergent dimension-eight insertion

The covariant direction-1 differences coincide with ordinary differences on this configuration. Since X_12=Y_aH, HJ5 implies

$$D_1 X_{12} = -\tau_a j_a\, H, \qquad D_1^2 X_{12} = -\tau_a D j_a\, H. \tag{HJ15}$$

The last field has magnitude 2 epsilon/a on exactly T jump planes and is zero elsewhere. Each plane has (T/a)ˆ3 fine sites. With physical volume and $\|H\| = 1$, this gives the exact value

$$a^4 \sum_x |D_1^2 X_{12}(x)|^2 = \frac{4\varepsilon^2 T^4}{a}. \tag{HJ16}$$

The insertion is a globally smooth gauge-invariant curvature/covariant- derivative polynomial, of engineering dimension eight. Its fine determining carrier is fixed. There is no local logarithm cutoff in its definition. A nonnegative sum over directions, or the finite reflection average of this entire squared insertion, also diverges, since it contains this term with a fixed positive weight. If the logarithmic curvature is used instead, the same conclusion persists: HJ8 gives $\|F_a - Y_a\|_\infty \le Ca^4\epsilon^3$, so a crude two-difference bound gives $\|D^2(F_a - Y_a)\|_\infty \le Ca^2\epsilon^3$. This is negligible compared with the L2 norm 2 epsilon Tˆ2 aˆ{-1/2} in HJ16.

Accordingly there is no cutoff-uniform estimate of the form

$$\|D_a^2 X_a(u)\|_2^2 \le C\,\mathcal{P}(\|u\|_{W^{1,p}}, \|u\|_{C^{1,\beta}}, \|\Delta u\|_\infty, \|F(u)\|_\infty) \tag{HJ17}$$

with a fixed polynomial or locally bounded right side, even when u is an actual same-target regular Wilson minimizer in the common chart.

This is consistent with FE, CC, and NL. A step current belongs uniformly to W^{s,p} for s<1/p and to H^s for s<1/2, but its next derivative has the jump singularity displayed here. It explains why those results cannot simply be iterated to arbitrary spatial jets at an insertion. The conditional step is created by the coarse constraint. A physical quantum insertion requires all fluctuation/covariance and interaction terms on the same Wilson measure, with its local mixing fixed. HJ16 does not estimate that completed expectation and gives no obstruction to its possible construction.